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**Where you are** CAS Formulas Functions Algebra Calculus Probability the same colour on the edge tab, the page head and the topic bar

**What it is** Tricky explained in full Trap Key idea Exam rule Definition `command` type it on the calculator

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## CALCULUS


| | | |
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## DATA ANALYSIS, PROBABILITY & STATISTICS


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CAS

START HERE

# TI-Nspire CX II CAS commands

Typed in a Calculator page on the TI-Nspire CX II CAS. Each example has a checked output (rounded: the screen shows 6 significant figures by default), so you can test your syntax against an answer you already know; the page number in each row is the notes page for that question type, and the row numbers (C1, C2, ...) are what the question finder points to. On paper, always write the equation you solved and the answer in the form asked, not just "by CAS".

## Set up

| To do this | Type or press |
|---|---|
| C1 Radians, real answers | doc ▸ Settings: Angle **Radian**, Real or Complex **Real**, Calculation Mode **Auto** (Home ▸ Settings ▸ Document Settings ▸ Make Default for new documents) |
| C2 Start each question clean | insert a new Problem (doc ▸ Insert ▸ Problem), or `ClearAZ` (menu ▸ Actions ▸ Clear a–z; single-letter names only) |
| C3 Decimal answer | **ctrl + enter**, or type a decimal point ( `2.` ): decimals in give decimals out. `exact(0.25)` gives 1/4; `approx(π/4)` gives 0.785398 |
| C4 Last answer; "such that" | `ans` (ctrl + (−)); `\|` from the ctrl + = palette, e.g. `f(x)\|x=2` |
| C5 Define; store a value | `Define f(x)=x^3-3·x` (menu ▸ Actions ▸ Define); store an unrounded result with **ctrl + var** (→), e.g. `normCdf(180,∞,170,8)→pr`, then use `pr` (pick a name you won't solve for later) |
| C6 Special symbols | `e^(x)` from the $e^x$ key (the letter e is a variable); ∞ from the catalog's symbols, or `1E99` (the E from the **EE** key, not the letter); `<=` and `>=` become ≤ and ≥ |

## Algebra and parameters

| To do this | Type | Gives |
|---|---|---|
| C7 Solve | `solve(x^2-5·x+6=0,x)` | x = 2 or x = 3 |
| C8 Solve on an interval → p. 38 | `solve(2·sin(2·x)=1,x)\|0≤x≤π` | x = π/12 or x = 5π/12 |
| C9 Solve numerically → p. 53 | `nSolve(x^3+x-1=0,x)\|0≤x≤1` or with a guess `nSolve(x^3+x-1=0,x=0.5)` | 0.682328 (one solution only) |
| C10 Simultaneous equations → p. 50 | `solve(2·x+y=5 and x-y=1,{x,y})` | x = 2 and y = 1 |
| C11 None or infinitely many solutions → p. 50 | `solve(det([a,3;1,a-2])=0,a)`, then solve the system with each value of a | a = −1 or a = 3; `false` means no solution, an answer in terms of c1 means infinitely many |
| C12 Factorise, expand; roots → p. 18 | `factor(2·x^3-3·x^2-11·x+6)`; `expand((x+1)^3)`; `zeros(x^3-2·x^2-5·x+6,x)` | (x − 3)(2x − 1)(x + 2); $x^3 + 3x^2 + 3x + 1$; {−2, 1, 3} |
| C13 Remainder; hyperbola form; turning-point form → pp. 18, 24 | `polyRemainder(x^3+2·x+1,x-1,x)`; `propFrac((2·x+3)/(x+1),x)`; `completeSquare(x^2-6·x+5,x)` | 4; 1/(x + 1) + 2; $(x - 3)^2 - 4$ |
| C14 Inequality → p. 92 | `solve(1-0.9^n>0.95,n)` | n > 28.4332, so n = 29 |

| To do this | Type | Gives |
|---|---|---|
| **C15** Sum of the solutions → p. 41 | `zeros(2·sin(2·x)-1,x)\|0≤x≤π`, then `sum(ans)` | {π/12, 5π/12}, then π/2 |
| **C16** Quadratic inequality (then use the range) → p. 39 | `solve(2·a^2+a-1>0,a)` with a = sin x | a < −1 or a > 1/2, so sin x > 1/2 |
| **C17** k for a number of solutions → pp. 21, 51 | `solve(k^2-16>0,k)` (Δ of $x^2 + kx + 4 > 0$) | k < −4 or k > 4 |
| **C18** Cubic from turning points (coefficients from conditions) → p. 20 | `Define p(x)=a·x^3+b·x^2+c·x+d`, `Define dp(x)=derivative(p(x),x)`, `solve(p(1)=5 and p(3)=1 and dp(1)=0 and dp(3)=0,{a,b,c,d})` | a = 1, b = −6, c = 9, d = 1 |
| **C19** Check a functional equation → p. 48 | `Define f(x)=e^(3·x)`, then `f(x+y)=f(x)·f(y)` | true (anything else isn't a proof that it fails: test values) |

## Functions and graphs

| To do this | Type | Gives |
|---|---|---|
| **C20** Maximal domain → p. 14 | `domain(ln(x-2)+√(5-x),x)` | 2 < x ≤ 5 |
| **C21** Hybrid function → p. 15 | `Define g(x)=piecewise(x+3,x<-1,x^2+1,-1≤x≤1, 4-2·x,x>1)` (or the template) | use g like any function |
| **C22** Restricted domain → p. 44 | `Define r(x)=(x-2)^2-3\|x≤2`, then `solve(r(x)=x,x)` | x = (5 − √21)/2 ≈ 0.209 (the other root is outside the domain) |
| **C23** Composite → p. 43 | `f(g(x))` | the rule of f∘g |
| **C24** Inverse, choosing the branch → p. 44 | `solve(x=(y-2)^2-3,y)\|y≤2` | y = 2 − √(x + 3) |
| **C25** Image under a transformation → p. 27 | y = √x under (x, y) → (1 − 2x, 3y − 2): `solve((y+2)/3=√((1-x)/2),y)` | y = 3√((1 − x)/2) − 2 |
| **C26** Where f meets its inverse → p. 45 | `Define f(x)=1-x^3`, then `solve(f(f(x))=x,x)` | all five x-values (`f(x)=x` finds only the one on y = x) |
| **C27** Tangency with a parameter → p. 30 | `solve(e^(x)=k·x and e^(x)=k,{x,k})` (equal values and equal gradients) | x = 1 and k = e |
| **C28** A family of graphs → p. 30 | on a Graphs page: `f1(x)=x^2-2·{1,2,3}·x+3` | draws all three members |
| **C29** Graph of f′ → p. 67 | on a Graphs page: `f2(x)=derivative(f1(x),x)` | f′ drawn under f, for "which graph could be f′" questions |
| **C30** Graphs page | menu ▸ Analyze Graph ▸ Zero / Minimum / Maximum / Intersection / Inflection; ctrl + T for a table; `f1(x)=x^2\|0≤x≤3` restricts the domain | readings are approximate: confirm in Calculator |
| **C31** Other functions | `abs(x)`, `ln(x)`, `log(x)` (base 10), `log(x,2)` (base 2), `root(x,3)` | |

# Calculus

The examples below use `Define f(x)=x^3-3·x^2-9·x+5` and `Define df(x)=derivative(f(x),x)` unless they say otherwise.

| To do this | Type | Gives |
|---|---|---|
| **C32** Derivative; at a point; second → p. 60 | `derivative(f(x),x)`; `df(2)`; `derivative(f(x),x,2)` | $3x^2 - 6x - 9$; −9; $6x - 6$ |
| **C33** Stationary points → p. 64 | `zeros(df(x),x)`, then `f(ans)` | {−1, 3}, then {10, −22}: (−1, 10) and (3, −22) |
| **C34** Where decreasing → p. 64 | `solve(df(x)<0,x)` | −1 < x < 3 (write [−1, 3] for "strictly decreasing") |
| **C35** Inflection point → p. 66 | `solve(derivative(f(x),x,2)=0,x)`, then `df(1)` | x = 1; gradient −12 there |
| **C36** Greatest rate of change on an interval → p. 66 | steepest descent `fMin(df(x),x)\|0≤x≤5`; steepest ascent `fMax(df(x),x)\|0≤x≤5`, then `df(x)\|ans` for each (periodic? add multiples of the period) | x = 1, −12; x = 5 (an endpoint), 36 |
| **C37** Tangent, normal → p. 63 | `tangentLine(x^3-2·x,x,1)`, `normalLine(x^3-2·x,x,1)` | x − 2 and −x: write y = x − 2, y = −x |
| **C38** Tangent through an outside point → p. 63 | for $y = x^2$ through (1, −3): `Define w(x)=x^2`, `Define dw(x)=derivative(w(x),x)`, then `solve(-3-w(a)=dw(a)·(1-a),a)` | a = −1 or a = 3 |
| **C39** Smooth join of a hybrid (join at x = 1) → p. 58 | `Define p(x)=a·x^2+b`, `Define q(x)=4·x-1`, `Define dp(x)=derivative(p(x),x)`, `Define dq(x)=derivative(q(x),x)`, then `solve(p(1)=q(1) and dp(1)=dq(1),{a,b})` | a = 2 and b = 1 |
| **C40** Max or min on an interval → p. 72 | `fMax(f(x),x)\|0≤x≤5` (or `fMin`), then `f(x)\|ans`; periodic? add multiples of the period for every time | x = 5 (an endpoint), then 10 |
| **C41** Minimum distance to a curve → p. 71 | from (3, 0) to y = √x: `fMin((x-3)^2+x,x)\|x≥0`, then `√((x-3)^2+x)\|ans` (minimise the squared distance) | x = 5/2, then √11/2 ≈ 1.6583 |
| **C42** Greatest difference of two models → p. 72 | with u(x) = 2sin(x) + 3 and v(x) = 0.5x + 2 defined: `fMax(abs(u(x)-v(x)),x)\|0≤x≤5`, then `abs(u(x)-v(x))\|ans` | x = 4.965, then 3.419 |
| **C43** Antiderivative; definite integral → p. 74 | `integral(f(x),x)` (add + c yourself); `integral(x^2-1,x,0,2)` | 2/3 (signed: split at the x-intercepts for area) |
| **C44** Area between curves → p. 78 | `integral(x-x^2,x,0,1)` (upper − lower, split where they cross) | 1/6 |
| **C45** Intersections as exact limits → p. 78 | `zeros(x^2-(x+1),x)→xs`, then `integral(x+1-x^2,x,xs[1],xs[2])` (check xs[1] < xs[2]) | {(1 − √5)/2, (1 + √5)/2}, then 5√5/6 ≈ 1.8634 |
| **C46** Area when curves cross (one integral) → p. 79 | for $f(x) = x^3 - x$ and g(x) = 3x: `integral(abs((x^3-x)-3·x),x,-2,2)` (ctrl + enter if it stays unevaluated) | 8 |
| **C47** Average value → p. 79 | `1/2·integral(e^(x),x,0,2)` | $(e^2 - 1)/2$ |
| **C48** Solve for an integral's limit → p. 80 | `solve(integral(10·t,t,0,k)=45,k)\|k>0` | k = 3 |

| To do this | Type | Gives |
|---|---|---|
| C49 Integrate a hybrid → p. 15 | with the hybrid g defined by `piecewise` (above): `integral(g(x),x,-2,2)`; or split the integral at each join | 31/6 |
| C50 Limit → p. 57 | `limit((x^2-9)/(x-3),x,3)`; from the left/right add `,-1` or `,1` | 6 |
| C51 Newton's method → p. 53 | `Define f(x)=x^3-2·x-5`; type `2.` and press enter; then `x-f(x)/derivative(f(x),x)\|x=ans` and re-run it (▲, enter, enter) for each step | 2.1, 2.09457, 2.09455 |
| C52 Left and right rectangles → p. 82 | on [0, 2], width 0.5, with `Define f(x)=x^2`: `0.5·sum(seq(f(0.5·i),i,0,3))`; `0.5·sum(seq(f(0.5·i),i,1,4))` | 1.75 (left); 3.75 (right) |
| C53 Trapezium rule → p. 82 | `Define f(x)=x^2`, then `0.5/2·(f(0)+2·sum(seq(f(0.5·i),i,1,3))+f(2))` | 2.75 |

## Probability and statistics

Distributions: menu ▸ Probability ▸ Distributions. Confidence intervals: menu ▸ Statistics ▸ Confidence Intervals.

| To do this | Type | Gives |
|---|---|---|
| C54 Independent trials, different probabilities → p. 87 | `expand((0.5·x+0.5)·(0.4·x+0.6)·(0.2·x+0.8))` (one bracket per trial: p·x + (1 − p)) | $0.04x^3 + 0.26x^2 + 0.46x + 0.24$: the coefficient of $x^k$ is P(k successes) |
| C55 DRV mean and variance → p. 89 | `sum({0,1,2,3}·{0.1,0.3,0.4,0.2})`; `sum({0,1,2,3}^2·{0.1,0.3,0.4,0.2})-1.7^2` | 1.7; 0.81 |
| C56 Binomial P(X = x); nCr; the whole distribution → p. 91 | `binomPdf(10,0.3,3)`; `nCr(4,3)`; `binomPdf(10,0.3)` | 0.2668; 4; {0.0282, 0.1211, 0.2335, 0.2668, 0.2001, ...}: the mode is 3 |
| C57 Binomial P(a ≤ X ≤ b), P(X ≤ b) → p. 92 | `binomCdf(10,0.3,3,10)`; `binomCdf(10,0.3,2)` | 0.6172 (P(X ≥ 3)); 0.3828 |
| C58 Conditional (binomial) → p. 92 | `binomCdf(10,0.3,3,10)/binomCdf(10,0.3,1,10)` | 0.6352 = P(X ≥ 3 \| X ≥ 1) |
| C59 Smallest n (table of values) → p. 92 | `seq(1-0.9^n,n,27,30)` | 0.9419, 0.9477, 0.9529, 0.9576: n = 29 |
| C60 Integer from a rounded probability → p. 92 | `seq(1-0.9^n,n,13,19)`, keep the values in [0.75, 0.85) for "0.8 to 1 d.p." | 0.7458, 0.7712, ..., 0.8499, 0.8649: n = 14 to 18 |
| C61 n and p from the mean and variance → p. 93 | `solve(n·p=6 and n·p·(1-p)=4.2,{n,p})` | n = 20 and p = 0.3 |
| C62 Unknown p → p. 93 | `solve((1-p)^10=0.2,p)\|0<p<1` (write the formula out) | p = 0.1487 |
| C63 CRV: k, mean, variance → p. 95 | `solve(integral(k·x^2,x,0,3)=1,k)`; `integral(x·x^2/9,x,0,3)`; `integral(x^2·x^2/9,x,0,3)-(9/4)^2` | k = 1/9; 9/4; 27/80 (= 0.3375) |
| C64 CRV: median, percentile → p. 95 | `solve(integral(t^2/9,t,0,m)=0.5,m)\|0≤m≤3` (0.25 for the 25th percentile) | m = 2.381 (1.890) |

| To do this | Type | Gives |
|---|---|---|
| **C65** A parameter inside an integral, tabulated → p. 95 | `seq(integral(t^2/9,t,k,3),k,0,3,0.5)` (low, high, step) | 1, 0.9954, 0.9630, 0.875, 0.7037, 0.4213, 0: P(X > k) for each k |
| **C66** pdf, then binomial (chain) → pp. 92, 95 | P(at least one of 2 values exceeds k), k = 1, 1.5, ..., 3: `seq(1-(1-integral(t^2/9,t,k,3))^2,k,1,3,0.5)` | 0.9986, 0.9844, 0.9122, 0.6651, 0 |
| **C67** Piecewise pdf, conditional → p. 97 | `Define f(x)=piecewise(x,0≤x≤1,2-x,1<x≤2,0)` (the last 0 is "otherwise"), then `integral(f(x),x,1.5, 2)/integral(f(x),x,1,2)` | 0.25 = P(X > 1.5 \| X > 1) |
| **C68** Normal probability → p. 100 | `normCdf(180,∞,170,8)` (lower, upper, μ, σ: not σ²) | 0.1056 |
| **C69** Conditional (normal) → p. 101 | `normCdf(180,∞,170,8)/normCdf(170,∞,170,8)` | 0.2113 |
| **C70** Normal, then binomial (keep it unrounded) → p. 101 | `normCdf(180,∞,170,8)→pr`, then `binomCdf(10,pr,2, 10)` | 0.2858 = P(at least 2 of 10 are over 180) |
| **C71** Inverse normal; z-value → p. 100 | `invNorm(0.9,170,8)` (area to the **left**); `invNorm(0.1)` | 180.25; −1.2816 |
| **C72** Unknown σ → p. 101 | `solve((40-50)/s=invNorm(0.1),s)` | s = 7.80 |
| **C73** Unknown μ and σ → p. 101 | `solve((30-m)/s=invNorm(0.2) and (60-m)/s=invNorm(0.9),{m,s})` | m = 41.89, s = 14.13 |
| **C74** Sample proportion, exact → p. 105 | `binomCdf(100,0.3,36,100)` for P(p̂ > 0.35), n = 100 | 0.1161 |
| **C75** Sample proportion, normal approximation → p. 105 | `normCdf(0.35,∞,0.3,√(0.3·0.7/100))` | 0.1376 |
| **C76** Simulate sample proportions → p. 104 | `randBin(100,0.3,50)/100` | 50 simulated values of p̂ |
| **C77** Confidence interval for p → p. 107 | `zInterval_1Prop 60,200,0.95` | stat.CLower = 0.2365, stat.CUpper = 0.3635 |
| **C78** z for a C% interval → p. 107 | `invNorm((1+0.9)/2)`; `invNorm(0.975)`; `invNorm(0.995)` | 1.6449 (90%); 1.9600 (95%); 2.5758 (99%) |
| **C79** Sample size for a margin of error → p. 108 | `solve(invNorm(0.975)·√(0.5·0.5/n)=0.03,n)` | n = 1067.07, so n = 1068 (always round up) |
| **C80** Confidence level from an interval → p. 108 | z = margin ÷ standard error, then `normCdf(-1.645,1.645)` | 0.90, a 90% interval |

**CAS traps that cost marks**

- **Type the multiplication.** `kx` is one variable called "kx": type `k·x`. And `x(x+1)` is read as a function: type `x·(x+1)`.
- **Old definitions leak.** A new Problem for each question is safest; `ClearAZ` clears only single-letter names. Define stores the expression as it was, so redefine anything built from f after changing f. `DelVar p` (menu ▸ Actions ▸ Delete Variable) clears one name you stored.
- **nSolve, fMax and fMin return one value.** Restrict the domain to find each one, and for periodic models add the period to list every time in the domain.
- **Check every solution against the domain.** solve can return endpoint or extra roots (e.g. t = 0 where two models both start): keep only the ones the question allows.
- **Trig without a domain** gives general solutions with `n1` (any integer): add the question's domain, e.g. `|0≤x≤2·π`.
- **normCdf and invNorm take σ**, not the variance, and invNorm wants the area to the left.
- **Rounding.** If Pr = 0.2 "correct to 1 decimal place", the true value is in [0.15, 0.25). Don't solve = 0.2 exactly for an integer like n or k: tabulate the candidates with `seq` and keep every one in the interval. Store unrounded values (→) and round only at the end.
- **No rounding stated means exact.** In Section B, if no decimal places are given, the answer must be exact: enter exact inputs (fractions, π, √, not decimals) so CAS stays exact; `exact()` only turns a decimal into a simple fraction and can't recover √ or π.
- **CAS answers come in their own form.** Convert to the exact form asked, and write the setting-out a marker needs.

START HERE

# Methods formula sheet & must-knows

VCAA provides a Methods formula sheet in both exams (mensuration, the derivative and antiderivative rules, and the probability and statistics formulas). Many things you use constantly aren't on it: exact values, log laws, transformations and the discriminant. Treat this page as a revision summary. Always check the current official sheet from the VCAA website before your exam.

Formulas already on the official VCAA sheet (derivative and antiderivative rules, the trapezium rule, Newton's method, probability rules, binomial, random-variable and sample-proportion formulas) are left out here: use the sheet. What's below is what the sheet doesn't give you.



## Algebra and functions

Quadratics

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \Delta = b^2 - 4ac$$

$\Delta > 0$: two roots; $= 0$: one; $< 0$: none. Turning point at $x = -\frac{b}{2a}$. Discriminant problems → p. 51

Polynomials

$P(a) = 0 \iff (x - a)$ is a factor

$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

Polynomials → p. 18

Index and log laws

$a^m a^n = a^{m+n} \quad (a^m)^n = a^{mn}$

$\log(xy) = \log x + \log y \quad \log x^n = n \log x \quad \log_a x = \frac{\log_e x}{\log_e a}$

Exponentials & logs → p. 32

Transformations

$y = af(n(x - h)) + k$

$(x, y) \to \left(\frac{x}{n} + h,\ ay + k\right)$

Dilation from the $x$-axis = vertical. Transformations → p. 27

Circular functions

$y = a\sin(n(x - h)) + k$ : period $\frac{2\pi}{|n|}$, amplitude $|a|$

$\tan$ : period $\frac{\pi}{|n|}$, asymptotes where $n(x - h) = \frac{\pi}{2} + m\pi \qquad \sin^2 x + \cos^2 x = 1$

$\sin x = \sin \alpha$ : $x = \alpha + 2m\pi$ or $\pi - \alpha + 2m\pi$

$\cos x = \cos \alpha$ : $x = \pm\alpha + 2m\pi \qquad \tan x = \tan \alpha$ : $x = \alpha + m\pi$

Circular functions → p. 37

Composite and inverse

$f \circ g$ exists $\iff \operatorname{ran} g \subseteq \operatorname{dom} f$

$\operatorname{dom} f^{-1} = \operatorname{ran} f$

Composite & inverse → p. 43

# Calculus

Tangents and normals

$y - f(a) = f'(a)(x - a)$ $\quad m_{\text{normal}} = -\frac{1}{f'(a)}\ (f'(a) \neq 0)$

Applications → p. 63

Integrals

$\int_a^b f(x)\,dx = F(b) - F(a)$ $\quad f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$

Area between curves: $\int(\text{upper} - \text{lower})$. Integration → p. 77

# Probability and statistics

Normal

$Z = \dfrac{X - \mu}{\sigma}$ $\quad 68\%/95\%/99.7\%$ within $1/2/3\sigma$

$z = 1.645\ (90\%),\quad 1.960\ (95\%),\quad 2.576\ (99\%)$

Normal → p. 100

# Exact values (not on the sheet)

| $\theta$ | 0 | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ |
|---|---|---|---|---|---|
| sin | 0 | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1 |
| cos | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | 0 |
| tan | 0 | $\frac{\sqrt{3}}{3}$ | 1 | $\sqrt{3}$ | undefined |

FUNCTIONS, RELATIONS & GRAPHS

# Functions: domain, range, hybrids & modulus



## Relations and functions

A **relation** is any set of ordered pairs $(x, y)$. A **function** is a relation where every $x$-value has *exactly one* $y$-value.

- **Vertical line test:** a graph is a function if no vertical line cuts it more than once.
- **One-to-one:** every $y$-value comes from exactly one $x$-value (passes the horizontal line test too). Examples: $y = x^3$, $y = e^x$, $y = \log_e x$.
- **Many-to-one:** some $y$-values come from more than one $x$-value. Examples: $y = x^2$, $y = \cos x$, $y = |x|$.

**Function notation**

$$f : D \to \mathbb{R}, \quad f(x) = \ldots$$

$D$ is the **domain** (allowed inputs). The **range** is the set of outputs actually produced. $\mathbb{R}$ here is the codomain, not the range.

Interval notation: $[a, b]$ includes both ends, $(a, b)$ excludes both, $[a, b)$ includes $a$ only. $\mathbb{R} \setminus \{2\}$ means "all reals except 2"; $\mathbb{R}^+ = (0, \infty)$.

## Maximal (implied) domain

If no domain is given, use the largest set of real numbers for which the rule makes sense:

| Rule contains | Condition |
|---|---|
| $\sqrt{g(x)}$ | $g(x) \geq 0$ |
| $\dfrac{1}{g(x)}$ | $g(x) \neq 0$ |
| $\log_e(g(x))$ | $g(x) > 0$ (strictly) |
| $\tan(x)$ | $x \neq \dfrac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$ |

When a rule is a sum or product of pieces, the maximal domain is the **intersection** of the pieces' domains.

**Worked example 1** **Maximal domain of a combination** 2 marks

Find the maximal domain of $f(x) = \sqrt{4 - x} + \dfrac{1}{x + 1}$.

1. **Square root:** $4 - x \geq 0 \Rightarrow x \leq 4$.
   USING $\sqrt{u}$ needs $u \geq 0$
2. **Fraction:** $x + 1 \neq 0 \Rightarrow x \neq -1$.
   USING $\dfrac{1}{u}$ needs $u \neq 0$

✓ $(-\infty, -1) \cup (-1, 4]$, which can also be written $(-\infty, 4] \setminus \{-1\}$.
USING $\text{dom}(f + g) = \text{dom} f \cap \text{dom} g$

## Finding the range on a restricted domain

Sketch the graph over the domain only, then read off the lowest and highest $y$-values. Check three things: **turning points** inside the domain, the values at the **endpoints**, and whether each endpoint is included.

**Worked example 2** TRICKY **Range with an open endpoint (trick)** 3 marks

Find the range of $f : [-1, 3) \to \mathbb{R},\ f(x) = x^2 - 2x$.

1. **Turning point:** $f(x) = (x-1)^2 - 1$, so the minimum is $-1$ at $x = 1$, which is in the domain.
   USING $ax^2 + bx + c = a(x-h)^2 + k$
2. **Endpoints:** $f(-1) = 1 + 2 = 3$ (included, since $-1$ is in the domain); $f(3) = 9 - 6 = 3$ ($x = 3$ is excluded).
   USING evaluate the endpoints
3. **Is 3 in the range?** Yes: even though $x = 3$ is excluded, the value 3 is still produced at $x = -1$.
   USING check whether the value occurs elsewhere

✓ **Range** $= [-1, 3]$**.** Many students write $[-1, 3)$ because of the open endpoint. Always check whether the "missing" value is produced somewhere else.

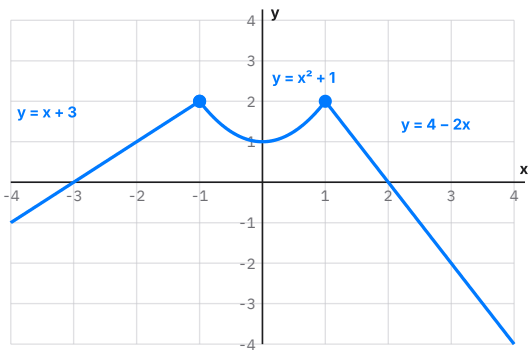
**The range on a restricted domain** TRICKY

The range is the set of $y$-values the graph actually reaches, using only the $x$-values you're allowed. Sketch it over the domain only. The lowest and highest heights are at **an endpoint or at a turning point inside the domain**, nowhere else. An endpoint whose $x$ is excluded gives a $y$-value that is **not reached** (round bracket), unless the graph reaches that height somewhere else too.

E.g. $f(x) = x^2$ on $(-1, 2]$: the turning point $(0, 0)$ is inside, so 0 is reached; the excluded end $x = -1$ would give 1, but 1 is reached anyway at $x = 1$; $f(2) = 4$ is reached. Range $[0, 4]$. And $g(x) = \dfrac{1}{x}$ on $[1, \infty)$ gets closer to 0 forever but never reaches it: range $(0, 1]$.

## Hybrid (piecewise) functions

A hybrid function uses different rules on different parts of the domain. Sketch each piece only over its own interval, with a closed dot where an endpoint is included and an open dot where it isn't.




$$f(x) = \begin{cases} x + 3, & x < -1 \\ x^2 + 1, & -1 \le x \le 1 \\ 4 - 2x, & x > 1 \end{cases}$$ : the pieces meet at both joins ($f = 2$ at $x = -1$ and at $x = 1$), so $f$ is continuous, but the corners there mean it isn't differentiable at $x = \pm 1$.

Functions

**Key idea**

A hybrid function is **continuous** at a join if the pieces give the same value there. Continuity problems usually become "set the two rules equal at the join and solve for the unknown constant". For differentiability (a smooth join), see Limits, continuity & differentiability → p. 58.

**Worked example 3** TRICKY **Making a hybrid continuous** — 2 marks

Find $a$ so that $g(x) = \begin{cases} x^2 + a, & x \leq 2 \\ 3x - 1, & x > 2 \end{cases}$ is continuous at $x = 2$.

1. Both rules must give the same value at $x = 2$: $2^2 + a = 3(2) - 1$.
   USING $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$

✓ $4 + a = 5 \Rightarrow a = 1$.

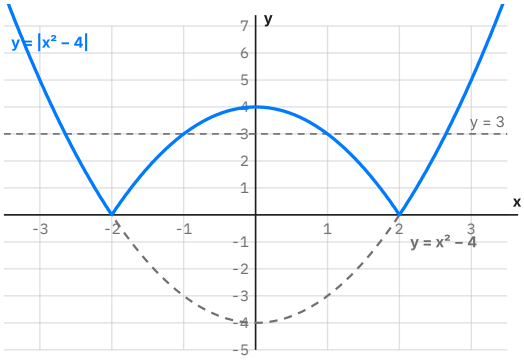
Functions

## The modulus function

**Modulus (absolute value)**

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$|x|$ is the distance from 0. The graph of $y = |f(x)|$ is the graph of $y = f(x)$ with every part below the $x$-axis reflected up.




$y = x^2 - 4$ (dashed) and $y = |x^2 - 4|$ (solid). The part between $x = -2$ and $x = 2$ is flipped up, creating sharp corners at $x = \pm 2$.

**Worked example 4** **Solving a modulus equation** — 2 marks

Solve $|x^2 - 4| = 3$.

1. Split into two equations: $x^2 - 4 = 3$ or $x^2 - 4 = -3$.
   USING $|u| = k \iff u = \pm k$

2. $x^2 = 7 \Rightarrow x = \pm\sqrt{7}$; $x^2 = 1 \Rightarrow x = \pm 1$.
   USING $x^2 = c \Rightarrow x = \pm\sqrt{c}$

✓ $x = \pm 1, \pm\sqrt{7}$ (four solutions: check on the graph, where the line $y = 3$ cuts the W shape four times).

**Trick alert: domain and range traps**

- **Logs need strictly positive arguments.** $\log_e(x^2)$ has domain $\mathbb{R} \setminus \{0\}$, but $2\log_e x$ has domain $(0, \infty)$. They are only equal for $x > 0$.
- **Open endpoints aren't automatically open in the range** (see the worked example above).
- **The basic graphs never reach their asymptotes.** The range of $y = 3 - \dfrac{2}{(x-1)^2}$ is $(-\infty, 3)$, not $(-\infty, 3]$. Combined functions can cross a horizontal asymptote, though: $y = xe^{-x}$ has asymptote $y = 0$ and equals 0 at $x = 0$, so check.

## Summary

- Function: one output per input. One-to-one: one input per output.
- Maximal domain: roots $\geq 0$, denominators $\neq 0$, log arguments $> 0$; intersect the conditions.
- Range on a restricted domain: check turning points, endpoints and whether excluded values are produced elsewhere.
- Hybrid functions: sketch each piece on its own interval; continuity means the pieces agree at the join.
- $|f(x)|$: reflect the negative parts up; corners appear at the $x$-intercepts.



### Hard exam questions

**2023 Exam 2 · Section A · Question 3** © VCAA — 47% correct

**The question (not reprinted in this copy):** Domain of the sum of two functions with domains [−2, 3) and (−1, 5]
Full question: 2023 Exam 2, Section A, Question 3 (page 3 of the paper).

**Answer: E** · A 22% · B 12% · C 11% · D 7% · E 47%

The sum's domain is the intersection: $(-1, 3)$.

**2024 Exam 2 · Section A · Question 8** © VCAA — 65% correct

**The question (not reprinted in this copy):** Where $h = f - g$ must have an x-intercept, from a table of values
Full question: 2024 Exam 2, Section A, Question 8 (page 4 of the paper).

**Answer: A** · A 65% · B 15% · C 16% · D 3%

h(1) = −3 and h(2) = 0, so (2, 0) is an intercept. Look for where f = g in the table.

FUNCTIONS, RELATIONS & GRAPHS

# Polynomials: factor theorem, cubics & quartics



## Division, remainders and factors

Dividing a polynomial $P(x)$ by $(x-a)$ gives a quotient $Q(x)$ and a remainder $R$ (a number):

**Division statement**

$$P(x) = (x-a)\,Q(x) + R$$

Substituting $x = a$ kills the first term, which gives the two theorems you use every week.

**Remainder theorem**

$$\text{The remainder when } P(x) \text{ is divided by } (x-a) \text{ is } P(a).$$

For a divisor $(bx-a)$ the remainder is $P\left(\frac{a}{b}\right)$.

**Factor theorem**

$$(x-a) \text{ is a factor of } P(x) \iff P(a) = 0.$$

**Key idea**

To factorise a cubic by hand: try the factors of the constant term ($\pm 1, \pm 2, \pm 3, \ldots$) until $P(a) = 0$, then divide by $(x-a)$ and factorise the quadratic that is left. With a non-monic leading coefficient, also try fractions $\pm\dfrac{\text{factor of constant}}{\text{factor of leading coeff}}$.

**Worked example 1** Factorising a cubic (tech-free) — 3 marks

Factorise $P(x) = 2x^3 - 3x^2 - 11x + 6$ and solve $P(x) = 0$.

1. **Hunt for a root:** $P(1) = -6$, $P(-1) = 12$, $P(3) = 54 - 27 - 33 + 6 = 0$. So $(x-3)$ is a factor.
   USING $P(a) = 0 \iff (x-a)$ is a factor
2. **Divide** (or match coefficients): $2x^3 - 3x^2 - 11x + 6 = (x-3)(2x^2 + 3x - 2)$.
   USING polynomial division
3. **Factorise the quadratic:** $2x^2 + 3x - 2 = (2x-1)(x+2)$.
   USING factorise the quadratic

✓ $P(x) = (x-3)(2x-1)(x+2)$, so $x = -2, \frac{1}{2}, 3$.

**Worked example 2** **Two unknowns from two factors** 3 marks

$P(x) = x^3 + ax^2 + bx - 6$ has factors $(x - 1)$ and $(x + 2)$. Find $a$ and $b$, then fully factorise $P(x)$.

1. $P(1) = 0$: $1 + a + b - 6 = 0 \Rightarrow a + b = 5$.
   USING $P(1) = 0$
2. $P(-2) = 0$: $-8 + 4a - 2b - 6 = 0 \Rightarrow 2a - b = 7$.
   USING $P(-2) = 0$
3. Add the equations: $3a = 12$, so $a = 4$, $b = 1$.
   USING solve simultaneously

✓ $a = 4$, $b = 1$ and $P(x) = x^3 + 4x^2 + x - 6 = (x - 1)(x + 2)(x + 3)$. (The third factor must make the constant term $-6$: $(-1)(2)(c) = -6 \Rightarrow c = 3$.)

**Sum and difference of cubes**

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

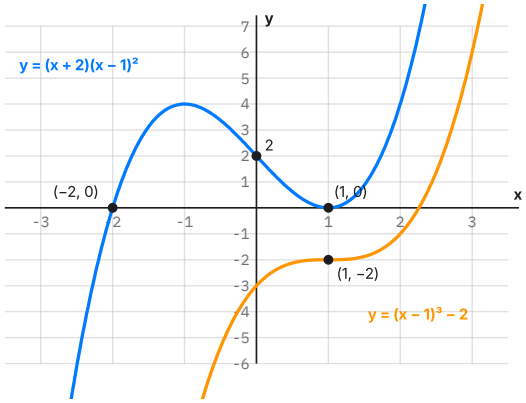
The quadratic factor never has real roots, so $x^3 - 27 = 0$ has only one real solution, $x = 3$.

## Shapes of polynomial graphs

A factor tells you what the graph does at that $x$-intercept:

| Factor | Behaviour at $x = a$ |
|---|---|
| $(x - a)$ | Cuts straight through the axis |
| $(x - a)^2$ | Touches the axis and turns (a turning point on the axis) |
| $(x - a)^3$ | Flattens out and passes through: a **stationary point of inflection** |

**End behaviour** depends on the degree and the sign of the leading coefficient. Odd degree, positive: from bottom-left to top-right. Even degree, positive: both ends up. A negative leading coefficient flips everything.




$y = (x + 2)(x - 1)^2$ cuts at $x = -2$ and touches at $x = 1$. $y = (x - 1)^3 - 2$ is the basic cubic translated: its stationary point of inflection is at $(1, -2)$.

**Cubic in "turning point" (inflection) form**

$$y = a(x - h)^3 + k$$

Stationary point of inflection at $(h, k)$. No other turning points.

Functions

**Worked example 3** Rule from a graph's features — 2 marks

A cubic touches the $x$-axis at $x = 2$, crosses it at $x = -1$ and passes through $(0, 8)$. Find its rule.

1. The touch gives a squared factor: $y = a(x + 1)(x - 2)^2$.
   USING touch at $q \Rightarrow (x - q)^2$
2. Substitute $(0, 8)$: $8 = a(1)(4)$, so $a = 2$.
   USING substitute a known point to find $a$

✓ $y = 2(x + 1)(x - 2)^2$ (expanded: $y = 2x^3 - 6x^2 + 8$).

**Worked example 4** TRICKY A cubic from its turning points (Exam 2 staple) — 4 marks

A cubic $p(x) = ax^3 + bx^2 + cx + d$ has a local maximum at $(1, 5)$ and a local minimum at $(3, 1)$. Find $a$, $b$, $c$ and $d$.

1. $p'(x) = 3ax^2 + 2bx + c$ is zero at $x = 1$ and $x = 3$, so $p'(x) = 3a(x - 1)(x - 3) = 3ax^2 - 12ax + 9a$. Matching coefficients: $b = -6a$, $c = 9a$.
   USING $p'(h) = 0$ at each turning point
2. $p(3) = 27a - 54a + 27a + d = d$, so $d = 1$. Then $p(1) = a - 6a + 9a + 1 = 4a + 1 = 5$, so $a = 1$.
   USING $p(h) = k$

✓ $a = 1,\ b = -6,\ c = 9,\ d = 1$, so $p(x) = x^3 - 6x^2 + 9x + 1$. On CAS: `Define p(x)=a·x^3+b·x^2+c·x+d`, `Define dp(x)=derivative(p(x),x)`, then `solve(p(1)=5 and p(3)=1 and dp(1)=0 and dp(3)=0,{a,b,c,d})`.

**Worked example 5** Biquadratic (quartic in disguise) — 2 marks

Solve $x^4 - 5x^2 + 4 = 0$.

1. Let $u = x^2$: $u^2 - 5u + 4 = 0 \Rightarrow (u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$.
   USING $u = x^2$

✓ $x = \pm 1, \pm 2$. Don't stop at $u$: the question asked for $x$.
USING $x^2 = u \Rightarrow x = \pm\sqrt{u}$

**Worked example 6** Sketching a quartic — 3 marks

Sketch $y = x^4 - 4x^3$, labelling the axis intercepts and stationary points.

1. $y = x^3(x - 4)$: intercepts $(0, 0)$ (a triple root, so a stationary point of inflection) and $(4, 0)$ (crosses).
   USING factorise: $x^3(x - 4)$
2. $y' = 4x^3 - 12x^2 = 4x^2(x - 3) = 0$: $x = 0$ and $x = 3$, $y(3) = -27$. The gradient table is in Applications of differentiation → p. 65.
   USING $y' = 0$

✓ **Positive $x^4$: up at both ends. Flattens through $(0, 0)$ going down, falls to a local minimum at $(3, -27)$, then rises through $(4, 0)$.**

**Every fact about a polynomial is an equation** TRICKY

To find a polynomial from its graph or from conditions, count the unknowns (a cubic $ax^3 + bx^2 + cx + d$ has four) and turn **each fact into one equation**. A point $(x_0, y_0)$ on the graph gives $P(x_0) = y_0$. A **turning point gives two equations**: it is on the graph *and* $P'(x_0) = 0$. A stationary point of inflection also has $P''(x_0) = 0$. Two turning points therefore give four equations, exactly enough for a cubic.

Let CAS solve them together (`solve(… and … ,{a,b,c,d})`), but write the equations out in your working: the marks are for the setup.

**Distinct roots when the roots depend on a parameter**

When the roots are in terms of $a$, set each pair of roots equal to find the values of $a$ where two of them merge into a repeated root. E.g. $f(x) = (x-1)(x-a)(x-2a)$ has roots $1, a, 2a$: pairs merge when $a = 1$, $2a = 1$ or $a = 2a$, so there are three distinct roots unless $a \in \{0, \frac{1}{2}, 1\}$, and two distinct roots (one repeated) at each of those. Check each special value: one value can merge more than one pair.



## How many solutions? Polynomials with a parameter

Exam 2 loves "find the values of $k$ for which the equation has exactly one / two / three solutions". Rewrite so the parameter sits alone, then think of it as a horizontal line $y = k$ moving up and down against a fixed graph.

**Worked example 7** **Number of solutions of $x^3 - 3x + k = 0$** 4 marks

Find the values of $k$ for which $x^3 - 3x + k = 0$ has (a) exactly three, (b) exactly two, (c) exactly one real solution.

1. Rearrange: $-k = x^3 - 3x$. Count intersections of $y = x^3 - 3x$ with the horizontal line $y = -k$.
   USING $f(x) = c \leftrightarrow$ intersections with $y = c$

2. $y = x^3 - 3x$ has turning points where $3x^2 - 3 = 0$, i.e. $x = \pm 1$: a local max at $(-1, 2)$ and a local min at $(1, -2)$.
   USING $f'(x) = 0$ at turning points

3. The line cuts three times when it is strictly between the turning points: $-2 < -k < 2$, i.e. $-2 < k < 2$. It touches a turning point (two solutions) when $-k = \pm 2$.

✓ **(a)** $-2 < k < 2$**; (b)** $k = \pm 2$**; (c)** $k < -2$ **or** $k > 2$**.** Check (b): $k = 2$ gives $x^3 - 3x + 2 = (x-1)^2(x+2)$, two distinct solutions.

**Worked example 8** TRICKY **How many solutions? A difference of squares (hard)** 4 marks

Let $f(x) = (x^2 - 1)^2$ and $k > 0$. Find the number of solutions of $f(x) = k^2$ for each value of $k$.

1. $(x^2 - 1)^2 - k^2 = 0 \Rightarrow (x^2 - 1 - k)(x^2 - 1 + k) = 0$, so $x^2 = 1 + k$ or $x^2 = 1 - k$.
   USING $A^2 - B^2 = (A - B)(A + B)$
2. $x^2 = 1 + k$: since $k > 0$, $1 + k > 0$, giving always two solutions $x = \pm\sqrt{1 + k}$.
   USING $x^2 = c$: 2 solutions if $c > 0$, 1 if $c = 0$, 0 if $c < 0$
3. $x^2 = 1 - k$: two solutions if $0 < k < 1$; one solution ($x = 0$) if $k = 1$; none if $k > 1$. The two equations never share a solution, because $1 + k \neq 1 - k$ for $k > 0$.

✓ **4 solutions for $0 < k < 1$; 3 solutions for $k = 1$; 2 solutions for $k > 1$.** Graphically: $y = (x^2 - 1)^2$ has minimums at $(\pm 1, 0)$ and a local maximum at $(0, 1)$. The line $y = k^2$ cuts four times below the maximum, touches it (3 solutions) when $k^2 = 1$, and cuts twice above it.
USING $f'(x) = 0$ at turning points

**Trick alert: polynomial traps**

- **"Solutions" means distinct solutions.** $(x - 1)^2(x + 2) = 0$ has two solutions, not three.
- **A factor isn't always a root you can see.** If $(x - 1)$ is a factor of $x^3 + x - 2$, the other factor $x^2 + x + 2$ has a negative discriminant, so the graph has only one $x$-intercept.
- **Remainder with $(x + a)$:** substitute $x = -a$, not $x = a$.
- **The turning point at a double root isn't always a minimum.** For $y = (x + 1)^2(x - 3)$ (positive leading coefficient) the double root is on the left, so it's a local *maximum*.

## Summary

- Remainder of $P(x) \div (x - a)$ is $P(a)$; $(x - a)$ is a factor iff $P(a) = 0$.
- Single factor: cut. Squared: touch and turn. Cubed: stationary point of inflection.
- Counting solutions with a parameter: isolate the parameter and slide a horizontal line past the turning points.
- Watch for quadratic factors with negative discriminants: they add no $x$-intercepts.

### Hard exam questions

**2023 Exam 2 · Section A · Question 19** © VCAA — 32% correct

**The question (not reprinted in this copy):** Values of k giving one positive and one negative root of a quadratic
Full question: 2023 Exam 2, Section A, Question 19 (page 9 of the paper).

**Answer: D** · A 27% · B 11% · C 13% · D 32% · E 16%

Opposite signs need the product of the roots to be negative: $4k^2 - \frac{9}{4} < 0$, so $-\frac{3}{4} < k < \frac{3}{4}$. A negative product already forces two real roots ($\Delta > 0$), so no separate discriminant check is needed.

**2023 Exam 2 · Section B · Question 1** © VCAA — state average 58% of marks

**The question (not reprinted in this copy):** Cubic $f(x) = x(x - 2)(x + 1)$: intercepts, stationary points, area with a line, matching $(x - a)(x - b)^2$
Full question: 2023 Exam 2, Section B, Question 1 (page 10 of the paper).

**a.** Coordinates of all axial intercepts (1 mark · state average 0.9 / 1, 90%)

- $(-1, 0), (0, 0), (2, 0)$

**b.** Coordinates of the stationary points (exact) (2 marks · state average 1.6 / 2, 80%)

- $f'(x) = 3x^2 - 2x - 2 = 0 \Rightarrow x = \dfrac{1 \pm \sqrt{7}}{3}$
- $\left(\dfrac{1-\sqrt{7}}{3}, \dfrac{14\sqrt{7}-20}{27}\right)$ and $\left(\dfrac{1+\sqrt{7}}{3}, \dfrac{-14\sqrt{7}-20}{27}\right)$

**Notes:** Give exact y-values as well as x-values.

**c.i.** Solve $f(x) = x - 2$ (1 mark · state average 0.9 / 1, 90%)

- $x = 2,\ \dfrac{-1 \pm \sqrt{5}}{2}$

**Notes:** $(x-2)(x^2+x-1) = 0$

**c.ii.** Integral expression for the total area between f and g (2 marks · state average 1.4 / 2, 70%)

- $\displaystyle\int_{\frac{-1-\sqrt{5}}{2}}^{\frac{-1+\sqrt{5}}{2}} (f-g)\,dx$
- $+\displaystyle\int_{\frac{-1+\sqrt{5}}{2}}^{2} (g-f)\,dx$

**Notes:** Split at the middle intersection, with the upper function first in each.

**c.iii.** Total area (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $\approx 5.95$

**d.** a and b with $(x-a)(x-b)^2 = f(x) + k$ (4 marks · state average 1.0 / 4, 25%)

- Expands and equates coefficients: $a + 2b = 1$, $2ab + b^2 = -2$
- $3b^2 - 2b - 2 = 0$, so $b = \dfrac{1 \pm \sqrt{7}}{3}$
- Matching $a = 1 - 2b$
- $(a, b) = \left(\dfrac{1-2\sqrt{7}}{3}, \dfrac{1+\sqrt{7}}{3}\right)$ or $\left(\dfrac{1+2\sqrt{7}}{3}, \dfrac{1-\sqrt{7}}{3}\right)$

**Notes:** The repeated root b must sit at a turning point of f, so b is a stationary x-value. Equating coefficients is cleanest. The state averaged 1/4.

## 2023 Exam 2 · Section A · Question 2 © VCAA

53% correct

**The question (not reprinted in this copy):** Axis of symmetry of $y = ax^2 + 2bx + c$
Full question: 2023 Exam 2, Section A, Question 2 (page 2 of the paper).

**Answer: A** · A 53% · B 29% · C 3% · D 6% · E 8%

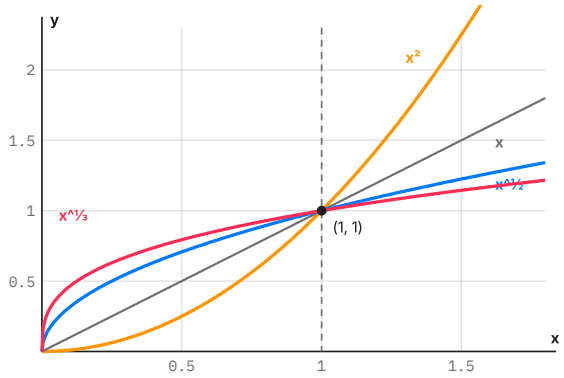
$x = -\dfrac{2b}{2a} = -\dfrac{b}{a}$. The coefficient of x is 2b, not b.

FUNCTIONS, RELATIONS & GRAPHS

# Power functions, hyperbolas & root functions

## The families at a glance

| Rule | Name | Domain | Range | Key features |
|---|---|---|---|---|
| $x^n$, $n$ even positive | even power ( $x^2, x^4$) | $\mathbb{R}$ | $[0, \infty)$ | U-shaped, symmetric in the $y$-axis; flatter near 0 and steeper outside $[-1, 1]$ as $n$ grows |
| $x^n$, $n$ odd positive | odd power ( $x^3, x^5$) | $\mathbb{R}$ | $\mathbb{R}$ | Stationary point of inflection at the origin, point-symmetric |
| $x^{-1} = \frac{1}{x}$ | hyperbola | $\mathbb{R} \setminus \{0\}$ | $\mathbb{R} \setminus \{0\}$ | Asymptotes $x = 0,\ y = 0$; branches in quadrants 1 and 3 |
| $x^{-2} = \frac{1}{x^2}$ | truncus | $\mathbb{R} \setminus \{0\}$ | $(0, \infty)$ | Both branches above the $x$-axis |
| $x^{1/2} = \sqrt{x}$ | square root | $[0, \infty)$ | $[0, \infty)$ | Starts at the origin (an endpoint), increasing |
| $x^{1/3} = \sqrt[3]{x}$ | cube root | $\mathbb{R}$ | $\mathbb{R}$ | Vertical tangent at the origin |
| $x^{2/3} = (\sqrt[3]{x})^2$ | "seagull" | $\mathbb{R}$ | $[0, \infty)$ | Sharp cusp at the origin |




On $(0, 1)$ the order flips: $x^{1/3} > x^{1/2} > x > x^2$. On $(1, \infty)$ it's the other way round. Every power function passes through $(1, 1)$.

## Transformed forms

**Hyperbola, truncus and root in transformed form**

$$y = \frac{a}{x - h} + k \qquad y = \frac{a}{(x - h)^2} + k \qquad y = a\sqrt{\pm(x - h)} + k$$

For the hyperbola and truncus: asymptotes $x = h$ and $y = k$. For the root: the endpoint is $(h, k)$.

**Key idea**

For every sketch, find: (1) asymptotes or endpoint, (2) the $y$-intercept (put $x = 0$), (3) the $x$-intercept(s) (put $y = 0$), (4) the domain and range. The sign of $a$ tells you which way the branches go.

**Worked example 1** Sketching a hyperbola — 3 marks

Sketch $y = \dfrac{2}{x-1} + 3$, labelling asymptotes and intercepts.

1. **Asymptotes:** $x = 1$ and $y = 3$.
   USING $y = \dfrac{a}{x-h} + k:\ x = h, y = k$
2. **$y$-intercept:** $y = \dfrac{2}{-1} + 3 = 1$, so $(0, 1)$.
   USING $x = 0$ for the $y$-intercept
3. **$x$-intercept:** $0 = \dfrac{2}{x-1} + 3 \Rightarrow x - 1 = -\dfrac{2}{3} \Rightarrow x = \dfrac{1}{3}$.
   USING $y = 0$ for the $x$-intercept

✓ Since $a = 2 > 0$, the branches sit top-right and bottom-left of the asymptote intersection $(1, 3)$.
**Asymptotes** $x = 1,\ y = 3$**; intercepts** $(0, 1)$ **and** $(\frac{1}{3}, 0)$**.** Domain $\mathbb{R} \setminus \{1\}$, range $\mathbb{R} \setminus \{3\}$.

**Worked example 2** Sketching a truncus — 3 marks

Find the intercepts, asymptotes and range of $y = -\dfrac{1}{(x+2)^2} + 4$.

1. Asymptotes $x = -2$, $y = 4$. $a < 0$, so both branches are *below* $y = 4$.
   USING $y = \dfrac{a}{(x-h)^2} + k:\ x = h, y = k$
2. $y$-intercept: $-\dfrac{1}{4} + 4 = \dfrac{15}{4}$.
   USING $x = 0$
3. $x$-intercepts: $(x+2)^2 = \dfrac{1}{4} \Rightarrow x + 2 = \pm\dfrac{1}{2} \Rightarrow x = -\dfrac{5}{2}, -\dfrac{3}{2}$.
   USING $y = 0$

✓ **Intercepts** $(-\frac{5}{2}, 0), (-\frac{3}{2}, 0), (0, \frac{15}{4})$**; range** $(-\infty, 4)$**.**

**Worked example 3** A reflected square root — 3 marks

Sketch $y = 2\sqrt{3-x} - 4$, stating the domain, range and intercepts.

1. Rewrite as $y = 2\sqrt{-(x-3)} - 4$: reflected in the $y$-axis, so the graph goes *left* from the endpoint $(3, -4)$. Domain: $3 - x \geq 0 \Rightarrow x \leq 3$.
   USING $\sqrt{u}$ needs $u \geq 0$
2. $x$-intercept: $\sqrt{3-x} = 2 \Rightarrow 3 - x = 4 \Rightarrow x = -1$. $y$-intercept: $2\sqrt{3} - 4 \approx -0.54$.
   USING $y = 0$ and $x = 0$

✓ **Domain** $(-\infty, 3]$**, range** $[-4, \infty)$**; intercepts** $(-1, 0)$ **and** $(0, 2\sqrt{3} - 4)$**.**

**Worked example 4** **Finding a rule from asymptotes and a point** 2 marks

A truncus has asymptotes $x = -1$ and $y = 2$ and passes through $(0, 5)$. Find its rule.

1. $y = \dfrac{a}{(x+1)^2} + 2$. Substitute $(0, 5)$: $5 = a + 2$.

   USING $y = \dfrac{a}{(x-h)^2} + k$

✓ $a = 3$, **so** $y = \dfrac{3}{(x+1)^2} + 2$.

**Trick alert: power function traps**

- **Cube roots of negatives exist.** $x^{1/3}$ and $x^{2/3}$ have domain $\mathbb{R}$. CAS sometimes shows only the $x \geq 0$ half. Don't trust it blindly.
- $x^{2/3} = 4$ **has two solutions:** $x^{1/3} = \pm 2$, so $x = \pm 8$.
- **Asymptotes are not in the range.** $y = \dfrac{a}{x-h} + k$ never equals $k$.
- $\sqrt{3 - x}$ **goes left**, not right. Factor out the negative to see the reflection.

## Summary

- Hyperbola $\dfrac{a}{x-h} + k$ and truncus $\dfrac{a}{(x-h)^2} + k$: asymptotes $x = h$, $y = k$.
- Root functions start at an endpoint; factor out a negative inside the root to spot a reflection.
- $x^{1/3}$ and $x^{2/3}$ are defined for all real $x$.
- Between 0 and 1 the powers swap order; all pass through $(1, 1)$.

FUNCTIONS, RELATIONS & GRAPHS

# Transformations of graphs

## The three moves

| Transformation | Mapping | Effect on the rule $y = f(x)$ |
|---|---|---|
| Dilation by factor $a$ **from the $x$-axis** (parallel to the $y$-axis) | $(x, y) \to (x, ay)$ | $y = af(x)$ |
| Dilation by factor $b$ **from the $y$-axis** (parallel to the $x$-axis) | $(x, y) \to (bx, y)$ | $y = f\left(\frac{x}{b}\right)$ |
| Reflection in the $x$-axis | $(x, y) \to (x, -y)$ | $y = -f(x)$ |
| Reflection in the $y$-axis | $(x, y) \to (-x, y)$ | $y = f(-x)$ |
| Translation $h$ right and $k$ up | $(x, y) \to (x + h, y + k)$ | $y = f(x - h) + k$ |

VCAA wording: "a translation of $h$ units in the positive direction of the $x$-axis" means $h$ right; "in the negative direction of the $y$-axis" means down.

**"From the x-axis" means vertical**

A dilation *from* the $x$-axis pushes points away from the $x$-axis, so it stretches vertically. VCAA also writes "parallel to the $y$-axis" for the same thing.

## The general form, and the order that works

**Standard form**

$$y = a\,f\big(n(x - h)\big) + k$$

Read it as: dilate by $|a|$ from the $x$-axis (reflect in the $x$-axis if $a < 0$); dilate by $\frac{1}{|n|}$ from the $y$-axis (reflect in the $y$-axis if $n < 0$); *then* translate $h$ right and $k$ up. The mapping is $(x, y) \to \left(\frac{x}{n} + h,\ ay + k\right)$.

**Key idea**

**Always factor out the coefficient of $x$ first.** $f(2x + 6) = f(2(x + 3))$ is a translation of 3 left, not 6, when the dilation is done first. With dilations first and translations last you never go wrong.

**Why f(x – 3) moves RIGHT, and why points and equations go opposite ways** TRICKY

The graph of $y = f(x - 3)$ does at $x = 5$ what $f$ did at $x = 2$, because $5 - 3 = 2$. Everything happens 3 units *later*, so the picture moves 3 to the **right**. In the same way $f(2x)$ does at $x = 1$ what $f$ did at $x = 2$: it gets there in half the distance, so the graph is squashed to half the width (dilation by factor $\frac{1}{2}$ from the $y$-axis).

So **points and equations move oppositely**. A dilation by factor 3 from the $y$-axis sends the point $(x, y)$ to $(3x, y)$ (multiply), but the image curve is $y = f\left(\frac{x}{3}\right)$ (divide inside). When you need the image equation, don't guess: write $x' = 3x$, $y' = y$, solve for $x$ and $y$, and substitute (the dash method).

## Finding the image equation: the dash method

This is the method VCAA's solutions use, and it works for any transformation (even ones you can't picture):

1. Write the mapping as $x' = \ldots$ and $y' = \ldots$.
2. Rearrange to get $x$ and $y$ in terms of $x'$ and $y'$.
3. Substitute into the original rule, then drop the dashes.

**Worked example 1** **Image of a square-root graph** 2 marks

Find the image of $y = \sqrt{x}$ under the transformation $(x, y) \to (-2x + 1,\ 3y - 2)$.

1. $x' = -2x + 1 \Rightarrow x = \dfrac{1 - x'}{2}$, and $y' = 3y - 2 \Rightarrow y = \dfrac{y' + 2}{3}$.
   USING $x' = ax + b \Rightarrow x = \dfrac{x' - b}{a}$

2. Substitute into $y = \sqrt{x}$: $\dfrac{y' + 2}{3} = \sqrt{\dfrac{1 - x'}{2}}$.
   USING substitute into $y = f(x)$

✓ $y = 3\sqrt{\dfrac{1 - x}{2}} - 2$, with domain $x \leq 1$.

**Worked example 2** **Describing a sequence of transformations** 3 marks

Describe a sequence of transformations that maps $y = e^x$ onto $y = 3e^{2x-4} + 1$, and give the mapping.

1. Factor the exponent: $y = 3e^{2(x-2)} + 1$. So $a = 3,\ n = 2,\ h = 2,\ k = 1$.
   USING $y = af(n(x - h)) + k$

2. Dilation by factor 3 from the $x$-axis; dilation by factor $\dfrac{1}{2}$ from the $y$-axis; translation 2 units right and 1 unit up.
   USING dilate, then translate

✓ **Mapping** $(x, y) \to \left(\dfrac{x}{2} + 2,\ 3y + 1\right)$. Check with the dash method: $x = 2x' - 4$, $y = \dfrac{y' - 1}{3}$, giving $\dfrac{y' - 1}{3} = e^{2x'-4}$. ✓
USING $(x, y) \to \left(\dfrac{x}{n} + h,\ ay + k\right)$

**Worked example 3** **Transforming a point, domain and range** 3 marks

The function $f$ has domain $[-2, 4]$ and range $[0, 6]$, and $f(2) = 5$. For $g(x) = -f(2x) + 3$, find (a) the domain and range of $g$, and (b) the image of the point $(2, 5)$.

1. **Domain:** need $2x \in [-2, 4]$, so $x \in [-1, 2]$.
   USING $2x \in \text{dom} f$

2. **Range:** $f \in [0, 6] \Rightarrow -f \in [-6, 0] \Rightarrow -f + 3 \in [-3, 3]$.
   USING apply $-f + 3$ to the range

3. **Point:** the mapping is $(x, y) \to \left(\dfrac{x}{2},\ -y + 3\right)$, so $(2, 5) \to (1, -2)$.
   USING $(x, y) \to \left(\dfrac{x}{2},\ -y + 3\right)$

✓ **Domain** $[-1, 2]$**, range** $[-3, 3]$**; image point** $(1, -2)$**.** Check: $g(1) = -f(2) + 3 = -5 + 3 = -2$. ✓

**Worked example 4** TRICKY **Choosing transformations to match points** 3 marks

The graph of $y = f(x)$ has turning points at $(0, 0)$ and $(2, -4)$. Find a dilation from the $y$-axis and a translation that map them to $(1, 3)$ and $(7, -1)$ respectively.

1. Horizontal gap: 2 before, 6 after, so dilate by factor 3 from the $y$-axis. Vertical gap: 4 before and after, so no vertical dilation.

   USING horizontal gap scales by the dilation factor

2. After the dilation $(0, 0) \to (0, 0)$ and $(2, -4) \to (6, -4)$. Translating 1 right and 3 up gives $(1, 3)$ ✓ and $(7, -1)$ ✓.

   USING $(x, y) \to (3x + h, y + k)$

✓ **Dilation by factor 3 from the $y$-axis, then a translation 1 unit right and 3 units up**: $(x, y) \to (3x + 1,\ y + 3)$, so the image is $y = f\left(\dfrac{x - 1}{3}\right) + 3$.

**When the order of the points flips, reflect**

If the image points are in the opposite left–right order, the map needs a **reflection in the $y$-axis**; opposite up–down order needs a **reflection in the $x$-axis**. E.g. $A(1, 2)$, $B(3, -2)$ to $A'(4, 2)$, $B'(0, -2)$: $A$ was left of $B$ and is now right of it, and the gap doubles, so reflect in the $y$-axis, dilate by 2 from the $y$-axis and translate 6 right: $(x, y) \to (6 - 2x,\ y)$, image $y = f\left(\dfrac{6 - x}{2}\right)$.

**Worked example 5** TRICKY **The inverse transformation: mapping g back to f** 2 marks

The graph of $y = g(x)$, where $g(x) = 2f(3x - 6) + 1$, is the image of $y = f(x)$. Find the transformation that maps $g$ back to $f$.

1. $f \to g$: $y' = 2y + 1$ and $3x' - 6 = x$, so $(x, y) \to \left(\dfrac{x + 6}{3},\ 2y + 1\right)$.

   USING $y' = 2y + 1$, $x = 3x' - 6$

2. Reverse: $x = 3x' - 6$, $y = \dfrac{y' - 1}{2}$, so $g \to f$ is $(x, y) \to \left(3x - 6,\ \dfrac{y - 1}{2}\right)$.

   USING make $x, y$ the subjects

✓ **Dilate by factor 3 from the $y$-axis, then translate 6 units left; translate 1 unit down, then dilate by factor $\dfrac{1}{2}$ from the $x$-axis.** Check: $f(x) = \dfrac{g\left(\frac{x+6}{3}\right) - 1}{2}$.

# Families of functions with a parameter

A rule with a parameter, such as $y = x^2 - 2kx + 3$, describes a whole family of graphs. Typical questions ask where a feature (a turning point, an intercept) lies as $k$ varies, or which values of $k$ give a property (one-to-one, no $x$-intercepts, a given number of solutions).

**Worked example 6** TRICKY **Where the turning points go** — 3 marks

For $y = x^2 - 2kx + 3$, find the turning point in terms of $k$ and the curve it lies on as $k$ varies. For which $k$ are there no $x$-intercepts?

1. Turning point $(k,\ 3 - k^2)$.
   USING $x^2 - 2kx + 3 = (x - k)^2 + 3 - k^2$
2. $x = k$ and $y = 3 - k^2$, so $y = 3 - x^2$.
   USING eliminate $k$

✓ **Turning point $(k, 3 - k^2)$, on the parabola $y = 3 - x^2$.** No $x$-intercepts when the turning point is above the axis: $3 - k^2 > 0$, so $-\sqrt{3} < k < \sqrt{3}$ (the same as $\Delta = 4k^2 - 12 < 0$).

**Worked example 7** TRICKY **Parameter conditions: increasing, common points, stationary points** — 4 marks

(a) For which $k$ is $f(x) = x^3 + kx^2 + 3x$ strictly increasing for all $x$? (b) Find the points that every member of $y = kx^2 - (k + 2)x + 2$ passes through. (c) How many stationary points does $g(x) = x^3 - 3kx$ have, as $k$ varies?

1. (a) $f'(x) = 3x^2 + 2kx + 3$; $\Delta = 4k^2 - 36 \leq 0 \Rightarrow -3 \leq k \leq 3$.
   USING $f'(x) \geq 0$ for all $x \iff \Delta \leq 0$
2. (b) $y = k(x^2 - x) + 2 - 2x$. Independent of $k$ when $x^2 - x = 0$: $x = 0, y = 2$ and $x = 1, y = 0$.
   USING $k(\ldots) + (\ldots)$ : coefficient of $k = 0$
3. (c) $g'(x) = 3x^2 - 3k = 0 \Rightarrow x^2 = k$.
   USING $g'(x) = 0$

✓ **(a) $-3 \leq k \leq 3$ (the ends are included: $f'$ touches 0 once). (b) $(0, 2)$ and $(1, 0)$. (c) Two if $k > 0$, one (a stationary point of inflection at the origin) if $k = 0$, none if $k < 0$.**

**Worked example 8** TRICKY **When does f meet its inverse? (tangency)** — 3 marks

For which values of $k$ does the graph of $f(x) = e^x + k$ intersect the graph of its inverse?

1. $e^x = 1 \Rightarrow x = 0$; then $e^0 + k = 0 \Rightarrow k = -1$. CAS: `solve(e^(x)+k=x and e^(x)=1,{x,k})`.
   USING $f(x) = x$ and $f'(x) = 1$
2. Lowering the curve (smaller $k$) makes it cross $y = x$ twice; raising it lifts it clear of the line.
   USING decide which side of the boundary works

✓ $k \leq -1$ (one meeting point at $k = -1$, two for $k < -1$). The same "touch $y = x$" idea finds the smallest or largest parameter in Section B questions on inverses.

**Trick alert: transformation traps**

- **Horizontal changes act "backwards".** $f(x-3)$ moves right 3; $f(2x)$ squashes by factor $\frac{1}{2}$ (dilation factor $\frac{1}{2}$ from the $y$-axis, not 2).
- **Order matters when dilations and translations mix.** "Translate 1 right, then dilate by 2 from the $y$-axis" gives $f\left(\frac{x}{2}-1\right)$, not $f\left(\frac{x-1}{2}\right)$.
- **Sequences aren't unique.** $y=(2x)^2=4x^2$: a dilation by $\frac{1}{2}$ from the $y$-axis gives the same graph as a dilation by 4 from the $x$-axis. Similarly $e^{x-2}=e^{-2}e^{x}$.
- **Transform the asymptotes too.** The image of $y=\log_e x$ under a translation 3 left has asymptote $x=-3$.

## Summary

- $y=af(n(x-h))+k$: mapping $(x,y)\to\left(\frac{x}{n}+h,\ ay+k\right)$.
- "From the $x$-axis" = vertical; "from the $y$-axis" = horizontal.
- Dash method: express $x,y$ in terms of $x',y'$ and substitute.
- Domains, ranges, points and asymptotes all transform with the graph.

### Hard exam questions

**2025 Exam 2 · Section A · Question 20** © VCAA — 36% correct

**The question (not reprinted in this copy):** Which transformation sequence does not map $a^x$ to $a^{2x+2}$
Full question: 2025 Exam 2, Section A, Question 20 (page 12 of the paper).

**Answer: C** · A 24% · B 18% · C 36% · D 21%

Apply each sequence in order. The sequence "dilate by a from the x-axis, dilate by ½ from the y-axis, then translate 1 right" gives $a^{2x-1}$.

**2025 Exam 2 · Section A · Question 15** © VCAA — 44% correct

**The question (not reprinted in this copy):** Image of the point (1, 3) on $y=1-g(2x-3)$
Full question: 2025 Exam 2, Section A, Question 15 (page 9 of the paper).

**Answer: B** · A 25% · B 44% · C 22% · D 8%

You need $2x-3=1$, so $x=2$ and $y=1-3=-2$.

FUNCTIONS, RELATIONS & GRAPHS

# Exponential & logarithmic functions



## Laws you must know cold

**Index laws**

$$a^m a^n = a^{m+n} \qquad \frac{a^m}{a^n} = a^{m-n} \qquad (a^m)^n = a^{mn}$$

$$a^0 = 1 \qquad a^{-n} = \frac{1}{a^n}$$

**Log laws** ($a, x, y > 0,\ a \neq 1$)

$$\log_a(xy) = \log_a x + \log_a y \qquad \log_a \frac{x}{y} = \log_a x - \log_a y$$

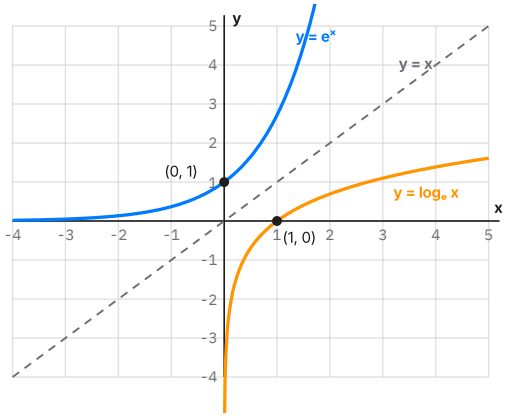
$$\log_a x^n = n \log_a x$$

**Definition and inverse pair**

$$y = a^x \iff x = \log_a y \qquad e^{\log_e x} = x\ (x > 0) \qquad \log_e(e^x) = x$$

$\log_e$ is written $\ln$ on most calculators. Change of base: $\log_a x = \dfrac{\log_e x}{\log_e a}$.

## Graphs




$y = e^x$ and $y = \log_e x$ are inverses: each is the reflection of the other in $y = x$. $e^x$ has horizontal asymptote $y = 0$ and passes through $(0, 1)$; $\log_e x$ has vertical asymptote $x = 0$ and passes through $(1, 0)$. A log graph has **no horizontal asymptote**: it keeps rising (slowly) without bound.

**Worked example 1** Sketching a transformed log graph — 3 marks

For $y = 2\log_e(3 - x)$, state the maximal domain and the asymptote, and find the axis intercepts.

1. Need $3 - x > 0$: domain $(-\infty, 3)$, asymptote $x = 3$. It's reflected in the $y$-axis, so the graph rises to the *left*.

   USING $\log_e u$ needs $u > 0$

2. $x$-intercept: $\log_e(3 - x) = 0 \Rightarrow 3 - x = 1 \Rightarrow x = 2$.

   USING $\log_e 1 = 0$

✓ **Domain** $(-\infty, 3)$**, asymptote** $x = 3$**, intercepts** $(2, 0)$ **and** $(0, 2\log_e 3)$.

# Solving equations

**Worked example 2** Hidden quadratic in $e^x$ (tech-free classic) — 3 marks

Solve $e^{2x} - 5e^x + 6 = 0$.

1. Let $u = e^x$, so $e^{2x} = u^2$: $u^2 - 5u + 6 = 0 \Rightarrow (u-2)(u-3) = 0$.
   USING $u = e^x$, $e^{2x} = u^2$
2. $e^x = 2$ or $e^x = 3$.

✓ $x = \log_e 2$ **or** $x = \log_e 3$. (If a factor gave $e^x = -1$, you'd reject it: $e^x > 0$.)
USING $e^x = a \iff x = \log_e a$

**Worked example 3** Log equation with a rejected solution — 3 marks

Solve $\log_2 x + \log_2(x-2) = 3$.

1. Combine: $\log_2\big(x(x-2)\big) = 3 \Rightarrow x^2 - 2x = 2^3 = 8$.
   USING $\log_a m + \log_a n = \log_a(mn)$, $\log_a x = y \iff x = a^y$
2. $x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$, so $x = 4$ or $x = -2$.
   USING null factor law

✓ $x = 4$ **only.** $x = -2$ makes $\log_2 x$ undefined in the original equation. Checking the original equation is worth a mark.
USING reject values outside the domain

**Worked example 4** Different bases — 2 marks

Solve $2^x = 3^{x-1}$, giving an exact answer.

1. Take $\log_e$ of both sides: $x \log_e 2 = (x-1)\log_e 3$.
   USING $\log_e(a^x) = x \log_e a$
2. $x(\log_e 3 - \log_e 2) = \log_e 3$.
   USING collect $x$ terms

✓ $x = \dfrac{\log_e 3}{\log_e\left(\frac{3}{2}\right)} \approx 2.71$.
USING $\log_e a - \log_e b = \log_e \dfrac{a}{b}$

Functions

**Worked example 5** **Finding an inverse** 3 marks

Find the inverse of $f : \mathbb{R} \to \mathbb{R},\ f(x) = 2e^{x-1} + 3$, stating its domain.

1. Swap: $x = 2e^{y-1} + 3 \Rightarrow e^{y-1} = \dfrac{x-3}{2}$.

   USING swap $x$ and $y$

2. $y - 1 = \log_e\left(\dfrac{x-3}{2}\right)$.

   USING $e^u = v \iff u = \log_e v$

✓ $f^{-1}(x) = \log_e\left(\dfrac{x-3}{2}\right) + 1$, with domain = range of $f$ = $(3, \infty)$.

USING $\text{dom} f^{-1} = \text{ran} f$

## Growth and decay models

**Exponential model**

$$A = A_0 e^{kt}$$

$k > 0$: growth; $k < 0$: decay. Half-life (decay) $= \dfrac{\log_e 2}{|k|}$.

**Worked example 6** TRICKY **Newton's law of cooling (Exam 2 style)** 4 marks

A cup of coffee cools according to $T = 20 + 70e^{-kt}$ (°C, $t$ in minutes). After 10 minutes it is 55 °C. (a) Find $k$ exactly. (b) How long until it reaches 30 °C?

1. (a) $55 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \dfrac{1}{2} \Rightarrow k = \dfrac{\log_e 2}{10}$.

   USING $e^u = v \iff u = \log_e v$

2. (b) $30 = 20 + 70e^{-kt} \Rightarrow e^{-kt} = \dfrac{1}{7} \Rightarrow t = \dfrac{\log_e 7}{k} = \dfrac{10 \log_e 7}{\log_e 2}$.

✓ **(a)** $k = \dfrac{\log_e 2}{10} \approx 0.0693$**; (b)** $t \approx 28.1$ **minutes.** The long-run temperature is 20 °C (room temperature), the horizontal asymptote.

**Worked example 7** **Fitting y = Ae$^{kx}$ through two points** 3 marks

The curve $y = Ae^{kx}$ passes through $(2, 12)$ and $(5, 96)$. Find $A$ and $k$ exactly.

1. $12 = Ae^{2k}$ and $96 = Ae^{5k}$. Divide: $e^{3k} = 8$, so $k = \dfrac{1}{3}\log_e 8 = \log_e 2$.

   USING divide to eliminate $A$

2. $A = \dfrac{12}{e^{2\log_e 2}} = \dfrac{12}{4} = 3$.

   USING substitute back

✓ $A = 3,\ k = \log_e 2$, so $y = 3e^{x \log_e 2} = 3 \times 2^x$.

**What each part of a model y = A·e^kx + c means** TRICKY

$c$ is the **horizontal asymptote**: the value the model settles to (a cooling drink's room temperature). $A$ is the *distance from the asymptote at* $x = 0$, so the starting value is $A + c$, not $A$. The sign of $k$ decides whether that distance grows or shrinks: $k < 0$ means the gap to $c$ shrinks, and the time for it to halve is $\frac{\log_e 2}{|k|}$ whatever the starting gap (for growth, $k > 0$, the time to double is $\frac{\log_e 2}{k}$).

To fit a model through two points when $c = 0$, write one equation per point and **divide them** to cancel $A$: $\frac{Ae^{kx_2}}{Ae^{kx_1}} = e^{k(x_2 - x_1)} = \frac{y_2}{y_1}$, then take logs. With a $+c$, subtract $c$ from each $y$ first.

**Changing the base**

$a^x = e^{x \log_e a}$, and $e^{kx} = (e^k)^x$. To write $e^{kx}$ as $2^{cx}$, match exponents: $kx = cx \log_e 2$, so $c = \frac{k}{\log_e 2}$.

**Base-10 logs**

$\log_{10} x$ obeys the same laws, and $10^{\log_{10} x} = x$. On the TI-Nspire, `log(x)` is base 10 and `ln(x)` is base $e$. Base-10 models: pH $= -\log_{10}[\mathrm{H}^+]$, so pH 3 means $[\mathrm{H}^+] = 10^{-3}$ and each unit of pH is a factor of 10; sound level $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB, so doubling $I$ adds $10 \log_{10} 2 \approx 3$ dB.

**Trick alert: exp and log traps**

- $e^x = 0$ **has no solution.** In $e^{2x} = 3e^x$, divide by $e^x$ (it's never 0): $x = \log_e 3$ only.
- $\log_e(x^2) = 2$ **has two solutions** ($x = \pm e$), but $2 \log_e x = 2$ has one ($x = e$). Log laws quietly change domains.
- $\log_e(a + b) \neq \log_e a + \log_e b$. The product law is for products only.
- $\log_e(2x) = \log_e 2 + \log_e x$: a horizontal dilation of a log graph is the same as a vertical translation.

## Summary

- $e^x$ and $\log_e x$ are inverses: domain and range swap; asymptotes $y = 0$ and $x = 0$ swap.
- Hidden quadratics: let $u = e^x$ (or $2^x$), solve, reject impossible values.
- After combining logs, check every solution in the original equation.
- Models: find $k$ from one data point, then use it.

### Hard exam questions

**2023 Exam 2 · Section B · Question 3** © VCAA — state average 44% of marks

**The question (not reprinted in this copy):** $g(x) = 2^x + 5$: limit, derivative, tangents through the origin; $h(x) = 2^x - x^2$: inflection, decreasing interval, Newton's method
Full question: 2023 Exam 2, Section B, Question 3 (page 16 of the paper).

**a.** $\lim_{x \to -\infty} g(x)$ (1 mark · state average 0.7 / 1, 70%)

- 5

**b.** k with $g'(x) = k \times 2^x$ (1 mark · state average 0.9 / 1, 90%)

- $k = \log_e 2$

Functions

**c.i.** Tangent at $(a, g(a))$ (1 mark · state average 0.5 / 1, 50%)

- $y = 2^a \log_e 2\,(x - a) + 2^a + 5$

**Notes:** Point–gradient form.

**c.ii.** Tangent to g through the origin (3 d.p.) (2 marks · state average 0.6 / 2, 30%)

- Sets $0 = -a\,2^a \log_e 2 + 2^a + 5$, giving $a \approx 2.618$
- $y \approx 4.255x$

**d.** Point of inflection of h (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $\approx (2.06, -0.07)$

**Notes:** $h''(x) = (\log_e 2)^2 2^x - 2 = 0$

**e.** Largest interval on which h is strictly decreasing (2 d.p.) (1 mark · state average 0.4 / 1, 40%)

- $[0.49, 3.21]$

**Notes:** Between the two roots of h′, with square brackets (the endpoints are included).

**f.** Newton's method from $x_0 = 0$: $x_1, x_2, x_3$ (3 d.p.) (2 marks · state average 1.2 / 2, 60%)

- $x_1 \approx -1.443$
- $x_2 \approx -0.897,\ x_3 \approx -0.773$

**g.** Why a solution of $\log_e(2)2^x - 2x = 0$ is a bad starting point (1 mark · state average 0.2 / 1, 20%)

- That equation is $h'(x) = 0$, so the tangent is horizontal and Newton's formula divides by zero

**Notes:** A stationary point makes the tangent horizontal, so it never meets the x-axis.

**h.** Positive n for which $n^x - x^n$ has a local minimum on the x-axis (2 marks · state average 0.2 / 2, 10%)

- Requires $f(x) = 0$ and $f'(x) = 0$ at the same x, giving $x = \dfrac{n}{\log_e n}$
- $n = e$

**Notes:** $n = e$: the curve $e^x - x^e$ touches the axis at x = e. An exact answer was required.

FUNCTIONS, RELATIONS & GRAPHS

# Circular (trigonometric) functions

## Radians, exact values and symmetry

$\pi$ radians $= 180°$. On the unit circle, the point at angle $\theta$ is $(\cos\theta, \sin\theta)$, and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

| $\theta$ | 0 | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ |
|---|---|---|---|---|---|
| $\sin\theta$ | 0 | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1 |
| $\cos\theta$ | 1 | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | 0 |
| $\tan\theta$ | 0 | $\frac{1}{\sqrt{3}}$ | 1 | $\sqrt{3}$ | undefined |

**Symmetry (quadrants 2, 3, 4)**

$$\sin(\pi-\theta)=\sin\theta \quad \sin(\pi+\theta)=-\sin\theta \quad \sin(2\pi-\theta)=-\sin\theta$$

$$\cos(\pi-\theta)=-\cos\theta \quad \cos(\pi+\theta)=-\cos\theta \quad \cos(2\pi-\theta)=\cos\theta$$

$$\sin\left(\tfrac{\pi}{2}-\theta\right)=\cos\theta \qquad \sin^2\theta+\cos^2\theta=1$$

Memory aid for the positive ratio in each quadrant, going anticlockwise from quadrant 1: **A**ll, **S**in, **T**an, **C**os.

## Graphs

| $m \in \mathbb{Z}$ | $y=\sin x$ | $y=\cos x$ | $y=\tan x$ |
|---|---|---|---|
| Period | $2\pi$ | $2\pi$ | $\pi$ |
| Range | $[-1,1]$ | $[-1,1]$ | $\mathbb{R}$ |
| Zeros | $m\pi$ | $\frac{\pi}{2}+m\pi$ | $m\pi$ |
| Maximum 1 at | $\frac{\pi}{2}+2m\pi$ | $2m\pi$ | none |
| Minimum −1 at | $\frac{3\pi}{2}+2m\pi$ | $\pi+2m\pi$ | none |
| Vertical asymptotes | none | none | $x=\frac{\pi}{2}+m\pi$ |
| Points of inflection | at the zeros | at the zeros | at the zeros (gradient 1) |

**Transformed tan**

$$y=a\tan\big(n(x-h)\big)+k: \quad \text{asymptotes } x=h+\frac{\pi}{2n}+\frac{m\pi}{n}, \quad \text{inflections } \left(h+\frac{m\pi}{n},\, k\right), \quad m\in\mathbb{Z}$$

Period $\frac{\pi}{|n|}$. E.g. $y=3\tan(2x)+1$: asymptotes $x=\frac{\pi}{4}+\frac{m\pi}{2}$, inflections at $\left(\frac{m\pi}{2},1\right)$, period $\frac{\pi}{2}$.

**Transformed sin, cos and tan**

$$y=a\sin\big(n(x-h)\big)+k$$

Amplitude $|a|$, period $\frac{2\pi}{|n|}$, horizontal shift $h$, centre line $y=k$, range $[k-|a|,\ k+|a|]$. For $y=a\tan(n(x-h))+k$: period $\frac{\pi}{|n|}$, with vertical asymptotes wherever $n(x-h)=\frac{\pi}{2}+m\pi$ and points of inflection at $x=h+\frac{m\pi}{n}$ on $y=k$, halfway between the asymptotes. The period of a sum of trig terms is the lowest common multiple of their periods: $\sin\left(\frac{\pi t}{6}\right)+\cos\left(\frac{\pi t}{3}\right)$ has periods 12 and 6, so its period is 12.

Functions

**Worked example 1** **Sketch over a restricted domain** 4 marks

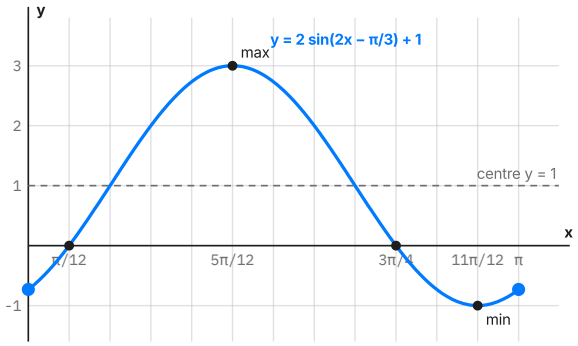
Sketch $y = 2\sin\left(2x - \frac{\pi}{3}\right) + 1$ for $x \in [0, \pi]$, labelling endpoints and $x$-intercepts with exact coordinates.

1. Factor: $y = 2\sin\left(2\left(x - \frac{\pi}{6}\right)\right) + 1$. Amplitude 2, period $\pi$, shifted $\frac{\pi}{6}$ right, centre $y = 1$, range $[-1, 3]$.
   **USING** $\text{period} = \frac{2\pi}{n}$, $\text{range} = [k - |a|, k + |a|]$

2. Endpoints: $y(0) = 2\sin\left(-\frac{\pi}{3}\right) + 1 = 1 - \sqrt{3}$; $y(\pi) = 2\sin\left(\frac{5\pi}{3}\right) + 1 = 1 - \sqrt{3}$.
   **USING** substitute the endpoints

3. $x$-intercepts: $\sin\left(2x - \frac{\pi}{3}\right) = -\frac{1}{2}$. With $2x - \frac{\pi}{3} \in \left[-\frac{\pi}{3}, \frac{5\pi}{3}\right]$: $2x - \frac{\pi}{3} = -\frac{\pi}{6}$ or $\frac{7\pi}{6}$.
   **USING** $\sin\theta = -\frac{1}{2} \Rightarrow \theta = -\frac{\pi}{6}, \frac{7\pi}{6}, \ldots$

✓ **Endpoints** $(0, 1 - \sqrt{3})$ **and** $(\pi, 1 - \sqrt{3})$**;** $x$**-intercepts** $\left(\frac{\pi}{12}, 0\right)$ **and** $\left(\frac{3\pi}{4}, 0\right)$; maximum $\left(\frac{5\pi}{12}, 3\right)$, minimum $\left(\frac{11\pi}{12}, -1\right)$.




$y = 2\sin\left(2x - \frac{\pi}{3}\right) + 1$ on $[0, \pi]$. Always find where the argument $2x - \frac{\pi}{3}$ starts and ends. That's the step that finds all the solutions.

## Solving trig equations

1. Isolate the trig function: $\sin(\ldots) = c$.
2. **Adjust the domain** to match the argument (e.g. $x \in [0, 2\pi] \Rightarrow 2x \in [0, 4\pi]$).
3. Find the reference angle from $|c|$, then use the quadrants for the sign.
4. List every solution in the adjusted domain, then solve for $x$.

**Worked example 2** **A double-angle argument** 3 marks

Solve $2\cos(2x) + \sqrt{2} = 0$ for $x \in [0, 2\pi]$.

1. $\cos(2x) = -\frac{\sqrt{2}}{2}$. Reference angle $\frac{\pi}{4}$; cosine is negative in quadrants 2 and 3.
   **USING** $\cos\theta < 0$ in quadrants 2 and 3

2. $2x \in [0, 4\pi]$: $2x = \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{11\pi}{4}, \frac{13\pi}{4}$.
   **USING** $x \in [0, 2\pi] \Rightarrow 2x \in [0, 4\pi]$

✓ $x = \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}$. A doubled argument means twice as many solutions. Four, not two.

**The interval is for x, but you solve for the whole argument** TRICKY

For $\sin(2x) = \frac{1}{2}$ on $0 \le x \le \pi$, let $\theta = 2x$. The interval for $x$ becomes $0 \le \theta \le 2\pi$ (multiply every part), so solve $\sin\theta = \frac{1}{2}$ *over that whole $\theta$ interval*: $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$, then $x = \frac{\pi}{12}, \frac{5\pi}{12}$.

On $0 \le x \le 2\pi$ the argument runs over $0 \le \theta \le 4\pi$: two full cycles, so **four** solutions. A dilation by $n$ packs in $n$ times as many cycles, so (typically) $n$ times as many solutions. With a translation as well, $\sin(2x - \frac{\pi}{3})$, the $\theta$ interval starts at $-\frac{\pi}{3}$ rather than 0: write the $\theta$ interval out before you solve.

**Worked example 3** **sin = cos, and general solutions** 3 marks

(a) Solve $\sin(2x) = \cos(2x)$ for $x \in [0, \pi]$. (b) Find the general solution of $\sqrt{3}\sin(2x) = \cos(2x)$.

1. (a) Divide by $\cos(2x)$: $\tan(2x) = 1$. $2x \in [0, 2\pi]$: $2x = \frac{\pi}{4}, \frac{5\pi}{4}$, so $x = \frac{\pi}{8}, \frac{5\pi}{8}$.

   USING $\tan\theta = \frac{\sin\theta}{\cos\theta}$

2. (b) $\tan(2x) = \frac{1}{\sqrt{3}}$, so $2x = \frac{\pi}{6} + n\pi$, $n \in \mathbb{Z}$.

   USING $\tan\theta = a \Rightarrow \theta = \tan^{-1}a + n\pi$

✓ **(a)** $x = \frac{\pi}{8}, \frac{5\pi}{8}$; **(b)** $x = \frac{\pi}{12} + \frac{n\pi}{2}$, $n \in \mathbb{Z}$.

**General solutions ($n \in \mathbb{Z}$)**

$$\sin x = a:\ x = 2n\pi + \sin^{-1}a \ \text{ or } \ x = (2n+1)\pi - \sin^{-1}a$$

$$\cos x = a:\ x = 2n\pi \pm \cos^{-1}a \qquad \tan x = a:\ x = n\pi + \tan^{-1}a$$

**Worked example 4** **A hidden quadratic in sin x** 3 marks

Solve $2\sin^2 x - \sin x - 1 = 0$ for $x \in [0, 2\pi]$.

1. $2a^2 - a - 1 = (2a+1)(a-1) = 0 \Rightarrow \sin x = -\frac{1}{2}$ or $\sin x = 1$.

   USING let $a = \sin x$

2. $\sin x = 1$: $x = \frac{\pi}{2}$. $\sin x = -\frac{1}{2}$: third and fourth quadrants, $x = \frac{7\pi}{6}, \frac{11\pi}{6}$.

   USING solve each on $[0, 2\pi]$

✓ $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$. The same move works for $e^{2x} - 3e^x + 2 = 0$ ($a = e^x > 0$) and for $\cos^2 x$ or $\tan x$ quadratics.

**Inequalities in a hidden quadratic**

Solve the quadratic inequality in $a$ first, then intersect with the range of $a$. E.g. $2\sin^2 x + \sin x - 1 > 0$: $(2a-1)(a+1) > 0$ gives $a < -1$ or $a > \frac{1}{2}$. Since $-1 \le \sin x \le 1$, only $\sin x > \frac{1}{2}$ is possible, so on $[0, 2\pi]$: $\frac{\pi}{6} < x < \frac{5\pi}{6}$. CAS: `solve(2·a^2+a-1>0,a)` gives a < −1 or a > 1/2.

Functions

**Showing there are no solutions: use the range**

To show an equation has no solutions, show the two sides can never be equal. E.g. $\cos x = x^2 + 1.5$ has none, because $\cos x \le 1 < 1.5 \le x^2 + 1.5$. Similarly, $\sin(\sin 2x) = 1$ has none: $|\sin 2x| \le 1 < \frac{\pi}{2}$, and $\sin\theta = 1$ needs $\theta = \frac{\pi}{2} + 2n\pi$.

## Periodic models

**Worked example 5** TRICKY **Tide model (Exam 2 style)** — 4 marks

The depth of water in a harbour is $D(t) = 3 + 2\cos\left(\frac{\pi t}{6}\right)$ metres, $t$ hours after midnight. (a) State the maximum and minimum depths and the period. (b) For how long in the first 12 hours is the depth at least 4 m?

1. (a) Max $3 + 2 = 5$ m, min $3 - 2 = 1$ m, period $\frac{2\pi}{\pi/6} = 12$ hours.
   USING $\max = k + |a|$, $\min = k - |a|$, $\text{period} = \frac{2\pi}{n}$
2. (b) $\cos\left(\frac{\pi t}{6}\right) \ge \frac{1}{2}$ with $\frac{\pi t}{6} \in [0, 2\pi]$: $\frac{\pi t}{6} \in \left[0, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 2\pi\right]$.
   USING $\cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$
3. So $t \in [0, 2] \cup [10, 12]$.

✓ **(a) 5 m, 1 m, 12 h. (b) 4 hours in total** (midnight to 2 am and 10 am to noon).

**Trend plus a periodic part**

If $f(t) = mt + g(t)$ where $g$ has period $P$, then $f(t + P) = f(t) + mP$: each period the graph repeats, shifted up by $mP$. E.g. $f(t) = 5t + 20\cos\left(\frac{\pi t}{4}\right)$ has period-8 wiggles and rises 40 every 8 units. To sketch, draw one period and repeat it, stepping up by $mP$ each time; label the endpoints.

**Worked example 6** **Fitting a sinusoidal model to its features** — 3 marks

The temperature in a town varies sinusoidally over a day, from a minimum of 8 °C at 3 am ($t = 3$) to a maximum of 18 °C at 3 pm ($t = 15$). Find a rule $T(t) = k - a\cos\big(n(t - h)\big)$, $0 \le t \le 24$.

1. $k = \frac{18 + 8}{2} = 13$, $a = \frac{18 - 8}{2} = 5$.
   USING $k = \frac{\max + \min}{2}$, $a = \frac{\max - \min}{2}$
2. Min to max takes 12 h, so the period is 24 h and $n = \frac{2\pi}{24} = \frac{\pi}{12}$. The minimum is at $t = 3$, so $h = 3$.
   USING $\text{period} = \frac{2\pi}{|n|}$

✓ $T(t) = 13 - 5\cos\left(\frac{\pi}{12}(t - 3)\right)$. Check: $T(15) = 13 - 5\cos\pi = 18$ ✓. The same curve is $13 + 5\sin\left(\frac{\pi}{12}(t - 9)\right)$: VCAA accepts any correct form.

**Worked example 7** **Sum of the solutions, by symmetry** 2 marks

Find the sum of the solutions of $\sin x = 0.3$ for $x \in [0, 4\pi]$.

1. Two cycles, so four solutions: one pair sums to $2 \times \frac{\pi}{2} = \pi$, the other to $2 \times \frac{5\pi}{2} = 5\pi$.

   USING pairs symmetric about $x = \frac{\pi}{2} + 2m\pi$

✓ $6\pi$. CAS check: `sum(zeros(sin(x)-0.3,x)|0≤x≤4·π)` gives 18.8496 ≈ $6\pi$.

**Trick alert: trig traps**

- **Adjust the domain for the argument.** Forgetting this is the number-one way students lose solutions.
- $\sin^2 x = \frac{1}{4}$ **means** $\sin x = \pm\frac{1}{2}$: four solutions in $[0, 2\pi]$, not two.
- **Endpoints count.** $\sin x = 0$ on $[0, 2\pi]$ has three solutions: $0, \pi, 2\pi$.
- **Calculator in radians!** Degree mode is an instant zero in Methods.
- **Tan's period is** $\frac{\pi}{|n|}$, not $\frac{2\pi}{|n|}$.

## Summary

- $y = a\sin(n(x-h)) + k$: amplitude $|a|$, period $\frac{2\pi}{|n|}$ (tan: $\frac{\pi}{|n|}$), range $[k-|a|, k+|a|]$.
- Solve: isolate, adjust the domain for the argument, reference angle, quadrants, then solve for $x$.
- General solutions: cos: $2n\pi \pm \alpha$; tan: $n\pi + \alpha$.

### Hard exam questions

**2023 Exam 2 · Section A · Question 18** © VCAA — 29% correct

**The question (not reprinted in this copy):** Number of local minima of $\sin(ax)$ on $[-a\pi, a\pi]$
Full question: 2023 Exam 2, Section A, Question 18 (page 9 of the paper).

**Answer: E** · A 12% · B 9% · C 22% · D 26% · E 29%

Interval length $2a\pi$ ÷ period $\frac{2\pi}{a}$ = $a^2$ periods, each with one minimum.

**2023 Exam 2 · Section B · Question 2** © VCAA — state average 45% of marks

**The question (not reprinted in this copy):** Observation wheel modelled by h(t) = −60cos(bt) + c; then it stops and restarts at double speed
Full question: 2023 Exam 2, Section B, Question 2 (page 12 of the paper).

**a.** Show $b = \frac{\pi}{15}$ and $c = 75$ (2 marks · state average 1.5 / 2, 75%)

- Period 30: $\frac{2\pi}{b} = 30 \Rightarrow b = \frac{\pi}{15}$
- $h(0) = -60 + c = 15 \Rightarrow c = 75$

**Notes:** Use the period and the lowest height.

**b.** Average height from A (bottom) to B (level with the centre) (2 d.p.) (2 marks · state average 0.9 / 2, 45%)

- $\frac{1}{7.5}\int_0^{7.5} h(t)\,dt$
- $= 75 - \frac{120}{\pi} \approx 36.80$ m

Functions

**c.** Average rate of change of height from A to B (1 mark · state average 0.5 / 1, 50%)

- $\frac{75-15}{7.5}=8$ m/min

**d.i.** k and m in the piecewise model (1 mark · state average 0.4 / 1, 40%)

- $k=135,\ m=2$

**d.ii.** All possible values of n (2 marks · state average 0.4 / 2, 20%)

- Continuity at t = 20: $h(40+n)=135$, so $\cos\left(\frac{\pi(40+n)}{15}\right)=-1$
- $n=30j-25,\ j\in Z$

**Notes:** $n=30j-25$ for integer j (the same set as $n=5+30j$). A general solution was required.

**d.iii.** Sketch w with the coordinates of the endpoints (3 marks · state average 1.2 / 3, 40%)

- Rising cosine from $(0,15)$ to $(15,135)$
- Flat from $(15,135)$ to $(20,135)$
- Half a (faster) cycle down to $(27.5,15)$

## 2025 Exam 2 · Section A · Question 2 © VCAA

51% correct

**The question (not reprinted in this copy):** All asymptotes of $y=2\tan\left(\pi\left(x+\frac{1}{2}\right)\right)$

Full question: 2025 Exam 2, Section A, Question 2 (page 2 of the paper).

**Answer: A** · A 51% · B 10% · C 21% · D 18%

Set $\pi\left(x+\frac{1}{2}\right)=\frac{\pi}{2}+k\pi$, so $x=k,\ k\in Z$.

FUNCTIONS, RELATIONS & GRAPHS

# Composite, inverse & combined functions

## Combining functions

For $f+g$, $f-g$ and $f\times g$, the domain is the **intersection** of the two domains: $\text{dom}(f+g)=\text{dom}f\cap\text{dom}g$. To sketch $y=f(x)+g(x)$ by **addition of ordinates**, add the heights of the two graphs at key $x$-values (intercepts, asymptotes, where one graph is zero).

## Composite functions

**Composite**

$$(f\circ g)(x)=f(g(x))$$

$f\circ g$ exists if and only if $\text{ran}\,g\subseteq\text{dom}\,f$. Then $\text{dom}(f\circ g)=\text{dom}\,g$.

**Key idea**

Think of $g$ as the first machine and $f$ as the second. Every output of $g$ must be an acceptable input for $f$. If it isn't, restrict the domain of $g$ until it is.

**Worked example 1** **Does the composite exist?** 4 marks

Let $f(x)=\sqrt{x}$ (maximal domain) and $g:\mathbb{R}\to\mathbb{R},\ g(x)=x^2-4$. (a) Explain why $f\circ g$ does not exist. (b) Find the maximal domain of $g$ for which $f\circ g$ exists. (c) Does $g\circ f$ exist? If so, state its rule and domain.

1. (a) $\text{ran}\,g=[-4,\infty)$ but $\text{dom}\,f=[0,\infty)$. Since $[-4,\infty)\not\subseteq[0,\infty)$, $f\circ g$ does not exist.
   USING $f\circ g$ exists $\iff$ ran $g\subseteq$ dom $f$
2. (b) Need $g(x)\geq 0$: $x^2\geq 4\Rightarrow x\leq -2$ or $x\geq 2$.
   USING $g(x)\geq 0$
3. (c) $\text{ran}\,f=[0,\infty)\subseteq\mathbb{R}=\text{dom}\,g$, so yes. $g(f(x))=(\sqrt{x})^2-4=x-4$, with domain $=\text{dom}\,f=[0,\infty)$.
   USING $\text{dom}(g\circ f)=\text{dom}\ f$

✓ **(b)** $(-\infty,-2]\cup[2,\infty)$**. (c)** $g\circ f(x)=x-4$**, domain** $[0,\infty)$**.** It's not the whole line $y=x-4$: it's only the part with $x\geq 0$.

**Worked example 2** TRICKY **Range of a composite** 2 marks

Let $f:[-1,2]\to\mathbb{R},\ f(x)=2-x^2$ and $g(x)=e^x$. Find the range of $g\circ f$.

1. $f$ has its maximum 2 at $x=0$ (inside the domain) and its minimum at an endpoint: $f(-1)=1$, $f(2)=-2$. So $\text{ran}\,f=[-2,2]$.
   USING ran $f$ on its domain
2. $g(x)=e^x$ is increasing, so it maps $[-2,2]$ onto $[e^{-2},e^2]$.
   USING $g$ increasing $\Rightarrow$ apply $g$ to the ends

✓ $\text{ran}(g\circ f)=[e^{-2},e^2]$. Using only the endpoints of the domain gives $g(f(-1))=e$ and $g(f(2))=e^{-2}$, which misses the maximum $e^2$ from $x=0$. If the outer function is **decreasing**, the ends swap: $e^{-x}$ maps $[-4,\infty)$ onto $(0,e^4]$, and an end that's never reached stays open.

**A linear inside is a transformation**

If $f(x) = mx + c$, then $g(f(x)) = g(mx + c)$ is just a transformation of $g$: a dilation by factor $\frac{1}{|m|}$ from the $y$-axis (a reflection in the $y$-axis too if $m < 0$) and a horizontal translation. For example, $y = g(c - x) = g\big(-(x - c)\big)$ is $g$ reflected in the $y$-axis, then moved $c$ units right.

## Inverse functions

Only **one-to-one** functions have inverse functions. To find $f^{-1}$: swap $x$ and $y$, make $y$ the subject, and pick the correct branch using the range of $f^{-1}$. If $f$ is continuous and $f'$ takes both signs on an interval in the domain, $f$ turns around, so it isn't one-to-one and has no inverse function (until the domain is restricted). The inverse of a hyperbola $y = \frac{a}{x-h} + k$ is another hyperbola with the asymptotes swapped: $y = \frac{1}{x-3} - 2$ (asymptotes $x = 3$, $y = -2$) has inverse $y = \frac{1}{x+2} + 3$ (asymptotes $x = -2$, $y = 3$).

**Inverse facts**

$$\operatorname{dom} f^{-1} = \operatorname{ran} f \qquad \operatorname{ran} f^{-1} = \operatorname{dom} f \qquad f\big(f^{-1}(x)\big) = x$$

The graph of $f^{-1}$ is the reflection of $f$ in the line $y = x$.

**Which sign (±) do you keep when inverting?** TRICKY

Swapping $x$ and $y$ in $y = (x-2)^2 - 3$ gives $y = 2 \pm \sqrt{x+3}$. Only one sign can be the inverse, and the deciding fact is: **the range of $f^{-1}$ is the domain of $f$**. If $f$ was restricted to $x \le 2$, then $f^{-1}$ must output values $\le 2$, so take $2 - \sqrt{x+3}$. If $x \ge 2$, take $2 + \sqrt{x+3}$. The domain of $f^{-1}$ is the range of $f$ (here $[-3, \infty)$).

The same idea works for any restricted function: read the branch from where $f$ lives, not from which looks nicer. On CAS, `solve(x=(y-2)^2-3,y)|y≤2` applies the domain for you.

**Worked example 3** TRICKY **Restrict, then invert** — 4 marks

Let $f : (-\infty, a] \to \mathbb{R},\ f(x) = (x-2)^2 - 3$. (a) Find the largest value of $a$ for which $f^{-1}$ exists. (b) With this $a$, find $f^{-1}$ and its domain.

1. (a) The turning point is at $x = 2$. To the left of it, $f$ is one-to-one (decreasing). So $a = 2$.
   USING restrict to one side of the turning point
2. (b) Swap: $x = (y-2)^2 - 3 \Rightarrow (y-2)^2 = x + 3 \Rightarrow y - 2 = \pm\sqrt{x+3}$.
   USING swap $x$ and $y$
3. The range of $f^{-1}$ is the domain of $f$: $y \le 2$. So take the **negative** root.
   USING $\operatorname{ran} f^{-1} = \operatorname{dom} f$

✓ $a = 2$; $f^{-1}(x) = 2 - \sqrt{x+3}$, **domain** $[-3, \infty)$ (the range of $f$).

# Where does a function meet its inverse?

If $f$ is **increasing**, every intersection of $y = f(x)$ and $y = f^{-1}(x)$ lies on $y = x$, so you can solve $f(x) = x$ instead. That's much easier.

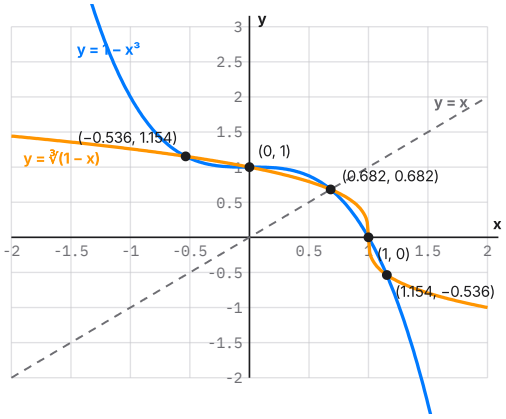
**Worked example 4** TRICKY **The decreasing-function trap** 3 marks

Find all points of intersection of $f(x) = 1 - x^3$ and $y = f^{-1}(x)$.

1. Solving $f(x) = x$: $1 - x^3 = x$, which has one real solution $x \approx 0.682$. That's one point, $(0.682, 0.682)$.
   USING $f(x) = x$

2. But $f$ is *decreasing*, so there can be intersections off the line $y = x$. Look for pairs with $f(a) = b$ and $f(b) = a$: $f(0) = 1$ and $f(1) = 0$ is one pair. Solving $f(f(x)) = x$ on CAS finds another: $f(-0.536) \approx 1.154$ and $f(1.154) \approx -0.536$.
   USING $f(a) = b$ and $f(b) = a$

✓ **Five points:** $(0.682, 0.682)$**,** $(0, 1)$**,** $(1, 0)$**,** $(-0.536, 1.154)$ **and** $(1.154, -0.536)$ (to 3 d.p.). Here $f^{-1}(x) = \sqrt[3]{1 - x}$, and a CAS solve of $f(x) = f^{-1}(x)$ confirms all five. Spotting one easy pair isn't enough: solve $f(f(x)) = x$ to be sure you have them all.




$y = 1 - x^3$ and its inverse $y = \sqrt[3]{1 - x}$ meet at five points. Only one is on $y = x$; the other four are two pairs swapped across it.

**Worked example 5** TRICKY **Where two inverses meet** 2 marks

$f(x) = x^3$ and $g(x) = 4x$ meet at $(-2, -8)$, $(0, 0)$ and $(2, 8)$. Where do $f^{-1}$ and $g^{-1}$ meet?

1. Swap each point: $(-8, -2)$, $(0, 0)$, $(8, 2)$. Check: $f^{-1}(x) = \sqrt[3]{x}$ and $g^{-1}(x) = \dfrac{x}{4}$ give $\sqrt[3]{8} = 2 = \dfrac{8}{4}$ ✓.
   USING $(a, b)$ on both $\Rightarrow$ $(b, a)$ on both inverses

✓ $(-8, -2)$**,** $(0, 0)$ **and** $(8, 2)$**.**

**Worked example 6** **Integral of an inverse: the rectangle trick** 2 marks

Use $\displaystyle\int_0^1 e^x \, dx$ to find $\displaystyle\int_1^e \log_e y \, dy$.

1. $f(x) = e^x$, $a = 0$, $b = 1$, $c = 1$, $d = e$: $\displaystyle\int_1^e \log_e y \, dy = 1 \times e - 0 \times 1 - (e - 1)$.
   USING $\displaystyle\int_c^d f^{-1}(y) \, dy = bd - ac - \int_a^b f(x) \, dx$

✓ **1.** Check: $\left[y \log_e y - y\right]_1^e = 0 - (-1) = 1$ ✓. Sketch it: the trick is just areas of rectangles.

Functions

**Area between f and its inverse**

If $f$ and $f^{-1}$ meet only on $y = x$, at $x = a$ and $x = b$, the area between them is twice the area between $f$ and $y = x$: $2\int_a^b |x - f(x)|\,dx$.

**Trick alert: composite and inverse traps**

- $f^{-1}(x) \neq \dfrac{1}{f(x)}$. The inverse undoes $f$; the reciprocal is something else.
- **The domain of a composite comes from the inner function**, even if the simplified rule looks as if it's defined everywhere.
- **Check that the range fits inside the domain, value by value.** For $f(x) = \dfrac{1}{x-1}$ and $g(x) = x^2 + 1$, $\operatorname{ran} g = [1, \infty)$ includes 1, which isn't in $\operatorname{dom} f$. So $f \circ g$ doesn't exist on $\mathbb{R}$.
- **Choose the right branch** when you take a square root, using the range of the inverse.

## Summary

- $f \circ g$ exists iff $\operatorname{ran} g \subseteq \operatorname{dom} f$; $\operatorname{dom}(f \circ g) = \operatorname{dom} g$.
- Inverse: one-to-one first; swap, solve, choose the branch; domain and range swap.
- Increasing $f$: solve $f(x) = x$ for intersections with $f^{-1}$. Decreasing $f$: check for points off $y = x$.

### Hard exam questions

**2023 Exam 2 · Section A · Question 20** © VCAA — 30% correct

**The question (not reprinted in this copy):** Largest interval where both $f \circ g$ and $g \circ f$ exist (log and sine)
Full question: 2023 Exam 2, Section A, Question 20 (page 9 of the paper).

**Answer: A** · A 30% · B 19% · C 26% · D 14% · E 9%

$g \circ f$ needs $x > -\dfrac{1}{\sqrt{2}}$. $f \circ g$ needs $\sin x > -\dfrac{1}{\sqrt{2}}$, i.e. $x < \dfrac{5\pi}{4}$ here. Combined: $\left(-\dfrac{1}{\sqrt{2}}, \dfrac{5\pi}{4}\right)$.

**2023 Exam 2 · Section B · Question 5** © VCAA — state average 35% of marks

**The question (not reprinted in this copy):** $g(x) = \dfrac{1}{2}f(2 - x)$ with $f(x) = e^x + e^{-x}$: transformations, restricted inverses, areas with inverses, a family of curves
Full question: 2023 Exam 2, Section B, Question 5 (page 21 of the paper).

**a.** Complete the sequence of transformations from f to g (2 marks · state average 1.2 / 2, 60%)

- Reflection in the y-axis
- Then translation 2 units right (or just translate 2 right, since f is even)

**b.** Domain and range of the inverse of the increasing branch $g_1$ (2 marks · state average 1.0 / 2, 50%)

- Domain $[1, \infty)$
- Range $[2, \infty)$

**c.i.** Coordinates of P and Q (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $P \approx (1.27, 1.27)$, $Q \approx (4.09, 4.09)$

**Notes:** Solve g(x) = x. Both points lie on y = x.

**c.ii.** Area bounded by g and the two inverses (2 d.p.) (2 marks · state average 0.6 / 2, 30%)

- $2\int_{1.27}^{4.09}(x-g(x))\,dx$
- $\approx 5.56$

**Notes:** Use the symmetry about y = x: double the area between y = x and g. Forgetting the 2 gives 2.78.

**d.** Integer n such that the turning point of h always lies on $y = 2x^n$ (1 mark · state average 0.1 / 1, 10%)

- $n = -1$

**Notes:** The turning point is $\left(k, \frac{2}{k}\right)$, and $\frac{2}{k} = 2k^{-1}$.

**e.** Smallest k for which h meets the inverse of $h_1$ (2 d.p.) (1 mark · state average 0.0 / 1, 0%)

- $k \approx 1.27$

**Notes:** $h_1$ is increasing, so it meets its inverse on $y = x$. The smallest k is where $h$ just touches $y = x$: solve $h(x) = x$ and $h'(x) = 1$ simultaneously for x and k. (The left branch of h can't get there first: $h_1^{-1}$ starts at $\left(\frac{2}{k}, k\right)$, which the left branch only reaches once $\frac{2}{k} \leq k$, i.e. $k \geq \sqrt{2}$.) The state averaged 0.0/1.

**f.** Area between h and the inverse of $h_1$ when k = 5 (2 d.p.) (2 marks · state average 0.3 / 2, 15%)

- Intersections at x ≈ 1.45 and x ≈ 8.78
- Area $\approx 43.91$

**Notes:** Integrate $|h - h_1^{-1}|$ between the two intersection points.

## 2025 Exam 2 · Section A · Question 13 © VCAA

45% correct

**The question (not reprinted in this copy):** Graph of $g(f(x))$ given sketches of a line f and an even function g
Full question: 2025 Exam 2, Section A, Question 13 (page 8 of the paper).

**Answer: C** · A 12% · B 17% · C 45% · D 25%

f is $y = -x$ and g is even, so $g(-x) = g(x)$. The composite looks just like g.

FUNCTIONS, RELATIONS & GRAPHS

# Functional equations, odd & even functions



## What's a functional equation?

A functional equation describes a *property* a function has for all inputs, such as $f(x+y)=f(x)f(y)$. VCAA usually asks "which function satisfies..." in multiple choice, or "show that $f$ satisfies..." in short answer.

| Property (for all $x, y$ in the domain) | Satisfied by | Why |
|---|---|---|
| $f(x+y)=f(x)+f(y)$ | $f(x)=kx$ | $k(x+y)=kx+ky$ |
| $f(x+y)=f(x)\,f(y)$ | $f(x)=e^{kx}$ (or $a^x$) | index law $a^{x+y}=a^xa^y$ |
| $f(xy)=f(x)+f(y)$ | $f(x)=k\log_e x$ | log law $\log(xy)=\log x+\log y$ |
| $f(xy)=f(x)\,f(y)$ | $f(x)=x^n$ (also $\lvert x\rvert^n$) | $(xy)^n=x^ny^n$ |
| $f(2x)=2f(x)$ | $f(x)=kx$, and also $k\lvert x\rvert$ or anything linear on each side of 0 | $k(2x)=2(kx)$; $\lvert 2x\rvert=2\lvert x\rvert$ |
| $f(-x)=f(x)$ | **even**: $x^2, \lvert x\rvert, \cos x, e^x+e^{-x}$ | symmetric in the $y$-axis |
| $f(-x)=-f(x)$ | **odd**: $x^3, \sin x, \tan x, \dfrac{1}{x}$ | point symmetry about the origin |
| $f(x+p)=f(x)$ | **periodic** with period $p$: $\sin x, \cos x$ ($p=2\pi$), $\tan x$ ($p=\pi$) | the graph repeats every $p$ units |
| $f(f(x))=x$ | self-inverse: $\dfrac{1}{x}$, $c-x$, some hyperbolas | graph symmetric in $y=x$ |

**The number-test strategy**

In multiple choice, don't do algebra first. Pick easy values ($x=1,\ y=2$, or $x=y=0$) and test each option. Usually three options fail on the first test. Then prove the survivor in your head.

**Worked example 1** **Which function satisfies f(x + y) = f(x)f(y)?** 1 mark

Which one of $f(x)=3x$, $f(x)=x^3$, $f(x)=e^{3x}$, $f(x)=\log_e(3x)$ satisfies $f(x+y)=f(x)f(y)$ for all real $x, y$?

1. Test $x=y=0$: we need $f(0)=f(0)^2$, so $f(0)=0$ or 1. $\log_e(0)$ is undefined, so D is out.
   USING test $x=y=0$
2. Test $x=y=1$: $3x$ gives 6 vs 9 ×; $x^3$ gives 8 vs 1 ×; $e^{3x}$ gives $e^6$ vs $e^3\cdot e^3=e^6$ ✓.
   USING test $x=y=1$

✓ $f(x)=e^{3x}$: $e^{3(x+y)}=e^{3x}e^{3y}$ for all $x, y$.
USING $e^{a+b}=e^ae^b$

**Worked example 2** **Finding a parameter** 2 marks

$f(x) = e^{kx}$ satisfies $f(x+1) = 2f(x)$ for all $x$. Find $k$.

1. $e^{k(x+1)} = e^{kx}e^{k}$, so the condition is $e^{kx}e^{k} = 2e^{kx}$.
   USING $e^{a+b} = e^{a}e^{b}$

✓ Divide by $e^{kx}$ (never 0): $e^{k} = 2$, so $k = \log_e 2$.
USING $e^{k} = 2 \iff k = \log_e 2$

**Worked example 3** **Showing a symmetry property** 2 marks

Let $f(x) = x^2 - 2x$. Show that $f(1+h) = f(1-h)$ for all $h$, and interpret this.

1. $f(1+h) = (1+h)^2 - 2(1+h) = 1 + 2h + h^2 - 2 - 2h = h^2 - 1$.
   USING expand $f(1+h)$

2. $f(1-h) = (1-h)^2 - 2(1-h) = 1 - 2h + h^2 - 2 + 2h = h^2 - 1$. Equal. ✓
   USING expand $f(1-h)$

✓ Points the same distance either side of $x = 1$ have the same height: **the graph is symmetric about the line** $x = 1$ (its axis of symmetry).

**Worked example 4** TRICKY **Linear functions with f(f(x)) = 4x + 3 (hard)** 3 marks

Find all linear functions $f(x) = ax + b$ such that $f(f(x)) = 4x + 3$ for all $x$.

1. $f(f(x)) = a(ax+b) + b = a^2x + ab + b$.
   USING $f(f(x)) = a(ax+b) + b$

2. Equate coefficients: $a^2 = 4$ and $b(a+1) = 3$.
   USING equate coefficients

✓ $a = 2 \Rightarrow b = 1$; $a = -2 \Rightarrow -b = 3 \Rightarrow b = -3$. $f(x) = 2x + 1$ **or** $f(x) = -2x - 3$**.** Most students miss the second one.

**Trick alert**

- **"For all $x$" means one counter-example kills an option,** but one success doesn't prove it.
- **Watch the domain.** $f(x) = \log_e x$ satisfies $f(xy) = f(x) + f(y)$ only for $x, y > 0$.
- **Neither odd nor even is the usual case.** $e^x$, $x^2 + x$ and $\log_e x$ are neither.
- **A function can be odd and still have $f(0)$ undefined** (e.g. $\frac{1}{x}$). If an odd function is defined at 0, then $f(0) = 0$.

## Summary

- Sums inside → products outside: exponential. Products inside → sums outside: log. Products → products: power.
- Even: $f(-x) = f(x)$. Odd: $f(-x) = -f(x)$.
- Multiple choice: test numbers first; $x = y = 0$ and $x = y = 1$ are your friends.

ALGEBRA, NUMBER & STRUCTURE

# Simultaneous equations & parameters

## Two linear equations, two unknowns

Two lines in the plane either cross once (**unique solution**), are parallel (**no solution**) or are the same line (**infinitely many solutions**).

**The parameter test**

$$\begin{aligned} ax + by &= e \\ cx + dy &= f \end{aligned} \qquad \text{unique solution} \iff ad - bc \neq 0$$

If $ad - bc = 0$ the lines have the same gradient. Then check the constants: if one equation is a multiple of the other, there are infinitely many solutions; otherwise there are none.

**Key idea**

$ad - bc$ is the determinant of the coefficient matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$. You don't need matrix theory. Just remember "unique ⇔ determinant ≠ 0", then test the special values by substituting them back in.

**Why a zero determinant gives two different answers** TRICKY

Two lines $ax + by = e$ and $cx + dy = f$ have the same direction exactly when $ad - bc = 0$. Same direction means *parallel*, and parallel lines can mean two different things: the **same line** (infinitely many solutions) or **different lines** (no solution). The determinant can't tell you which, so substitute the special value of the parameter back in and compare: if the equations become multiples of each other (right-hand sides included) it is infinitely many; if the left sides match but the right sides don't, it is none.

Three equations work the same way. On CAS, an answer containing a letter such as `c1` means infinitely many solutions, and `false` means none.

**Worked example 1** TRICKY **Classifying by a parameter** 4 marks

Consider $\begin{aligned} ax + 3y &= 3 \\ x + (a-2)y &= 1 \end{aligned}$. Find the values of $a$ for which the system has (a) a unique solution, (b) no solution, (c) infinitely many solutions.

1. Determinant: $a(a-2) - 3 \cdot 1 = a^2 - 2a - 3 = (a-3)(a+1)$. Zero when $a = 3$ or $a = -1$.
   USING $\det = a_1 b_2 - a_2 b_1 = 0$
2. $a = 3$: $3x + 3y = 3$ and $x + y = 1$. The first is 3 × the second, so they're the same line: infinitely many solutions.
   USING same line ⇒ infinitely many
3. $a = -1$: $-x + 3y = 3$ and $x - 3y = 1$, i.e. $x - 3y = -3$ and $x - 3y = 1$. Parallel, different lines: no solution.
   USING parallel, distinct ⇒ none

✓ **(a)** $a \in \mathbb{R} \setminus \{-1, 3\}$; **(b)** $a = -1$; **(c)** $a = 3$.

Algebra

## Three equations, three unknowns

Use elimination to reduce to two equations in two unknowns, or use CAS in Exam 2 (solve the three equations together). Geometrically each equation is a plane; they may meet at a point, along a line (infinitely many solutions) or not at all.

**Worked example 2** **A 3 × 3 system by elimination (tech-free)** 3 marks

Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

1. Add equations 1 and 3: $2x + 3y = 8$. Add equations 2 and 3: $3x + y = 5$.
   USING eliminate $z$
2. From $3x + y = 5$: $y = 5 - 3x$. Substitute: $2x + 15 - 9x = 8 \Rightarrow x = 1$, so $y = 2$.

✓ Then $z = 6 - 1 - 2 = 3$. $(x, y, z) = (1, 2, 3)$. Check in equation 2: $2 - 2 + 3 = 3$. ✓

**Worked example 3** TRICKY **Infinitely many solutions in 3D (Exam 2 style)** 4 marks

For the system $x + y + z = 1$, $x + 2y + 3z = 2$, $x + 4y + kz = c$: (a) find $k$ for which there is no unique solution; (b) for that $k$, find $c$ so there are infinitely many solutions and give them in terms of a parameter.

1. (a) Subtract: (2) − (1) gives $y + 2z = 1$; (3) − (1) gives $3y + (k - 1)z = c - 1$.
   USING eliminate $x$
2. Triple the first: $3y + 6z = 3$. The two equations have the same left side when $k - 1 = 6$, i.e. $k = 7$.
   USING same left side $\Rightarrow$ no unique solution
3. (b) With $k = 7$: infinitely many solutions need $c - 1 = 3$, so $c = 4$ (otherwise no solution). Let $z = t$: $y = 1 - 2t$, $x = 1 - y - z = t$.
   USING let $z = t$

✓ **(a)** $k = 7$. **(b)** $c = 4$; $(x, y, z) = (t,\ 1 - 2t,\ t)$, $t \in \mathbb{R}$.

## Lines meeting curves: the discriminant

Substitute the line into the curve to get a quadratic. Its discriminant $\Delta = b^2 - 4ac$ counts the intersections: $\Delta > 0$ two, $\Delta = 0$ one (the line is a **tangent**), $\Delta < 0$ none.

**Worked example 4** **Tangent line condition** 3 marks

Find the values of $m$ for which the line $y = mx + 1$ is a tangent to $y = x^2 + 2x + 2$.

1. Equate: $x^2 + 2x + 2 = mx + 1 \Rightarrow x^2 + (2 - m)x + 1 = 0$.
   USING equate the two rules
2. Tangent $\Leftrightarrow \Delta = 0$: $(2 - m)^2 - 4 = 0 \Rightarrow 2 - m = \pm 2$.
   USING tangent $\iff \Delta = b^2 - 4ac = 0$

✓ $m = 0$ **or** $m = 4$. (Check $m = 0$: the line $y = 1$ touches the parabola's vertex at $(-1, 1)$. ✓)

Algebra

## Literal equations

A literal equation has letters as well as the unknown: solve $ax + b = cx + d$ for $x$ to get $x = \frac{d-b}{a-c}$, *provided* $a \neq c$. Always state the restriction: that's where the marks for "consider all cases" are.

**Trick alert: parameter traps**

- **Determinant = 0 doesn't tell you which case you're in.** Substitute each value back and test for "same line" vs "parallel".
- **Dividing by an expression containing a parameter** silently assumes it isn't 0. Handle that value separately.
- **"At least one solution"** means unique *or* infinitely many, so it includes the coincident case.
- **For $x^2$-coefficient parameters** (e.g. $kx^2 + 4x + 1 = 0$), check $k = 0$ separately: the equation becomes linear and has exactly one solution.



## Summary

- Unique solution ⇔ determinant ≠ 0. At determinant = 0, substitute and compare constants.
- 3 × 3: eliminate to 2 × 2; a parameter family of solutions means planes share a line.
- Line and curve: substitute, then use the discriminant.
- Always handle "divide by zero" values of parameters separately.

### Hard exam questions

**2023 Exam 2 · Section A · Question 4** © VCAA — 55% correct

**The question (not reprinted in this copy):** Values of k giving infinitely many solutions
Full question: 2023 Exam 2, Section A, Question 4 (page 3 of the paper).

**Answer: B** · A 15% · B 55% · C 10% · D 14% · E 4%

The determinant $k^2 + k - 20 = 0$ gives k = −5 or 4. Only k = −5 makes the equations identical. k = 4 gives parallel lines.

**2025 Exam 2 · Section A · Question 4** © VCAA — 55% correct

**The question (not reprinted in this copy):** Values of k for which a 2×2 linear system has no solution
Full question: 2025 Exam 2, Section A, Question 4 (page 4 of the paper).

**Answer: A** · A 55% · B 13% · C 19% · D 12%

The determinant $2k^2 + k - 6 = 0$ gives $k = -2$ or $\frac{3}{2}$. $k = \frac{3}{2}$ gives the same line twice (infinitely many solutions), so only $k = -2$.

ALGEBRA, NUMBER & STRUCTURE

# Newton's method, bisection & pseudocode

## Newton's method

Start with a guess $x_0$ near a root. Draw the tangent at $(x_0, f(x_0))$; where it hits the $x$-axis is the next, usually better, estimate. Repeat.

**Newton's method**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Derivation: the tangent $y - f(x_n) = f'(x_n)(x - x_n)$ crosses $y = 0$ at $x = x_n - \frac{f(x_n)}{f'(x_n)}$.

**Worked example 1** Two iterations by hand — 3 marks

Use Newton's method with $x_0 = 2$ to find $x_2$ for $f(x) = x^3 - 2x - 5$. Give $x_2$ correct to four decimal places.

1. $f'(x) = 3x^2 - 2$. $f(2) = 8 - 4 - 5 = -1$, $f'(2) = 10$. So $x_1 = 2 - \frac{-1}{10} = 2.1$.

   USING $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

2. $f(2.1) = 9.261 - 4.2 - 5 = 0.061$, $f'(2.1) = 13.23 - 2 = 11.23$. So $x_2 = 2.1 - \frac{0.061}{11.23} = 2.09457\ldots$

✓ $x_2 \approx 2.0946$. (The true root is 2.094551...: two steps already give four correct decimal places.)

**Worked example 2** Estimating √2 — 2 marks

Use Newton's method on $f(x) = x^2 - 2$ with $x_0 = 1$ to find $x_1$, $x_2$ and $x_3$.

1. $x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n}$. $x_1 = 1 - \frac{-1}{2} = 1.5$.

   USING $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

2. $x_2 = 1.5 - \frac{0.25}{3} = 1.41\overline{6}$; $x_3 = 1.41667 - \frac{0.0069444}{2.83333} \approx 1.414216$.

✓ $x_1 = 1.5$, $x_2 \approx 1.416667$, $x_3 \approx 1.414216$. The number of correct digits roughly doubles each step.

### When Newton's method fails

| Problem | What happens | Example |
|---|---|---|
| $f'(x_n) = 0$ | Horizontal tangent never meets the axis: division by zero | $x^2 - 2$ with $x_0 = 0$ |
| Near a stationary point | Tangent is nearly flat, so the next estimate shoots far away | $x^3 - 3x + 1$ with $x_0 = 1.01$ |
| Cycling | Estimates bounce between values forever | $x^3 - 2x + 2$ with $x_0 = 0$: 0, 1, 0, 1, ... |
| Wrong root | Converges, but to a different root from the one you wanted | Starting on the far side of a turning point |

**Worked example 3** Show that it cycles — 2 marks

Show that Newton's method applied to $f(x) = x^3 - 2x + 2$ with $x_0 = 0$ does not converge.

1. $f'(x) = 3x^2 - 2$. $x_1 = 0 - \dfrac{2}{-2} = 1$.

   USING $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

2. $x_2 = 1 - \dfrac{1 - 2 + 2}{3 - 2} = 1 - 1 = 0$.

✓ **The estimates alternate 0, 1, 0, 1, … and never approach the actual root** (which is near $x \approx -1.77$).

## The bisection method

If $f$ is continuous and $f(a)$ and $f(b)$ have opposite signs, there's a root in $[a, b]$. Evaluate the midpoint $m = \dfrac{a+b}{2}$ and keep the half where the sign still changes. It's slow but reliable: after $n$ steps the interval width is $\dfrac{b-a}{2^n}$.

## Pseudocode, VCAA style

VCAA exams show algorithms in a simple pseudocode and ask you to trace them (fill in a table of values), find an output, or complete a missing line. The conventions: `←` means "assign", `define` creates a function, and loops are `for … end for` and `while … end while`.

```
define f(x)
    return x^3 - 2x - 5
define df(x)
    return 3x^2 - 2

x ← 2
for i from 1 to 3
    x ← x - f(x) / df(x)
end for
print x
```

This prints $x_3 \approx 2.094551$ (three iterations of Newton's method from $x_0 = 2$).

**Worked example 4** TRICKY **Trace a while loop** 3 marks

Trace this algorithm and state the output.

```
a ← 1
b ← 2
while b - a > 0.2
    m ← (a + b) / 2
    if m^2 - 2 < 0 then
        a ← m
    else
        b ← m
    end if
end while
print a, b
```

1. It's bisection on $x^2 - 2$. Start $a = 1, b = 2$ (width 1).
2. $m = 1.5$: $2.25 - 2 > 0$, so $b = 1.5$. $m = 1.25$: $1.5625 - 2 < 0$, so $a = 1.25$. Width now $0.25 > 0.2$: continue.

   USING $m = \dfrac{a+b}{2}$
3. $m = 1.375$: $1.890625 - 2 < 0$, so $a = 1.375$. Width $0.125 \leq 0.2$: stop.

**Output: 1.375, 1.5** (an interval containing $\sqrt{2} \approx 1.414$).

**Worked example 5** TRICKY **An accumulator loop (trapezium sum)** 3 marks

State the output, and the purpose of line 7.

```
define f(x)
    return x^2
a ← 0, b ← 2, n ← 4
h ← (b − a) / n
sum ← f(a) + f(b)
for i from 1 to n − 1
    sum ← sum + 2 × f(a + i × h)
end for
print h / 2 × sum
```

1. $h = 0.5$; start sum $= f(0) + f(2) = 4$. $i = 1$: $+2(0.25) = 4.5$; $i = 2$: $+2(1) = 6.5$; $i = 3$: $+2(2.25) = 11$.

**Output** $0.25 \times 11 = 2.75$: the trapezium estimate of $\displaystyle\int_0^2 x^2\,dx$ with 4 strips. Line 7 adds twice each interior height. If an exam blanks out a line like this, ask what the formula needs that the loop isn't yet doing.

**Trick alert: algorithm traps**

- **Count loop iterations carefully.** `for i from 1 to 3` runs 3 times; a `while` loop checks its condition *before* each pass.
- **Use the updated value.** Once `x ← …` runs, every later line uses the new $x$.
- **Newton needs $f'$, bisection needs a sign change.** Don't mix up the input requirements.
- **Keep full precision** in the calculator between iterations; round only the final answer.

## Summary

- $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$: fast, but it can fail near stationary points, cycle, or find a different root.
- Bisection: halve the interval while keeping a sign change. Slow but always works.
- Pseudocode: trace line by line, update variables immediately, and count loop passes carefully.

> **Also examined in:** 2023 Exam 2 B3 f–g **(question p. 35; answer p. 36)**; 2025 Exam 2 B4 e **(question p. 68; answer p. 68)**

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### Hard exam questions

**2024 Exam 2 · Section A · Question 17** © VCAA — **27% correct**

> **The question (not reprinted in this copy):** Output of an algorithm that searches for integer roots of a cubic
> Full question: 2024 Exam 2, Section A, Question 17 (page 9 of the paper).

> **Answer: D** · A 14% · B 26% · C 32% · D 27%
>
> The roots are 3, −3 and 2. The loop counts c down from 18, checking c then −c, so it prints 3, −3, 2.

**2023 Exam 2 · Section A · Question 13** © VCAA — **52% correct**

> **The question (not reprinted in this copy):** Result of three Newton's method iterations on $x^3 + 3x - 3$ from $x_0 = 1$
> Full question: 2023 Exam 2, Section A, Question 13 (page 6 of the paper).

> **Answer: C** · A 16% · B 22% · C 52% · D 7% · E 3%
>
> 1 → 0.83333 → 0.81785 → 0.81773. Count exactly three iterations.

CALCULUS

# Limits, continuity & differentiability

## Limits

$\lim_{x \to a} f(x) = L$ means $f(x)$ gets as close as we like to $L$ as $x$ gets close to $a$ (from both sides). It doesn't matter what happens *at* $x = a$ itself.

**Worked example 1** **Removing a 0/0 hole** 1 mark

Evaluate $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$.

1. Direct substitution gives $\dfrac{0}{0}$: factorise. $\dfrac{(x-3)(x+3)}{x-3} = x + 3$ for $x \neq 3$.

   USING factorise and cancel

✓ $\lim = 3 + 3 = 6$. The graph is the line $y = x + 3$ with a hole at $(3, 6)$.

**Limits at infinity:** divide top and bottom by the highest power. $\lim_{x \to \infty} \dfrac{2x+1}{x-3} = \lim_{x \to \infty} \dfrac{2 + \frac{1}{x}}{1 - \frac{3}{x}} = 2$: the horizontal asymptote.

## The derivative from first principles

**Definition of the derivative**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

The fraction is the gradient of the **secant** through $(x, f(x))$ and $(x+h, f(x+h))$. As $h \to 0$, the secant becomes the **tangent**.

**Worked example 2** **First principles (assumed knowledge from Units 1–2)** 3 marks

Use first principles to find the derivative of $f(x) = x^2 - 3x$.

1. $f(x+h) = (x+h)^2 - 3(x+h) = x^2 + 2xh + h^2 - 3x - 3h$.

   USING $f(x+h)$

2. $f(x+h) - f(x) = 2xh + h^2 - 3h$, so $\dfrac{f(x+h) - f(x)}{h} = 2x + h - 3$ (for $h \neq 0$).

   USING $\dfrac{f(x+h) - f(x)}{h}$

✓ $f'(x) = \lim_{h \to 0}(2x + h - 3) = 2x - 3$. Write "$\lim_{h \to 0}$" on every line until the last one: examiners deduct marks when it disappears too early.

USING $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

## Continuity

$f$ is continuous at $x = a$ if $f(a)$ is defined and $\lim_{x \to a} f(x) = f(a)$. In practice: you can draw through $x = a$ without lifting your pen. For a hybrid function, the left and right rules must give the same value at the join.

## Differentiability

$f$ is differentiable at $x = a$ if the left and right gradients exist and are equal. A function that is **not continuous** at a point is **not differentiable** there. But continuity alone isn't enough:

| Feature | Continuous? | Differentiable? | Example |
|---|---|---|---|
| Jump or hole | No | No | Hybrid with mismatched pieces |
| Corner (sharp point) | Yes | No | $\lvert x \rvert$ at $x = 0$; $\lvert x^2 - 4 \rvert$ at $x = \pm 2$ |
| Cusp | Yes | No | $x^{2/3}$ at $x = 0$ |
| Vertical tangent | Yes | No | $x^{1/3}$ at $x = 0$ |
| Endpoint of a closed domain | Yes | No (only one-sided) | $\sqrt{x}$ at $x = 0$ |

**Exam tip**

VCAA convention: a function is differentiable on an **open** interval. When asked for the domain of $f'$, exclude endpoints and every corner, cusp, jump and vertical tangent.

**Continuous, smooth and differentiable are asked in a fixed order** TRICKY

At a join $x = c$ of a hybrid function there are two separate tests, and the second needs the first. **Continuous:** the left rule and the right rule give the same *value* at $c$ (one equation). **Smooth** (differentiable): the two rules also have the same *gradient* at $c$ (a second equation). Differentiable always means continuous too, but a continuous join can still be a corner, like $\lvert x \rvert$ at 0.

So "find $a$ and $b$ so that $f$ is continuous and smooth at $c$" always means **two equations** (values, then derivatives of each piece) in two unknowns. For the domain of $f'$, leave out every corner (continuous but gradients differ) and every endpoint, and keep a join only if it is smooth.

**Worked example 3** TRICKY **A smooth join (continuous and differentiable)** 3 marks

Find $a$ and $b$ such that $f(x) = \begin{cases} ax^2 + b, & x \leq 1 \\ 4x - 1, & x > 1 \end{cases}$ is differentiable at $x = 1$.

1. **Continuous:** $a(1)^2 + b = 4(1) - 1 \Rightarrow a + b = 3$.
   USING continuous: pieces meet
2. **Gradients match:** left derivative $2ax$ at $x = 1$ is $2a$; right derivative is 4. So $2a = 4$.
   USING differentiable: gradients match

$a = 2,\ b = 1$. Both conditions are needed. Matching gradients without continuity doesn't make the function differentiable.

**Worked example 4** **Domain of the derivative** 2 marks

State the domain of $f'$ for $f : [-3, 3] \to \mathbb{R},\ f(x) = |x^2 - 4|$.

1. Corners where the graph is reflected: $x^2 - 4 = 0 \Rightarrow x = \pm 2$. Endpoints $\pm 3$ are also excluded.
   USING $|u|$ has a corner where $u = 0$

✓ $(-3, -2) \cup (-2, 2) \cup (2, 3)$.

**Worked example 5** TRICKY **Writing the derivative of a hybrid** 2 marks

Let $f(x) = \begin{cases} x^2, & 0 \le x \le 2 \\ 8 - 2x, & 2 < x \le 4 \end{cases}$. Write $f'(x)$ as a hybrid function, with its domain.

1. Left piece: $2x$, giving 4 at $x = 2$. Right piece: $-2$. The pieces meet (both 4) but the gradients differ, so there's a corner at $x = 2$.
   USING differentiate each piece

✓ $f'(x) = \begin{cases} 2x, & 0 < x < 2 \\ -2, & 2 < x < 4 \end{cases}$, with domain $(0, 2) \cup (2, 4)$. If the join were smooth, you'd include $x = 2$ in one piece.

**Trick alert: limit and differentiability traps**

- **Continuous doesn't mean differentiable.** Every corner is continuous.
- **A limit can exist even where $f(a)$ doesn't** (a hole), and $f(a)$ can exist without the limit (a jump).
- **For hybrid gradients, differentiate each piece separately** and compare the values at the join. Don't differentiate across the join.
- **Endpoints:** derivatives are only defined on open intervals in VCAA's convention.

## Summary

- $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$: expand, cancel $h$, then let $h \to 0$.
- Continuous: no breaks. Differentiable: continuous *and* no corners, cusps or vertical tangents.
- Smooth hybrid join: match values *and* gradients.

### Hard exam questions

**2023 Exam 2 · Section A · Question 9** © VCAA 42% correct

**The question (not reprinted in this copy):** Value of a that makes a tan/sin hybrid continuous and smooth at x = 2π
Full question: 2023 Exam 2, Section A, Question 9 (page 5 of the paper).

**Answer: C** · A 7% · B 9% · C 42% · D 31% · E 10%

Continuity needs $\sin(2\pi a) = 0$, so $a = \dfrac{n}{2}$ for an integer $n$. Smoothness needs $a\cos(2\pi a) = \dfrac{1}{2}$ (the gradient of $\tan\dfrac{x}{2}$ at $2\pi$ is $\dfrac{1}{2}\sec^2\pi = \dfrac{1}{2}$), i.e. $\dfrac{n}{2}(-1)^n = \dfrac{1}{2}$, which holds only for $n = -1$. So $a = -\dfrac{1}{2}$ ($a = \dfrac{1}{2}$ gives $-\dfrac{1}{2}$).

CALCULUS

# Differentiation rules

## The standard derivatives

| $f(x)$ | $f'(x)$ | $f(x)$ | $f'(x)$ |
|---|---|---|---|
| $x^n$ | $nx^{n-1}$ | $\sin(kx)$ | $k\cos(kx)$ |
| $(ax+b)^n$ | $an(ax+b)^{n-1}$ | $\cos(kx)$ | $-k\sin(kx)$ |
| $e^{kx}$ | $ke^{kx}$ | $\tan(kx)$ | $\dfrac{k}{\cos^2(kx)} = k\sec^2(kx)$ |
| $\log_e(kx)$ | $\dfrac{1}{x}$ | $\log_e(g(x))$ | $\dfrac{g'(x)}{g(x)}$ |

Rewrite roots and fractions as powers before differentiating: $\sqrt{x} = x^{1/2}$ and $\dfrac{3}{x^2} = 3x^{-2}$.

## The three rules

**Chain, product and quotient rules**

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx} \qquad (uv)' = u'v + uv' \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Function notation for the chain rule: $\big(f(g(x))\big)' = f'(g(x))\cdot g'(x)$.

**Key idea**

Decide which rule to use from the **outermost** structure. $x^2e^{3x}$ is a product. $e^{\sin x}$ is a composite (chain). $\dfrac{\sin x}{x}$ is a quotient. Many questions need two rules in a row.

**Worked example 1** Product rule, then factorise — 2 marks

Differentiate $y = x^2e^{3x}$ and fully factorise the result.

1. $u = x^2,\ u' = 2x;\ v = e^{3x},\ v' = 3e^{3x}$.
   USING $u = x^2,\ v = e^{3x}$

✓ $\dfrac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x} = \boldsymbol{xe^{3x}(2+3x)}$. Factorised form makes stationary points easy: $x = 0$ or $x = -\dfrac{2}{3}$.
USING $\dfrac{d}{dx}(uv) = u'v + uv'$

**Worked example 2** Chain rule with a log — 2 marks

Find $\dfrac{dy}{dx}$ for (a) $y = \log_e(x^2+1)$, (b) $y = \log_e(\cos x)$.

1. Use $\dfrac{d}{dx}\log_e(g(x)) = \dfrac{g'(x)}{g(x)}$.
   USING $\dfrac{d}{dx}\log_e(g(x)) = \dfrac{g'(x)}{g(x)}$

✓ **(a)** $\dfrac{2x}{x^2+1}$; **(b)** $\dfrac{-\sin x}{\cos x} = -\tan x$.

Calculus

**Worked example 3** Quotient rule — 2 marks

Differentiate $f(x) = \dfrac{x}{x+1}$.

1. $u = x,\ u' = 1;\ v = x+1,\ v' = 1.$
   USING $u = x,\ v = x+1$

✓ $f'(x) = \dfrac{1\cdot(x+1) - x\cdot 1}{(x+1)^2} = \dfrac{\mathbf{1}}{\mathbf{(x+1)^2}}$, which is always positive, so $f$ is increasing on each branch.
USING $\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{u'v - uv'}{v^2}$

**Worked example 4** Evaluate a derivative exactly (tech-free) — 2 marks

Let $f(x) = x^2 \sin x$. Find $f'\left(\dfrac{\pi}{2}\right)$.

1. $f'(x) = 2x\sin x + x^2\cos x.$
   USING $\dfrac{d}{dx}(uv) = u'v + uv'$

✓ $f'\left(\dfrac{\pi}{2}\right) = 2\cdot\dfrac{\pi}{2}\cdot 1 + \dfrac{\pi^2}{4}\cdot 0 = \boldsymbol{\pi}.$
USING $\sin\dfrac{\pi}{2} = 1,\ \cos\dfrac{\pi}{2} = 0$

**F'(g(x)) means f' evaluated at g(x)** — TRICKY

For $h(x) = f(g(x))$ the chain rule is $h'(x) = f'(g(x))\cdot g'(x)$: the derivative of the outer function, **evaluated at the inner function's value**, times the derivative of the inner. With a table, at $x = 2$: first find $g(2)$, then look up $f'$ *at that number*, then multiply by $g'(2)$.

E.g. $g(2) = 5$, $g'(2) = 3$, $f'(5) = 4$: $h'(2) = f'(5)\times 3 = 12$. The common slip is to use $f'(2)$. The same pattern gives $\dfrac{d}{dx}\sqrt{f(x)} = \dfrac{f'(x)}{2\sqrt{f(x)}}$: the outer derivative $\dfrac{1}{2\sqrt{u}}$ at $u = f(x)$, times $f'(x)$.

**Worked example 5** Derivatives from a table (Exam 1 style) — 3 marks

The table gives values of $f$, $g$ and their derivatives.

| $x$ | $f(x)$ | $f'(x)$ | $g(x)$ | $g'(x)$ |
|---|---|---|---|---|
| 1 | 3 | −2 | 2 | 5 |
| 2 | 4 | 7 | 1 | −3 |

Find the derivatives of (a) $f(g(x))$ at $x = 1$, (b) $f(x)g(x)$ at $x = 2$, (c) $[f(x)]^2$ at $x = 1$.

1. (a) $f'(g(1))\cdot g'(1) = f'(2)\cdot 5 = 7\times 5 = 35.$
   USING $\dfrac{d}{dx}f(g(x)) = f'(g(x))\,g'(x)$

2. (b) $f'(2)g(2) + f(2)g'(2) = 7(1) + 4(-3) = -5.$
   USING $\dfrac{d}{dx}(uv) = u'v + uv'$

3. (c) $2f(1)f'(1) = 2(3)(-2) = -12.$
   USING $\dfrac{d}{dx}[f(x)]^2 = 2f(x)f'(x)$

✓ **(a) 35, (b) −5, (c) −12.** In (a), $f'$ is evaluated at $g(1) = 2$, not at 1. That's the trap.

**Trick alert: differentiation traps**

- $\frac{d}{dx}\log_e(2x) = \frac{1}{x}$, not $\frac{2}{x}$: the 2 cancels ($\frac{2}{2x}$).
- $\sin^2 x$ **vs** $\sin(x^2)$: $2\sin x\cos x$ vs $2x\cos(x^2)$.
- $e^{x^2}$ **differentiates to** $2xe^{x^2}$, not $x^2e^{x^2-1}$. The power rule doesn't apply to exponentials.
- **The derivative of a constant is 0:** $\frac{d}{dx}e^2 = 0$ and $\frac{d}{dx}\pi^2 = 0$.
- **Quotient rule sign:** it's $u'v - uv'$ on top, so the order matters.

## Summary

- Know the standard derivatives, including $\tan(kx) \to \frac{k}{\cos^2 kx}$ and $\log_e g \to \frac{g'}{g}$.
- Chain: derivative of outside (inside unchanged) × derivative of inside.
- Product: $u'v + uv'$. Quotient: $\frac{u'v - uv'}{v^2}$.
- Factorise derivatives: you'll usually need $f'(x) = 0$ next.

### Hard exam questions

**2025 Exam 2 · Section A · Question 16** © VCAA — 18% correct

**The question (not reprinted in this copy):** Conditions on a, b if $h(x) = a\log_e(bx)$ has $h'(x) > 0$ everywhere
Full question: 2025 Exam 2, Section A, Question 16 (page 9 of the paper).

**Answer: D** · A 20% · B 16% · C 46% · D 18%

$h'(x) = \frac{a}{x}$, with domain $bx > 0$. If b > 0 then x > 0 and you need a > 0. If b < 0 then x < 0 and you need a < 0. Either way ab > 0.

**2023 Exam 2 · Section A · Question 11** © VCAA — 22% correct

**The question (not reprinted in this copy):** Gradient of f × g at x = −2 from given values
Full question: 2023 Exam 2, Section A, Question 11 (page 5 of the paper).

**Answer: E** · A 6% · B 13% · C 8% · D 51% · E 22%

Product rule: $3 \times 8 + (-7) \times 2 = 10$.

CALCULUS

# Applications of differentiation

## Tangents and normals

**Tangent and normal at $x = a$**

$$\text{tangent: } y - f(a) = f'(a)(x - a) \qquad \text{normal: } y - f(a) = -\frac{1}{f'(a)}(x - a)$$

The normal is perpendicular to the tangent: $m_1 m_2 = -1$.

**Worked example 1** **Tangent and normal** 3 marks

Find the equations of the tangent and the normal to $y = x^3 - 2x$ at $x = 1$.

1. Point: $y(1) = 1 - 2 = -1$, so $(1, -1)$. Gradient: $y' = 3x^2 - 2$, so $y'(1) = 1$.
   USING $m_{\text{tangent}} = f'(a)$
2. Tangent: $y + 1 = 1(x - 1) \Rightarrow y = x - 2$. Normal gradient $= -1$: $y + 1 = -(x - 1) \Rightarrow y = -x$.
   USING $y - y_1 = m(x - x_1)$ $m_{\text{normal}} = -\frac{1}{f'(a)}$

✓ **Tangent $y = x - 2$; normal $y = -x$.**

**A tangent from an outside point: the point of contact is the unknown** TRICKY

If the given point isn't on the curve, you can't use it to find the gradient. Instead call the **point of contact** $(a, f(a))$. The tangent there is $y - f(a) = f'(a)(x - a)$. Now force this line through the given point: substitute its coordinates and you get an equation in $a$ alone. **Each solution $a$ is one tangent**, so "how many tangents" means "how many solutions for $a$".

**Worked example 2** TRICKY **Tangents through a point not on the curve (hard)** 4 marks

Find the equations of the tangents to $y = x^2$ that pass through the point $(1, -3)$.

1. Let the point of contact be $(a, a^2)$. The gradient there is $2a$, so the tangent is $y - a^2 = 2a(x - a)$, i.e. $y = 2ax - a^2$.
   USING $y - f(a) = f'(a)(x - a)$
2. It must pass through $(1, -3)$: $-3 = 2a - a^2 \Rightarrow a^2 - 2a - 3 = 0 \Rightarrow (a - 3)(a + 1) = 0$.
   USING substitute the external point

✓ $a = 3$: $y = 6x - 9$; $a = -1$: $y = -2x - 1$. The external point isn't on the curve, so you can't use it as the point of contact. Use a general point instead.

**Worked example 3** TRICKY **A common tangent to two curves (hard)** 4 marks

Find the equation of the straight line that is a tangent to both $y = x^2$ and $y = x^2 - 4x + 8$.

1. With $y = x^2$: $x^2 - mx - c = 0$, so $\Delta = m^2 + 4c = 0$, giving $c = -\frac{m^2}{4}$.
   USING $\Delta = b^2 - 4ac = 0 \Rightarrow$ one solution (tangent)
2. With $y = x^2 - 4x + 8$: $x^2 - (4+m)x + (8-c) = 0$, so $(4+m)^2 - 4(8-c) = 0$.
   USING $\Delta = b^2 - 4ac = 0$
3. Substitute $4c = -m^2$: $(4+m)^2 - 32 - m^2 = 0 \Rightarrow 16 + 8m - 32 = 0 \Rightarrow m = 2$, so $c = -1$.

✓ $y = 2x - 1$. Check: it touches $y = x^2$ at $(1, 1)$ and $y = x^2 - 4x + 8$ at $(3, 5)$. If both curves have a turning point at the same height, that horizontal line is a common tangent too (gradient 0), so look for it first.

### The angle a tangent makes with the x-axis

A line with gradient $m$ makes an angle $\theta$ with the positive direction of the $x$-axis, where $\tan\theta = m$. A negative gradient gives an obtuse angle ($\theta = 180^\circ + \tan^{-1} m$, or $\pi + \tan^{-1} m$ in radians).

## Rates of change

**Average** rate of change over $[a, b]$ = $\frac{f(b) - f(a)}{b - a}$ (gradient of a secant). The **instantaneous** rate at $x = a$ is $f'(a)$ (gradient of the tangent). Read the question carefully: VCAA asks for both, often in the same question.

**Worked example 4** TRICKY **Average vs instantaneous rate (Exam 2 style)** 4 marks

A drink cools so that its temperature after $t$ minutes is $T(t) = 80e^{-0.2t} + 20$ °C. Find (a) the average rate of change over the first 5 minutes, (b) the instantaneous rate at $t = 5$, and (c) when the instantaneous rate equals the average rate in (a).

1. (a) $\frac{T(5) - T(0)}{5} = \frac{80e^{-1} - 80}{5} = -16\left(1 - e^{-1}\right) \approx -10.11$ °C/min.
   USING $\frac{T(b) - T(a)}{b - a}$
2. (b) $T'(t) = -16e^{-0.2t}$, so $T'(5) = -16e^{-1} \approx -5.89$ °C/min.
   USING $T'(a)$
3. (c) $-16e^{-0.2t} = -16\left(1 - e^{-1}\right) \Rightarrow t = -5\log_e\left(1 - e^{-1}\right) \approx 2.29$.
   USING solve $T'(t) =$ average rate

✓ **(a) −10.11 °C/min; (b) −5.89 °C/min; (c)** $t \approx 2.29$ **min.** Units on rates: °C per minute.

## Stationary points and their nature

Stationary points occur where $f'(x) = 0$. Classify with a **sign diagram of** $f'$ or with the **second derivative**:

| Sign of $f'$ (left → right) | $f''(a)$ | Nature |
|---|---|---|
| $+ \to -$ | $< 0$ | local maximum |
| $- \to +$ | $> 0$ | local minimum |
| $+ \to +$ or $- \to -$ | usually 0, but $f''(a) = 0$ on its own decides nothing: use the sign test | stationary point of inflection |

Calculus

**f''(a) = 0 doesn't prove an inflection**

$f(x) = x^4$ has $f'(0) = 0$ and $f''(0) = 0$, but $(0, 0)$ is a local *minimum*. When the second derivative is 0, go back to the sign test.

**Worked example 5** TRICKY **A gradient table, VCAA style** 3 marks

For $f(x) = x^4 - 4x^3$, find the stationary points and use a gradient table to state their nature.

1. $f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3) = 0 \Rightarrow x = 0$ or $x = 3$. $f(0) = 0$, $f(3) = 81 - 108 = -27$.
   USING $f'(x) = 0$

2. Gradient table:

| $x$ | $-1$ | $0$ | $1$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $f'(x)$ | $-16$ | $0$ | $-8$ | $0$ | $64$ |
| slope | ↘ | → | ↘ | → | ↗ |

   USING evaluate $f'$ either side

✓ **$(0, 0)$ is a stationary point of inflection (negative on both sides); $(3, -27)$ is a local minimum.** Here $f''(0) = 0$ as well, which on its own proves nothing.

**Worked example 6** **Full curve sketch of a cubic** 5 marks

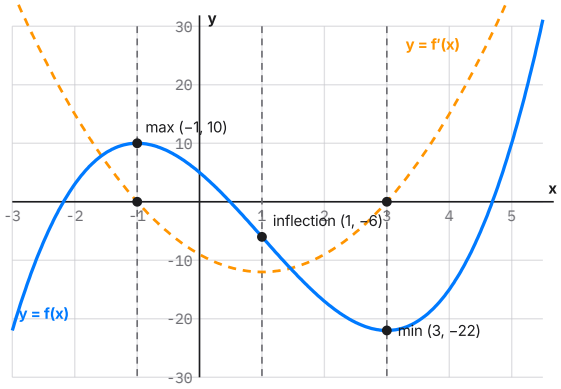
For $f(x) = x^3 - 3x^2 - 9x + 5$, find the stationary points and their nature, the point of inflection, and the intervals where $f$ is strictly decreasing.

1. $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1) = 0 \Rightarrow x = -1, 3$.
   USING $f'(x) = 0$

2. $f(-1) = -1 - 3 + 9 + 5 = 10$; $f(3) = 27 - 27 - 27 + 5 = -22$. $f''(x) = 6x - 6$: $f''(-1) = -12 < 0$ (max), $f''(3) = 12 > 0$ (min).
   USING $f''(a) < 0 \Rightarrow \max$ $f''(a) > 0 \Rightarrow \min$

3. Inflection: $f''(x) = 0 \Rightarrow x = 1$, and $f''$ changes sign there. $f(1) = -6$.
   USING $f''(x) = 0$ and changes sign

✓ **Local max $(-1, 10)$, local min $(3, -22)$, inflection $(1, -6)$. Strictly decreasing on $[-1, 3]$.** VCAA includes the endpoints of monotone intervals (the function is still strictly decreasing on the closed interval). But $f'(x) < 0$ only on the open interval $(-1, 3)$: when a question asks where the gradient is negative (or where $f'(x) > 0$), give the open interval.




$f(x) = x^3 - 3x^2 - 9x + 5$ (solid) and its gradient function $f'(x) = 3x^2 - 6x - 9$ (dashed). Where $f$ turns, $f'$ crosses the axis; where $f$ inflects, $f'$ has its turning point.

**"Hence verify there's a stationary point in the interval"**

If $f'$ is continuous on $[a, b]$ and $f'(a)$ and $f'(b)$ have **opposite signs**, then $f'(x) = 0$ somewhere in $(a, b)$, so $f$ has a stationary point there. It's the same sign-change idea as bisection, applied to $f'$. "Verify" means evaluate and conclude; don't solve $f'(x) = 0$ exactly.



## Concavity and points of inflection

$f''(x) > 0$: concave up (gradient increasing). $f''(x) < 0$: concave down. A **point of inflection** is where the concavity changes; it may or may not be stationary. At a non-stationary inflection point the gradient is at a local max or min, so it's where a quantity grows fastest or slowest (a favourite Exam 2 context).

**Worked example 7** Max and inflection of x e^(−x) — 4 marks

For $f(x) = xe^{-x}$, find the coordinates of the stationary point and the point of inflection.

1. $f'(x) = e^{-x} - xe^{-x} = (1 - x)e^{-x} = 0 \Rightarrow x = 1$. $f(1) = e^{-1}$; $f'$ goes $+ \to -$, so it's a maximum.
   USING $\frac{d}{dx}(uv) = u'v + uv'$ $\quad e^{-x} \neq 0$
2. $f''(x) = -e^{-x} - (1 - x)e^{-x} = (x - 2)e^{-x} = 0 \Rightarrow x = 2$, and it changes sign.
   USING $f''(x) = 0$ and changes sign

✓ **Local max $\left(1, \frac{1}{e}\right)$; inflection $\left(2, \frac{2}{e^2}\right)$.**

**Worked example 8** TRICKY Greatest rate of change: inflection or endpoint — 3 marks

The depth of water in a channel is $h(t) = -t^3 + 6t^2 + 15$ m, $0 \leq t \leq 5$ hours. When is the depth increasing most rapidly, and at what rate? When is it decreasing most rapidly?

1. $h'(t) = -3t^2 + 12t$, $h''(t) = -6t + 12 = 0 \Rightarrow t = 2$, and $h'(2) = 12$.
   USING max of $h'$ : $h''(t) = 0$
2. Endpoints: $h'(0) = 0$, $h'(5) = -75 + 60 = -15$. CAS: `fMax(derivative(h(t),t),t)|0≤t≤5` and `fMin(…)`.
   USING check the endpoints

✓ **Increasing most rapidly at $t = 2$ h, at 12 m/h; decreasing most rapidly at $t = 5$ h (the endpoint), at 15 m/h.**

## Sketching the gradient function

Read across a row to match a graph with its gradient graph (or, reading right to left, $f'$ back to $f$; the same table links an antiderivative $F$ to $f$).

| On the graph of $f$ | On the graph of $f'$ |
|---|---|
| increasing / decreasing | above / below the $x$-axis |
| local max / local min | crosses the axis $+ \to -$ / $- \to +$ |
| stationary point of inflection | touches the axis without crossing |
| non-stationary point of inflection | a turning point not on the axis |
| concave up / concave down | increasing / decreasing |
| straight-line section, gradient $m$ | horizontal section at height $m$ |
| corner | a jump (open circles), $f'$ undefined there |
| cusp | a vertical asymptote, with opposite signs on each side (e.g. $y = x^{2/3}$) |
| vertical tangent | vertical asymptote |
| vertical asymptote | vertical asymptote at the same $x$ |
| horizontal asymptote | $f' \to 0$ (approaches the $x$-axis) |

**"Which graph could be f'?"**

Mark the $x$-values of every stationary point of $f$ first and rule out any option that isn't zero exactly there. Then check the sign between them. That usually leaves one option; use concavity (where $f'$ turns) only to break a tie.

**Where is f increasing or decreasing? (endpoints in or out)** TRICKY

Find the stationary points, then test the sign of $f'$ between them. For $f(x) = x^3 - 3x^2 - 9x + 5$, $f'(x) = 3(x+1)(x-3) < 0$ for $-1 < x < 3$. VCAA includes the stationary points when asked where $f$ is **strictly increasing or decreasing**: $f$ is strictly decreasing on $[-1, 3]$, because it keeps falling right through the stationary ends. A question about where the **gradient is negative** ($f'(x) < 0$) wants the open interval $(-1, 3)$.

Wording decides the brackets, so copy the question's wording into your answer. Also, "strictly increasing for all $x$" allows $f' = 0$ at isolated points ($x^3$ is strictly increasing although $f'(0) = 0$), so the condition is $f' \geq 0$, not $f' > 0$.

**Trick alert: curve-sketching traps**

- **Stationary point ≠ turning point.** Turning points are only the maxima and minima; stationary inflections also have $f' = 0$.
- **Local ≠ absolute.** The absolute maximum on a closed interval might be at an endpoint.
- **Normal gradient:** negative reciprocal. If the tangent is horizontal, the normal is vertical ($x = a$).
- **A factor squared in $f'$** (e.g. $f'(x) = (x-2)^2(x+1)$) gives no sign change at $x = 2$: stationary inflection.

## Summary

- Tangent: $y - f(a) = f'(a)(x - a)$. For tangents through an external point, use a general contact point $(a, f(a))$.
- Stationary points: solve $f' = 0$; classify with a sign diagram (safest) or $f''$.
- Inflection: $f''$ changes sign; the gradient is at its greatest or least there.
- Monotone intervals are given as closed intervals in VCAA solutions.

## Hard exam questions

**2023 Exam 2 · Section A · Question 14** © VCAA — 29% correct

**The question (not reprinted in this copy):** Number of tangents to a cubic that pass through its positive x-intercept
Full question: 2023 Exam 2, Section A, Question 14 (page 7 of the paper).

**Answer: D** · A 6% · B 33% · C 22% · D 29% · E 10%

A tangent at $x = a$ passes through $\left(\frac{1}{3}, 0\right)$ when $-f(a) = f'(a)\left(\frac{1}{3} - a\right)$. This simplifies to $(3a-1)^2(3a^2+8a+3) = 0$, so $a = \frac{1}{3}$ (the tangent at the intercept itself counts) or $a = \frac{-4 \pm \sqrt{7}}{3}$: three tangents.

**2025 Exam 2 · Section B · Question 4** © VCAA — state average 45% of marks

**The question (not reprinted in this copy):** $f(x) = \sin x + 1$ on $[0, \frac{5\pi}{2}]$: values, periodicity, tangents, Newton's method, a family of tangents, matching a polynomial
Full question: 2025 Exam 2, Section B, Question 4 (page 23 of the paper).

**a.** $f\left(\frac{2\pi}{3}\right)$ (1 mark · state average 0.9 / 1, 90%)

- $\frac{\sqrt{3}}{2} + 1$

**b.** Exact x with $f(x) = \frac{3}{2}$ (1 mark · state average 0.8 / 1, 80%)

- $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$

**Notes:** $\sin x = \frac{1}{2}$. Three solutions lie in the domain.

**c.** k and the largest a with f(x + k) = f(x) on [0, a] (2 marks · state average 0.6 / 2, 30%)

- $k = 2\pi$
- Largest $a = \frac{\pi}{2}$

**Notes:** The period is $2\pi$, and $x + 2\pi$ must stay within $\frac{5\pi}{2}$.

**d.** Tangent at $x = \frac{2\pi}{3}$ (1 mark · state average 0.7 / 1, 70%)

- $y = -\frac{x}{2} + \frac{\pi}{3} + \frac{\sqrt{3}}{2} + 1$

**Notes:** Gradient $\cos\frac{2\pi}{3} = -\frac{1}{2}$.

**e.i.** Newton's method from $x_0 = \frac{2\pi}{3}$: $x_2$ to 1 d.p. (1 mark · state average 0.6 / 1, 60%)

- $x_2 \approx 5.2$

**Notes:** $x_1 = \frac{2\pi}{3} + \sqrt{3} + 2 \approx 5.83$, then $x_2 \approx 5.2$.

**e.ii.** Draw the tangent at $x = x_1$ on the graph (1 mark · state average 0.3 / 1, 30%)

- Tangent at $x_1 \approx 5.83$ drawn, meeting the x-axis at $x_2 \approx 5.2$

**f.i.** Show the tangent at $x = p$ is $t(x) = \cos(p)(x-p) + \sin(p) + 1$ (2 marks · state average 1.1 / 2, 55%)

- Gradient $f'(p) = \cos p$ through $(p, \sin p + 1)$
- Correct point–gradient form rearranged

**f.ii.** Minimum and maximum y-intercepts of t for $p \in (0, \frac{5\pi}{2})$ (2 marks · state average 0.5 / 2, 25%)

- y-intercept $= \sin p - p\cos p + 1$, derivative $p\sin p$, stationary at $\pi, 2\pi$
- Minimum $1 - 2\pi$, maximum $1 + \pi$

**f.iii.** p for which t has a unique x-intercept equal to f's x-intercept (2 d.p.) (2 marks · state average 0.4 / 2, 20%)

- Requires $t\left(\frac{3\pi}{2}\right) = 0$ with $p \neq \frac{3\pi}{2}$ (horizontal tangent excluded)
- $p \approx 2.38$ and $p \approx 7.04$

**Notes:** Solve $\cos p\left(\frac{3\pi}{2} - p\right) + \sin p + 1 = 0$: p ≈ 2.38, 7.04.

**g.i.** Show c = 1 and d = 1 given $g(0) = f(0)$, $g'(0) = f'(0)$ (2 marks · state average 1.4 / 2, 70%)

- $g(0) = d = f(0) = 1$
- $g'(0) = c = \cos 0 = 1$

**Notes:** Direct substitution.

**g.ii.** Area between f and g on $[0, 2\pi]$ when they also match at $2\pi$ (2 d.p.) (2 marks · state average 0.7 / 2, 35%)

- Solves for $a = \frac{1}{2\pi^2}$, $b = -\frac{3}{2\pi}$
- Area $\approx 1.53$

**Notes:** Use $\int_0^{2\pi} |f - g|\,dx \approx 1.53$.

**g.iii.** b and r with $g(r) = f(r)$, $g'(r) = f'(r)$ when a = 0, c = d = 1 (2 marks · state average 0.5 / 2, 25%)

- Sets up $br^2 + r = \sin r$ and $2br + 1 = \cos r$
- $r = \pi$, $b = -\frac{1}{\pi}$

**2024 Exam 2 · Section A · Question 10** © VCAA — 38% correct

**The question (not reprinted in this copy):** What must be true if $f'(2) < 0$ and $f'(4) > 0$
Full question: 2024 Exam 2, Section A, Question 10 (page 4 of the paper).

**Answer: B** · A 26% · B 38% · C 16% · D 20%

$f'$ is continuous and changes sign, so f is not one-to-one on [0, 5] and has no inverse. The others might not hold.

CALCULUS

# Optimisation (max/min problems)

## The method

1. **Draw a diagram** and label the variable(s).
2. **Write the quantity to optimise** as a function of *one* variable (use the constraint to eliminate the others).
3. **State the domain** (lengths positive, can't cut more than you have...).
4. **Differentiate and solve** $= 0$; check the nature.
5. **Check the endpoints** and answer the question actually asked (the maximum volume? the dimensions? the time?).

**Exam tip**

VCAA marking schemes award a mark for a correct expression in one variable and often for the domain, before any calculus. Show them.

**Worked example 1** The open box — 4 marks

Squares of side $x$ cm are cut from each corner of a 24 cm × 24 cm sheet, and the sides are folded up to make an open box. Find the maximum volume.

1. $V = x(24 - 2x)^2$, domain $0 < x < 12$.
   USING $V = lwh$
2. $V'(x) = (24 - 2x)^2 + x \cdot 2(24 - 2x)(-2) = (24 - 2x)(24 - 6x)$.
   USING $\frac{d}{dx}(uv) = u'v + uv'$ $\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x)$
3. $V' = 0 \Rightarrow x = 12$ (outside the domain; zero volume) or $x = 4$. Sign: $V' > 0$ before 4, $< 0$ after, so it's a maximum.
   USING $V'(x) = 0$

✓ $V_{\max} = 4(16)^2 = 1024 \text{ cm}^3$ when $x = 4$.

**Worked example 2** Fence against a wall — 3 marks

A farmer has 60 m of fencing to enclose a rectangular pen against a long straight wall (the wall forms one side). Find the maximum area.

1. Let the two sides perpendicular to the wall be $w$. The side parallel to the wall is $60 - 2w$. $A = w(60 - 2w)$, $0 < w < 30$.
   USING $A = lw$
2. $A'(w) = 60 - 4w = 0 \Rightarrow w = 15$. $A'' = -4 < 0$: maximum.
   USING $A'(w) = 0$ $A'' < 0 \Rightarrow \max$

✓ $A_{\max} = 15 \times 30 = 450 \text{ m}^2$.

**Worked example 3** **Minimum surface area of a can** 4 marks

A closed cylindrical can must hold 500 cm³. Find the radius that minimises its surface area, correct to two decimal places, and the minimum area.

1. Constraint: $\pi r^2 h = 500 \Rightarrow h = \dfrac{500}{\pi r^2}$.

   USING $V = \pi r^2 h$

2. $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \dfrac{1000}{r}$, $r > 0$.

   USING $S = 2\pi r^2 + 2\pi r h$

3. $S'(r) = 4\pi r - \dfrac{1000}{r^2} = 0 \Rightarrow r^3 = \dfrac{250}{\pi}$.

   USING $S'(r) = 0$

✓ $r = \left(\dfrac{250}{\pi}\right)^{1/3} \approx 4.30$ **cm**, giving $S_{\min} \approx 348.73\ \text{cm}^2$. (At the optimum $h = 2r$: the height equals the diameter.)

**Set up before you differentiate** TRICKY

Most marks in a context problem are lost *before* any calculus. (1) Name the quantity to optimise (area, time, cost). (2) If it involves two variables, use the constraint to eliminate one. (3) Write the **domain** from the situation (lengths are positive; the point must lie on the track). (4) Differentiate, solve, and compare the stationary values with the **endpoints** of the domain. (5) Answer the question asked (the maximum area, not the $x$ that gives it), with units.

For "closest point to $(p, q)$", minimise the **squared** distance $(x-p)^2 + (y-q)^2$: it is smallest at the same place and avoids the square root.

**Worked example 4** **Closest point on a curve** 4 marks

Find the minimum distance from the point $(0, 3)$ to the parabola $y = x^2$.

1. A point on the curve is $(x, x^2)$. Minimise the *square* of the distance (easier, same minimising $x$): $D = x^2 + (x^2 - 3)^2$.

   USING $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

2. $D'(x) = 2x + 4x(x^2 - 3) = 2x(2x^2 - 5) = 0 \Rightarrow x = 0$ or $x^2 = \dfrac{5}{2}$.

   USING $D'(x) = 0$

3. $D(0) = 9$ (local max of $D$); $D$ at $x^2 = \dfrac{5}{2}$: $\dfrac{5}{2} + \dfrac{1}{4} = \dfrac{11}{4}$.

✓ **Minimum distance** $= \sqrt{\dfrac{11}{4}} = \dfrac{\sqrt{11}}{2}$, at $\left(\pm\sqrt{\dfrac{5}{2}}, \dfrac{5}{2}\right)$.

**Worked example 5** TRICKY **Swim then run (Exam 2 style)** 4 marks

A swimmer is 3 km offshore, directly opposite point $P$ on a straight beach. She wants to reach a café on the beach 8 km from $P$. She swims at 2 km/h and runs at 5 km/h. Where should she land to minimise the total time?

1. Land $x$ km from $P$ towards the café. $T(x) = \dfrac{\sqrt{9+x^2}}{2} + \dfrac{8-x}{5}$, $0 \le x \le 8$.
   USING $t = \dfrac{\text{distance}}{\text{speed}}$ $d = \sqrt{a^2+b^2}$
2. $T'(x) = \dfrac{x}{2\sqrt{9+x^2}} - \dfrac{1}{5} = 0 \Rightarrow 5x = 2\sqrt{9+x^2} \Rightarrow 21x^2 = 36$.
   USING $T'(x) = 0$
3. $x = \dfrac{6}{\sqrt{21}} \approx 1.31$ km, giving $T \approx 2.9748$ h. Endpoints: $T(0) = 3.1$ h, $T(8) \approx 4.27$ h.
   USING compare with the endpoints

✓ **Land about 1.31 km from $P$; the minimum time is about 2.97 h** (2 h 58.5 min).

## Endpoints: absolute max and min

On a closed interval, the absolute maximum and minimum are among the stationary points **and the endpoints**. Evaluate all of them and compare.

**Worked example 6** **Absolute extrema on [−2, 3]** 3 marks

Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$.

1. $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$. $f(-1) = 2$, $f(1) = -2$.
   USING $f'(x) = 0$
2. Endpoints: $f(-2) = -8 + 6 = -2$, $f(3) = 27 - 9 = 18$.
   USING evaluate the endpoints

✓ **Absolute max 18 (at $x = 3$); absolute min $-2$ (at $x = -2$ and $x = 1$).** The local max of 2 isn't the absolute max.

**Worked example 7** TRICKY **Greatest difference between two models** 3 marks

Two models are $u(x) = 2\sin x + 3$ and $v(x) = 0.5x + 2$ for $0 \le x \le 5$. Find the greatest difference between them.

1. $d(x) = 2\sin x - 0.5x + 1$, $d'(x) = 2\cos x - 0.5 = 0 \Rightarrow \cos x = 0.25$: $x \approx 1.318$, $d \approx 2.277$; $x \approx 4.965$, $d \approx -3.419$.
   USING $d(x) = u(x) - v(x)$, $d'(x) = 0$
2. Endpoints: $d(0) = 1$, $d(5) \approx -3.418$.
   USING check the endpoints

✓ **The greatest difference is about 3.419, at $x \approx 4.965$** ($v$ above $u$ there). The endpoint gives 3.418, so close that skipping either check could cost the mark. CAS: `fMax(abs(u(x)-v(x)),x)|0≤x≤5`.

Calculus

**Trick alert: optimisation traps**

- **Forgetting the endpoints.** Especially when a stationary point is outside the domain.
- **Answering the wrong question.** "Find the maximum area" wants the area, not $x$.
- **Two variables left in the function.** Use the constraint to eliminate one before differentiating.
- **Rounding too early.** Keep the exact $x$ (or full calculator value) until the final answer.

## Summary

- One-variable function + domain → derivative = 0 → check nature → check endpoints.
- Distance problems: minimise the distance squared.
- Give the quantity asked for, with units, to the precision asked for.

### Hard exam questions

**2025 Exam 2 · Section A · Question 19** © VCAA — 14% correct

**The question (not reprinted in this copy):** Value of c when the shortest distance from $y = x + c$ to $y = \log_e(x - 1)$ is $\sqrt{2}$
Full question: 2025 Exam 2, Section A, Question 19 (page 12 of the paper).

**Answer: D** · A 35% · B 25% · C 24% · D 14%

The closest point on the curve is where its gradient equals the line's: $\frac{1}{x-1} = 1$ at (2, 0). The normal there is y = 2 − x, and the points $\sqrt{2}$ units along it are (1, 1) and (3, −1). Putting each on y = x + c gives c = 0 or c = −4, but y = x − 4 crosses the curve (distance 0), so c = 0.

**2023 Exam 2 · Section A · Question 17** © VCAA — 28% correct

**The question (not reprinted in this copy):** Volume of a cylinder cut from a sheet, in terms of x and y
Full question: 2023 Exam 2, Section A, Question 17 (page 8 of the paper).

**Answer: B** · A 6% · B 28% · C 26% · D 26% · E 13%

The circumference $y = 2\pi r$ and $h = x - 4r$, so $V = \pi r^2 h = \frac{\pi x y^2 - 2y^3}{4\pi^2}$.

CALCULUS

# Antidifferentiation

## Standard antiderivatives

Antidifferentiation reverses differentiation: if $F'(x) = f(x)$ then $\int f(x)\,dx = F(x) + c$.

| $f(x)$ | $\int f(x)\,dx$ | $f(x)$ | $\int f(x)\,dx$ |
|---|---|---|---|
| $x^n,\ n \neq -1$ | $\frac{x^{n+1}}{n+1} + c$ | $e^{kx}$ | $\frac{1}{k}e^{kx} + c$ |
| $(ax+b)^n,\ n \neq -1$ | $\frac{(ax+b)^{n+1}}{a(n+1)} + c$ | $\sin(kx)$ | $-\frac{1}{k}\cos(kx) + c$ |
| $\frac{1}{x}$ | $\log_e \lvert x \rvert + c$ | $\cos(kx)$ | $\frac{1}{k}\sin(kx) + c$ |
| $\frac{1}{ax+b}$ | $\frac{1}{a}\log_e \lvert ax+b \rvert + c$ | $\frac{1}{\cos^2(kx)}$ | $\frac{1}{k}\tan(kx) + c$ |

**Key idea**

Linear inside, divide by the coefficient of $x$. That's the reverse chain rule, and it only works when the inside is **linear**. $\int e^{x^2}\,dx$ is *not* $\frac{e^{x^2}}{2x}$.

**Worked example 1** A mixed antiderivative — 2 marks

Find $\int \left(6x^2 - \frac{4}{x^3} + \frac{2}{2x+1} + e^{-3x}\right) dx$.

1. Term by term: $2x^3$; $-4 \cdot \frac{x^{-2}}{-2} = \frac{2}{x^2}$; $\frac{2}{2}\log_e \lvert 2x+1 \rvert = \log_e \lvert 2x+1 \rvert$; $-\frac{1}{3}e^{-3x}$.

   USING $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$ $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\log_e \lvert ax+b \rvert + c$ $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + c$

✓ $2x^3 + \frac{2}{x^2} + \log_e \lvert 2x+1 \rvert - \frac{1}{3}e^{-3x} + c$.

**Worked example 2** Function from derivative and a point — 2 marks

$f'(x) = 3x^2 - 4x + 1$ and $f(1) = 2$. Find $f(x)$.

1. $f(x) = x^3 - 2x^2 + x + c$.

   USING $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$

✓ $f(1) = 1 - 2 + 1 + c = 2 \Rightarrow c = 2$. $f(x) = x^3 - 2x^2 + x + 2$.

USING substitute the point to find $c$

**Worked example 3** Trig with a boundary condition — 3 marks

The gradient of a curve is $\frac{dy}{dx} = 2\cos(2x) + \sin x$, and the curve passes through $\left(\frac{\pi}{2}, 3\right)$. Find $y$.

1. $y = \sin(2x) - \cos x + c$.
   USING $\int \cos(kx)\,dx = \frac{1}{k}\sin(kx) + c$ $\int \sin x\,dx = -\cos x + c$
2. $3 = \sin\pi - \cos\frac{\pi}{2} + c = 0 - 0 + c$.
   USING substitute the point to find $c$

✓ $y = \sin(2x) - \cos x + 3$.

## Antidifferentiation by recognition

This is an Exam 1 staple: "Differentiate $g(x)$. Hence find an antiderivative of $h(x)$." Differentiate, rearrange so the thing you want is on its own, then antidifferentiate both sides.

**How a "hence" antiderivative works** — TRICKY

"Differentiate $x\log_e x$. Hence find $\int \log_e x\,dx$" is one idea in two steps. Differentiating gives $\log_e x + 1$, so $\int (\log_e x + 1)\,dx = x\log_e x$. That contains the thing you want *plus extra*. Split the integral and move the extra over: $\int \log_e x\,dx = x\log_e x - \int 1\,dx = x\log_e x - x + c$.

The plan is always: differentiate what they give you, spot the target integrand inside the result, rearrange. Don't reach for a technique you haven't been given.

**Worked example 4** TRICKY Hence find ∫ log x dx — 3 marks

Find $\frac{d}{dx}(x\log_e x)$. Hence find $\int \log_e x\,dx$.

1. Product rule: $\frac{d}{dx}(x\log_e x) = \log_e x + x\cdot\frac{1}{x} = \log_e x + 1$.
   USING $\frac{d}{dx}(uv) = u'v + uv'$
2. Antidifferentiate both sides: $x\log_e x = \int \log_e x\,dx + \int 1\,dx$.
   USING $\int \frac{d}{dx}F(x)\,dx = F(x) + c$

✓ $\int \log_e x\,dx = x\log_e x - x + c$.

**Worked example 5** TRICKY Hence find ∫ x cos x dx — 3 marks

Differentiate $x\sin x$ and hence find an antiderivative of $x\cos x$.

1. $\frac{d}{dx}(x\sin x) = \sin x + x\cos x$, so $x\cos x = \frac{d}{dx}(x\sin x) - \sin x$.
   USING $\frac{d}{dx}(uv) = u'v + uv'$

✓ $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = \boldsymbol{x\sin x + \cos x}$ (plus $c$ if they ask for the general antiderivative).
USING $\int \frac{d}{dx}F(x)\,dx = F(x)$ $\int \sin x\,dx = -\cos x$

Calculus

**One constant, one condition**

An antiderivative has one constant $c$, so it can usually be made to pass through only one given point. For two points, find $c$ from each: if the values differ, no antiderivative of $f$ passes through both, and the question will change something else (a dilation, or a parameter in $f$) to make it work.

## Graphs of antiderivatives

If you're given the graph of $f$ and asked to sketch $F$ (where $F' = f$): where $f$ is positive, $F$ increases; where $f$ crosses the axis, $F$ has a stationary point; turning points of $f$ are inflection points of $F$. There's a whole family of possible $F$ graphs, each a vertical translation of the others.

**Trick alert: antiderivative traps**

- $\int \frac{1}{x}\,dx \neq \frac{x^0}{0}$. The power rule excludes $n = -1$; use $\log_e |x|$.
- **Sign flip for sine:** $\int \sin x\,dx = -\cos x + c$.
- $\int \frac{1}{3x}\,dx = \frac{1}{3}\log_e |x| + c$, which equals $\frac{1}{3}\log_e |3x| + c'$: both are correct (the constants differ).
- **Don't forget** $+c$ when a question asks for "the general antiderivative" or "$\int \ldots dx$". When it asks for "an antiderivative", $c = 0$ is fine.

## Summary

- Reverse each derivative rule; with a linear inside, divide by its coefficient.
- $\int \frac{1}{x}\,dx = \log_e |x| + c$; $\int \sin x\,dx = -\cos x + c$.
- Use a known point to find $c$.
- "Hence" questions: differentiate, rearrange, antidifferentiate.

CALCULUS

# Definite integrals, areas & average value

## The fundamental theorem

**Fundamental theorem of calculus**

$$\int_a^b f(x)\,dx = F(b) - F(a) \quad \text{where } F' = f \qquad \frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

The definite integral gives the **signed area** between the graph and the $x$-axis: area above counts positive, area below counts negative.

**Properties**

$$\int_a^a f = 0 \qquad \int_b^a f = -\int_a^b f \qquad \int_a^b f + \int_b^c f = \int_a^c f \qquad \int_a^b (kf+g) = k\int_a^b f + \int_a^b g$$

Constants integrate to (constant) × (interval width): $\int_a^b k\,dx = k(b-a)$.

**Worked example 1** Using properties (Exam 1 classic) — 2 marks

Given $\int_1^4 f(x)\,dx = 5$, find $\int_1^4 (2f(x)+1)\,dx$ and $\int_4^1 f(x)\,dx$.

1. $2\int_1^4 f + \int_1^4 1\,dx = 2(5) + (4-1) = 13.$

   USING $\int_a^b (kf+g) = k\int_a^b f + \int_a^b g \quad \int_a^b 1\,dx = b-a$

**13 and $-5$.** The common error is $2(5)+1=11$: the 1 must be integrated over an interval of width 3.

USING $\int_b^a f = -\int_a^b f$

## Signed area vs actual area

**Worked example 2** When part of the region is below the axis — 3 marks

For $f(x) = x^2 - 1$ on $[0, 2]$, find (a) $\int_0^2 f(x)\,dx$ and (b) the total area between the graph and the $x$-axis.

1. (a) $\left[\frac{x^3}{3} - x\right]_0^2 = \frac{8}{3} - 2 = \frac{2}{3}.$

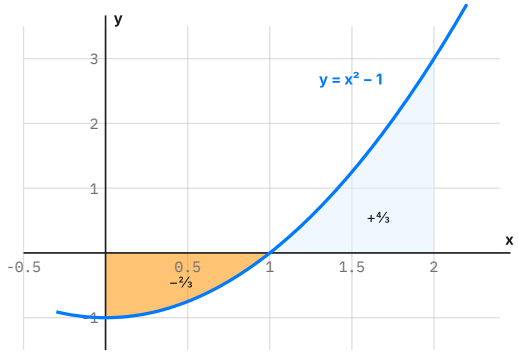
   USING $\int_a^b f(x)\,dx = F(b) - F(a)$

2. (b) The graph crosses at $x = 1$. Area $= -\int_0^1 (x^2-1)\,dx + \int_1^2 (x^2-1)\,dx = \frac{2}{3} + \frac{4}{3}.$

   USING $\text{area} = \left|\int_a^c f\right| + \left|\int_c^b f\right|$

**(a) $\frac{2}{3}$ (signed); (b) 2 square units.** Always sketch first and split at the $x$-intercepts.

Calculus




$y = x^2 - 1$ on $[0, 2]$: the region below the axis has area $\frac{2}{3}$ and the region above has area $\frac{4}{3}$. The integral subtracts them ($\frac{4}{3} - \frac{2}{3} = \frac{2}{3}$); the area adds them (2).

## Area between curves

**Area between curves**

$$A = \int_a^b (\text{upper} - \text{lower})\, dx$$

Find the intersection points first. If the curves swap over, split the integral where they cross. Upper − lower works whether the region is above or below the axis. When a question asks for **one** integral (Exam 2), write $A = \int_a^b |f(x) - g(x)|\, dx$: the modulus handles every crossing at once.

**Worked example 3** Area enclosed by y = x and y = x² — 2 marks

Find the area of the region enclosed by $y = x$ and $y = x^2$.

1. Intersections: $x = x^2 \Rightarrow x = 0, 1$. On $(0, 1)$, $x > x^2$.
   USING solve $f(x) = g(x)$

✓ $A = \int_0^1 (x - x^2)\, dx = \frac{1}{2} - \frac{1}{3} = \mathbf{\frac{1}{6}}$ square units.
USING $A = \int_a^b (\text{upper} - \text{lower})\, dx$

**Worked example 4** Between sin and cos (tech-free) — 3 marks

Find the area enclosed between $y = \sin x$ and $y = \cos x$ from $x = \frac{\pi}{4}$ to $x = \frac{5\pi}{4}$.

1. They meet at $\frac{\pi}{4}$ and $\frac{5\pi}{4}$ (where $\tan x = 1$); in between, $\sin x \geq \cos x$.
   USING $\tan x = 1$ at intersections

2. $\int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\, dx = \left[ -\cos x - \sin x \right]_{\pi/4}^{5\pi/4} = \left( \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - \left( -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right)$.
   USING $A = \int_a^b (\text{upper} - \text{lower})\, dx$ $\int \sin x\, dx = -\cos x$ $\int \cos x\, dx = \sin x$

✓ **$2\sqrt{2}$ square units.**

Calculus

**Why integrals cancel, and what to do about it** TRICKY

$\int f$ adds *signed* area: parts below the axis count as negative. So $\int_a^b f$ can be 0 even when there is plenty of area ($\int_{-\pi}^{\pi} \sin x\,dx = 0$). The same happens to $\int (f-g)$ when the curves cross, because $f-g$ changes sign. For the **actual area**, find where the curves cross (or where $f=0$), integrate each piece with the higher curve first and add; or integrate $|f-g|$ in one go (CAS: `integral(abs(f(x)-g(x)),x,a,b)`). Sketch first: the picture shows where to split and which curve is higher in each piece.

**Worked example 5** A region where the curves swap over — 3 marks

Find the total area enclosed between $y = x^3 - x$ and $y = 3x$.

1. Intersections: $x^3 - x = 3x \Rightarrow x(x^2-4) = 0 \Rightarrow x = -2, 0, 2$.
   USING solve $f(x) = g(x)$

2. On $(0,2)$, $3x$ is on top; on $(-2,0)$, the cubic is on top. By symmetry (both are odd functions), the two regions are equal.
   USING odd functions: $f(-x) = -f(x)$

✓ $A = 2\int_0^2 \left(3x - (x^3 - x)\right)dx = 2\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2(8-4) = \mathbf{8}$. $\int_{-2}^{2}(3x - x^3 + x)\,dx = 0$: don't integrate straight across a crossing. On CAS, one integral does it: `integral(abs(x^3-x-3·x),x,-2,2)` gives 8.

USING $A = \int_a^b (\text{upper} - \text{lower})\,dx$

## Average value

**Average value of $f$ on $[a,b]$**

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

The height of the rectangle on $[a,b]$ with the same (signed) area as the region under $f$.

**Worked example 6** Average value of $e^x$ — 2 marks

Find the average value of $f(x) = e^x$ on $[0,2]$.

1. $\frac{1}{2}\int_0^2 e^x\,dx = \frac{1}{2}\left[e^x\right]_0^2$.
   USING average value $= \frac{1}{b-a}\int_a^b f(x)\,dx$

✓ $\frac{e^2-1}{2} \approx 3.19$. This is not the same as the average of $f(0)$ and $f(2)$ (≈ 4.19).

**Periodic functions: any full period gives the same integral**

If $f(x+p) = f(x)$ for all $x$ (period $p$), then $\int_a^{a+p} f(x)\,dx = \int_0^p f(x)\,dx$ for any $a$, and over $n$ full periods the integral is $n$ times that. So the average value over any whole number of periods is the same: e.g. $\sin^2 x$ (period $\pi$) averages $\frac{1}{2}$ over $[0,\pi]$, $[3,3+\pi]$ or $[0,5\pi]$.

**Average value vs average rate of change**

**Average value** of $f$ on $[a,b]$ is $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$: a height, and it needs an integral. **Average rate of change** of $f$ on $[a,b]$ is $\dfrac{f(b)-f(a)}{b-a}$: a gradient, with no integral. If $f$ is itself a rate (litres per minute, say), its average value is the average rate.

## From rates to totals

If $R(t)$ is a rate (litres/min, cars/hour, m/s), then $\displaystyle\int_a^b R(t)\,dt$ is the **net change** in the quantity from $t=a$ to $t=b$.

**Worked example 7** TRICKY **Total change, then solve for the time (Exam 2 style)** 3 marks

A tank holds 50 L. Water flows in at $R(t)=30e^{-0.1t}$ L/min. (a) How much water is in the tank after 10 minutes? (b) When does it hold 200 L?

1. (a) $V(10)=50+\displaystyle\int_0^{10}30e^{-0.1t}\,dt=50+300\left(1-e^{-1}\right)\approx 239.6$ L.

   USING $V(T)=V(0)+\displaystyle\int_0^T R(t)\,dt$

2. (b) $50+300\left(1-e^{-0.1T}\right)=200\Rightarrow e^{-0.1T}=\dfrac{1}{2}\Rightarrow T=10\log_e 2$. CAS: `solve(50+integral(30·e^(-0.1·t),t,0,T)=200,T)`.

   USING solve for the upper limit

✓ **(a) about 239.6 L; (b)** $T=10\log_e 2\approx 6.93$ **min.** Don't forget the starting 50 L.

**Trick alert: integration traps**

- **"Area" is never negative.** Split at the $x$-intercepts and add magnitudes.
- **Constant terms integrate to (constant) × width**, not the constant itself.
- **Upper − lower**, not "curve − axis", for regions between curves.
- **Average value divides by** $b-a$, not by 2.
- **Exact vs approximate:** if no rounding is asked for, an area like $\dfrac{e^2-1}{2}$ must stay exact.

## Summary

- $\displaystyle\int_a^b f=F(b)-F(a)$ gives signed area; split at the $x$-intercepts for actual area.
- Between curves: $\displaystyle\int(\text{upper}-\text{lower})$, split where they cross.
- Average value: $\dfrac{1}{b-a}\displaystyle\int_a^b f$.
- $\dfrac{d}{dx}\displaystyle\int_a^x f(t)\,dt=f(x)$.

**Also examined in:** 2023 Exam 2 B1 c **(question p. 22; answer p. 23)**; 2023 Exam 2 B5 c.ii and f **(question p. 46; answer p. 47)**; 2025 Exam 2 B4 g.ii **(question p. 68; answer p. 69)**

## Hard exam questions

**2025 Exam 2 · Section A · Question 17** © VCAA — 38% correct

**The question (not reprinted in this copy):** Which graph fits $\int_1^2 f\,dx > \int_1^3 f\,dx$

Full question: 2025 Exam 2, Section A, Question 17 (page 10 of the paper).

**Answer: A** · A 38% · B 20% · C 17% · D 24%

The condition means $\int_2^3 f(x)\,dx < 0$: the net signed area on [2, 3] is negative, so there is more area below the axis than above it there.

**2024 Exam 2 · Section A · Question 20** © VCAA — 41% correct

**The question (not reprinted in this copy):** $\int_2^6 f(x)\,dx$ for a period-2 function with average value k on [0, 2]

Full question: 2024 Exam 2, Section A, Question 20 (page 10 of the paper).

**Answer: C** · A 20% · B 23% · C 41% · D 14%

Each period contributes 2k, and [2, 6] is two periods, so 4k.

CALCULUS

# Numerical integration: trapezium & rectangle rules

## Rectangles and trapeziums

Split $[a, b]$ into $n$ strips of width $h = \dfrac{b-a}{n}$, with $x_0 = a, x_1, \ldots, x_n = b$.

**The three estimates**

$$L = h\big[f(x_0) + f(x_1) + \cdots + f(x_{n-1})\big] \qquad R = h\big[f(x_1) + \cdots + f(x_n)\big]$$

$$T = \frac{h}{2}\big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\big] = \frac{L+R}{2}$$

The trapezium rule is on the formula sheet. $L$ and $R$ aren't, but they're easy: left uses every height except the last, right every height except the first.

**Study design check**

The study design names only the trapezium rule; left and right rectangles are background that make over/under reasoning easier.

**Worked example 1** All three estimates for $x^2$ on [0, 2] — 3 marks

Estimate $\displaystyle\int_0^2 x^2\,dx$ with four strips using (a) left rectangles, (b) right rectangles, (c) the trapezium rule. Compare with the exact value.

1. $h = 0.5$. Heights at $x = 0, 0.5, 1, 1.5, 2$: $0, 0.25, 1, 2.25, 4$.
   USING $h = \dfrac{b-a}{n}$

2. (a) $L = 0.5(0 + 0.25 + 1 + 2.25) = 1.75$. (b) $R = 0.5(0.25 + 1 + 2.25 + 4) = 3.75$.
   USING $L = h\left[f(x_0) + \cdots + f(x_{n-1})\right] \quad R = h\left[f(x_1) + \cdots + f(x_n)\right]$

3. (c) $T = \dfrac{0.5}{2}\big(0 + 2(0.25 + 1 + 2.25) + 4\big) = 0.25 \times 11 = 2.75$.
   USING $T = \dfrac{h}{2}[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]$

✓ $L = 1.75$, $R = 3.75$, $T = 2.75$; **exact** $\dfrac{8}{3} \approx 2.667$. $x^2$ is increasing, so $L$ under- and $R$ over-estimates. It's concave up, so the trapeziums sit above the curve and $T$ is an overestimate.
USING $f'' > 0 \Rightarrow T$ overestimates

## Over or under?

| Function on the interval | Left rectangles | Right rectangles | Trapezium rule |
|---|---|---|---|
| Increasing | under | over | depends on concavity |
| Decreasing | over | under | depends on concavity |
| Concave up ($f'' > 0$) | – | – | over |
| Concave down ($f'' < 0$) | – | – | under |

**Key idea**

Picture the tops: a trapezium's slanted top is a chord. Chords lie *above* a concave-up curve and *below* a concave-down one.

| Function | Concavity | Trapezium rule |
|---|---|---|
| $e^{kx}$, $x^2$, $x^n$ $(n \geq 2, x > 0)$, $\dfrac{1}{x}$ $(x > 0)$ | concave up | overestimates |
| $\log_e x$, $\sqrt{x}$, $\sqrt[3]{x}$ $(x > 0)$ | concave down | underestimates |
| $\sin x$ | down on $(0, \pi)$, up on $(\pi, 2\pi)$ | depends on the interval |
| $\cos x$ | down on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, up on $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ | depends on the interval |

**Worked example 2** TRICKY **Trapezium rule for ln 3** 3 marks

Use the trapezium rule with two strips to estimate $\displaystyle\int_1^3 \frac{1}{x}\,dx$. Is it an over- or under-estimate?

1. $h = 1$; heights $f(1) = 1,\ f(2) = \dfrac{1}{2},\ f(3) = \dfrac{1}{3}$.

   USING $h = \dfrac{b-a}{n}$

2. $T = \dfrac{1}{2}\left(1 + 2 \times \dfrac{1}{2} + \dfrac{1}{3}\right) = \dfrac{1}{2} \times \dfrac{7}{3} = \dfrac{7}{6} \approx 1.167.$

   USING $T = \dfrac{h}{2}[f(x_0) + 2f(x_1) + \cdots + f(x_n)]$

✓ $\dfrac{7}{6}$**, an overestimate**: $\dfrac{1}{x}$ is concave up for $x > 0$ $\left(f''(x) = \dfrac{2}{x^3} > 0\right)$. The exact value is $\log_e 3 \approx 1.099$.

USING $f'' > 0 \Rightarrow$ overestimate

**Worked example 3** **From a table of data** 2 marks

A car's speed (m/s) is recorded every 2 seconds: $t = 0, 2, 4, 6, 8$; $v = 0, 6, 10, 12, 13$. Estimate the distance travelled in 8 s using the trapezium rule.

1. $h = 2$: $T = \dfrac{2}{2}\big(0 + 2(6 + 10 + 12) + 13\big) = 56 + 13.$

   USING $T = \dfrac{h}{2}[f(x_0) + 2f(x_1) + \cdots + f(x_n)]$

✓ **About 69 m.** Distance is the area under a speed–time graph.

USING $\text{distance} = \displaystyle\int v\,dt$

**Trick alert: numerical-integration traps**

- **Count strips, not points.** Five $x$-values make four strips.
- **End heights appear once** in the trapezium rule; the middle ones are doubled.
- **Left rectangles don't use the last height**, and right rectangles don't use the first.
- **Over/under for a function that changes concavity** can't be decided by the table above. Compute both, or compare with the exact value.

## Summary

- $T = \dfrac{h}{2}[\text{ends} + 2(\text{middles})]$; $T = \dfrac{L+R}{2}$.
- Monotonicity decides $L$ and $R$; concavity decides $T$.
- More strips give a better estimate.

## Hard exam questions

**2025 Exam 2 · Section A · Question 6** © VCAA — 50% correct

**The question (not reprinted in this copy):** For which function the trapezium rule overestimates the area on [0, 1]

Full question: 2025 Exam 2, Section A, Question 6 (page 4 of the paper).

**Answer: B** · A 18% · B 50% · C 19% · D 11%

The trapezium rule overestimates when the curve is concave up on the interval. $x^3 + 1$ is concave up on [0, 1].

DATA ANALYSIS, PROBABILITY & STATISTICS

# Probability: conditional & independent events

## The core rules

**Addition, conditional and multiplication rules**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A') = 1 - P(A)$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad P(A \cap B) = P(A \mid B)\, P(B)$$

**Mutually exclusive**

$A$ and $B$ can't both happen: $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$.

**Independent**

One happening doesn't change the other's probability: $P(A \mid B) = P(A)$, or equivalently $P(A \cap B) = P(A)P(B)$.

**Mutually exclusive ≠ independent**

If $A$ and $B$ are mutually exclusive and both have non-zero probability, they're *dependent*: knowing $A$ happened tells you $B$ definitely didn't. $P(A \cap B) = 0 \neq P(A)P(B)$.

**What "given" actually does** TRICKY

$P(A \mid B)$ asks: *among only the cases where $B$ happened*, what fraction also have $A$? The "given" part **becomes the new total**, so it goes on the bottom: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$. With counts it is just "of those who ...": 120 of the 200 who passed were girls, so $P(\text{girl} \mid \text{passed}) = \frac{120}{200}$.

That is why $P(A \mid B)$ and $P(B \mid A)$ differ: the denominators are different groups. "**Given that** ...", "**of those who** ..." and "if it is known that ..." all mean the thing after it goes underneath.

# Karnaugh maps (probability tables)

A 2 × 2 table with totals is the fastest tool when there are two events. Fill in what you know; every row and column must add up.

**Worked example 1** **Karnaugh map and independence** 3 marks

$P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B') = 0.45$. (a) Find $P(A \cap B)$. (b) Are $A$ and $B$ independent? (c) Find $P(B \mid A)$.

1

| | $B$ | $B'$ | total |
|---|---|---|---|
| $A$ | **0.15** | 0.45 | 0.6 |
| $A'$ | 0.15 | 0.25 | 0.4 |
| total | 0.3 | 0.7 | 1 |

USING $P(A) = P(A \cap B) + P(A \cap B')$

2 (b) $P(A)P(B) = 0.18 \neq 0.15 = P(A \cap B)$: not independent.

USING independent $\iff P(A \cap B) = P(A)P(B)$

✓ **(a) 0.15; (b) not independent; (c)** $P(B \mid A) = \dfrac{0.15}{0.6} = 0.25$.

USING $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$

**Worked example 2** **Finding an unknown from independence** 2 marks

$A$ and $B$ are independent, $P(A) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(B)$.

1 $P(A \cup B) = P(A) + P(B) - P(A)P(B)$: $0.7 = 0.4 + p - 0.4p$.

USING $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A \cap B) = P(A)P(B)$

✓ $0.6p = 0.3$, so $P(B) = 0.5$.

# Tree diagrams and the law of total probability

**Law of total probability**

$$P(B) = P(B \mid A)P(A) + P(B \mid A')P(A')$$

Multiply along branches; add the branches that give the outcome you want.

**Trees with letters**

With $P(A) = m$, $P(B \mid A) = p$ and $P(B \mid A') = q$: $P(B) = mp + (1 - m)q$ and $P(A \mid B) = \dfrac{mp}{mp + (1 - m)q}$. Multiple-choice options are often these expressions, so build the tree with letters rather than guessing.

**Worked example 3** TRICKY **The false-positive problem (hard)** 4 marks

2% of a population has a disease. A test gives a positive result for 95% of people with the disease and for 10% of people without it. A randomly chosen person tests positive. Find the probability that they have the disease, correct to four decimal places.

1. Tree: $P(D \cap +) = 0.02 \times 0.95 = 0.019$; $P(D' \cap +) = 0.98 \times 0.10 = 0.098$.
   USING $P(D \cap +) = P(D)P(+ \mid D)$

2. $P(+) = 0.019 + 0.098 = 0.117$.
   USING $P(+) = P(D \cap +) + P(D' \cap +)$

3. $P(D \mid +) = \dfrac{P(D \cap +)}{P(+)} = \dfrac{0.019}{0.117}$.
   USING $P(D \mid +) = \dfrac{P(D \cap +)}{P(+)}$

✓ $\approx 0.1624$. Only about 16% of positives actually have the disease, because the disease is rare and the false positives swamp the true ones. This is the classic "reverse conditional".

**Worked example 4** **Without replacement** 2 marks

A bag holds 5 red and 3 blue marbles. Two are drawn without replacement. Find the probability of getting one of each colour.

1. $P(RB) + P(BR) = \dfrac{5}{8} \times \dfrac{3}{7} + \dfrac{3}{8} \times \dfrac{5}{7}$.
   USING $P(A \cap B) = P(A)P(B \mid A)$ add the separate orders

✓ $= \dfrac{30}{56} = \dfrac{15}{28}$. Don't forget both orders.

**Worked example 5** **Independent trials with different probabilities** 3 marks

Three independent traffic lights are green with probabilities 0.5, 0.4 and 0.2. $Y$ is the number of green lights. Find the probability distribution of $Y$ and $E(Y)$.

1. $P(Y = 0) = 0.5 \times 0.6 \times 0.8 = 0.24$; $P(Y = 3) = 0.5 \times 0.4 \times 0.2 = 0.04$.
   USING $P(A \cap B \cap C) = P(A)P(B)P(C)$

2. $P(Y = 1) = 0.5 \times 0.6 \times 0.8 + 0.5 \times 0.4 \times 0.8 + 0.5 \times 0.6 \times 0.2 = 0.24 + 0.16 + 0.06 = 0.46$; then $P(Y = 2) = 1 - 0.24 - 0.46 - 0.04 = 0.26$.
   USING add the arrangements

✓ $P(Y = 0, 1, 2, 3) = 0.24,\ 0.46,\ 0.26,\ 0.04$; $E(Y) = 0.5 + 0.4 + 0.2 = 1.1$ (expected values add even though $Y$ isn't binomial). CAS check: `expand((0.5+0.5·t)·(0.6+0.4·t)·(0.8+0.2·t))` gives $0.24 + 0.46t + 0.26t^2 + 0.04t^3$: the coefficients are the distribution.

**Trick alert: probability traps**

- $P(A \mid B) \neq P(B \mid A)$ in general, as the disease example shows.
- **"Given" shrinks the sample space.** Divide by the probability of the condition.
- **Independence must be checked with numbers**, not intuition.
- **Without replacement:** the second-draw probabilities change.

## Summary

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$; independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$.
- Karnaugh maps for two events; trees for multi-stage experiments.
- Reverse conditionals: build the tree, find the total, then divide.

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### Hard exam questions

**2024 Exam 2 · Section A · Question 11** © VCAA — 40% correct

**The question (not reprinted in this copy):** Choosing 3 of 12 students: probability exactly two are from the front row of seven
Full question: 2024 Exam 2, Section A, Question 11 (page 5 of the paper).

**Answer: B** · A 12% · B 40% · C 10% · D 37%

$$\frac{\binom{7}{2}\binom{5}{1}}{\binom{12}{3}} = \frac{105}{220} = \frac{21}{44}$$

**2025 Exam 2 · Section A · Question 9** © VCAA — 57% correct

**The question (not reprinted in this copy):** Conditional probability that a student walked, given they took at least 30 minutes
Full question: 2025 Exam 2, Section A, Question 9 (page 6 of the paper).

**Answer: A** · A 57% · B 19% · C 16% · D 7%

$$\frac{0.2m}{0.2m + 0.4n} = \frac{m}{m + 2n}$$

DATA ANALYSIS, PROBABILITY & STATISTICS

# Discrete random variables

## Distributions

A **random variable** assigns a real number to each outcome in the sample space, e.g. the number of heads when three coins are tossed. A discrete random variable $X$ takes separate values $x_1, x_2, \ldots$ with probabilities $p_1, p_2, \ldots$. Every probability satisfies $0 \leq p_i \leq 1$, and together they sum to 1.

**Mean, variance, standard deviation**

$$E(X) = \mu = \sum x\,p(x) \qquad \mathrm{Var}(X) = E\left[(X-\mu)^2\right] = E(X^2) - \mu^2 \qquad \mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}$$

$$E(aX+b) = aE(X) + b \qquad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$$

**Worked example 1** Full workout on a distribution table — 4 marks

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X=x)$ | 0.1 | 0.3 | 0.4 | 0.2 |

Find $E(X)$, $\mathrm{Var}(X)$, $\mathrm{sd}(X)$, and the mean and standard deviation of $Y = 3X - 2$.

1. $E(X) = 0 + 0.3 + 0.8 + 0.6 = 1.7$.
   USING $E(X) = \Sigma x\,p(x)$

2. $E(X^2) = 0 + 0.3 + 1.6 + 1.8 = 3.7$, so $\mathrm{Var}(X) = 3.7 - 1.7^2 = 0.81$ and $\mathrm{sd}(X) = 0.9$.
   USING $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$ $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}$

3. $E(Y) = 3(1.7) - 2 = 3.1$; $\mathrm{Var}(Y) = 9 \times 0.81 = 7.29$, so $\mathrm{sd}(Y) = 2.7$.
   USING $E(aX+b) = aE(X) + b$ $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$

✓ $E(X) = 1.7,\ \mathrm{Var}(X) = 0.81,\ \mathrm{sd}(X) = 0.9$; $E(Y) = 3.1,\ \mathrm{sd}(Y) = 2.7$. The $-2$ shifts the mean but doesn't change the spread.

**Conditional probability from a distribution table**

Add the probabilities of the values in both events, then divide by the probability of the given event. With the table above: $P(X \geq 2 \mid X \geq 1) = \dfrac{0.4 + 0.2}{0.9} = \dfrac{2}{3}$, and $P(X < 3 \mid X > 0) = \dfrac{P(1 \leq X \leq 2)}{P(X \geq 1)} = \dfrac{0.7}{0.9} = \dfrac{7}{9}$. The numerator is the overlap, not the first event on its own.

**Worked example 2** Unknown probabilities — 3 marks

$P(X = x) = kx$ for $x = 1, 2, 3, 4$. Find $k$, $E(X)$ and $P(X > E(X))$.

1. $k(1 + 2 + 3 + 4) = 1 \Rightarrow k = 0.1$. Probabilities: 0.1, 0.2, 0.3, 0.4.
   USING $\Sigma p(x) = 1$

2. $E(X) = 0.1 + 0.4 + 0.9 + 1.6 = 3$.
   USING $E(X) = \Sigma x\,p(x)$

✓ $k = 0.1$, $E(X) = 3$, $P(X > 3) = P(X = 4) = 0.4$. "Greater than" is strict: don't include $X = 3$.

**Worked example 3** **Drawing without replacement** 3 marks

Two balls are drawn without replacement from a bag with 3 red and 2 blue balls. $X$ is the number of red balls drawn. Find the distribution of $X$ and $E(X)$.

1. $P(X=0)=\frac{2}{5}\cdot\frac{1}{4}=\frac{1}{10}$; $P(X=2)=\frac{3}{5}\cdot\frac{2}{4}=\frac{3}{10}$.
   USING $P(A\cap B)=P(A)P(B\mid A)$
2. $P(X=1)=1-\frac{1}{10}-\frac{3}{10}=\frac{6}{10}$ (or $2\times\frac{3}{5}\times\frac{2}{4}$).
   USING $\Sigma p(x)=1$

✓ $E(X)=0.6+0.6=1.2$ (which is $2\times\frac{3}{5}$, as intuition says).
USING $E(X)=\Sigma x\,p(x)$

**Worked example 4** **Is the game fair?** 3 marks

It costs \$5 to play a game: roll a fair die; a 6 pays \$20, a 5 pays \$5, anything else pays nothing. Find the expected profit per game.

1. Let $W$ = amount won. $E(W)=20\cdot\frac{1}{6}+5\cdot\frac{1}{6}+0\cdot\frac{4}{6}=\frac{25}{6}\approx 4.17$.
   USING $E(X)=\Sigma x\,p(x)$
2. Profit $=W-5$, so $E(\text{profit})=\frac{25}{6}-5=-\frac{5}{6}$.
   USING $E(X-b)=E(X)-b$

✓ **About −\$0.83 per game**, so the game is unfair to the player. A fair game has expected profit 0.

**Trick alert: random variable traps**

- $\text{Var}(aX+b)=a^2\,\text{Var}(X)$. A negative $a$ never makes the variance negative: $\text{Var}(-2X)=4\,\text{Var}(X)$, not $-2\,\text{Var}(X)$.
- $E(X^2)\neq[E(X)]^2$. Their difference *is* the variance.
- **sd scales by** $|a|$, variance by $a^2$.
- **The expected value needn't be a possible value** (1.7 children...).
- **Symmetric distribution?** The mean is the centre of symmetry: no calculation needed.

## Summary

- Probabilities sum to 1; $E(X)=\sum xp$; $\text{Var}=E(X^2)-\mu^2$.
- $E(aX+b)=aE(X)+b$; $\text{Var}(aX+b)=a^2\,\text{Var}(X)$.
- Fair game: expected profit 0.

### Hard exam questions

**2023 Exam 2 · Section A · Question 12** © VCAA 29% correct

**The question (not reprinted in this copy):** Maximum possible mean of a pmf with parameter k
Full question: 2023 Exam 2, Section A, Question 12 (page 6 of the paper).

**Answer: E** · A 11% · B 14% · C 26% · D 18% · E 29%

All probabilities must lie in [0, 1], so $0\le k\le\sqrt{5}-2$. The mean $-3k^2-7k+2$ is largest at k = 0, giving 2.

DATA ANALYSIS, PROBABILITY & STATISTICS

# The binomial distribution

## Bernoulli trials

A **Bernoulli trial** has exactly two outcomes: success, scored $X = 1$, with probability $p$, and failure, scored $X = 0$, with probability $1 - p$.

**Bernoulli random variable**

$$P(X = 1) = p \qquad P(X = 0) = 1 - p \qquad E(X) = p \qquad \mathrm{Var}(X) = p(1 - p)$$

A binomial variable counts the successes in $n$ independent Bernoulli trials with the same $p$, which is why its mean and variance are $n$ times these: $np$ and $np(1 - p)$.

## When is it binomial?

$X \sim \mathrm{Bi}(n, p)$ counts the number of successes when:

1. there's a fixed number $n$ of trials;
2. each trial has two outcomes (success/failure);
3. the probability of success $p$ is the same every trial;
4. the trials are independent.

**Binomial distribution**

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \ldots, n \qquad E(X) = np \qquad \mathrm{Var}(X) = np(1 - p)$$

$\binom{n}{x} = \dfrac{n!}{x!(n - x)!}$ counts the arrangements of $x$ successes among $n$ trials. For example $\binom{5}{2} = 10$.

## How $n$ and $p$ change the graph

- $p = 0.5$: symmetric about the mean $np$.
- $p < 0.5$: positively skewed (tail to the right). $p > 0.5$: negatively skewed (tail to the left).
- Larger $n$: the peak moves right (mean $np$), the spread grows (sd $\sqrt{np(1 - p)}$) and the shape becomes more symmetric.
- The most likely value (the mode) is close to the mean $np$.

### Worked example 1 Exactly, at least, at most — 4 marks

30% of customers buy a coffee. Ten customers are chosen at random. Find the probability that (a) exactly 3 buy coffee, (b) at least 3 buy coffee, (c) given at least one buys coffee, at least 3 do.

1. (a) $\binom{10}{3}(0.3)^3(0.7)^7 = 120 \times 0.027 \times 0.0823543 \approx 0.2668$.
   USING $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$

2. (b) $P(X \geq 3) = 1 - P(X \leq 2) \approx 1 - 0.3828 = 0.6172$ (CAS: binomCdf).
   USING $P(X \geq a) = 1 - P(X \leq a - 1)$

3. (c) $P(X \geq 3 \mid X \geq 1) = \dfrac{P(X \geq 3)}{P(X \geq 1)} = \dfrac{0.61722}{1 - 0.7^{10}} = \dfrac{0.61722}{0.97175} \approx 0.6352$ (keep unrounded values until the end).
   USING $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ $P(X \geq 1) = 1 - (1-p)^n$

✓ **(a) 0.2668, (b) 0.6172, (c) 0.6352.** In (c), "at least 3" is already inside "at least 1", so the intersection is just $P(X \geq 3)$.

### Worked example 2 TRICKY Find the smallest n (Exam 2 staple) — 3 marks

A machine produces faulty items with probability 0.1. How many items must be tested so that the probability of finding at least one faulty item exceeds 0.95?

1. $P(X \geq 1) = 1 - 0.9^n > 0.95 \Rightarrow 0.9^n < 0.05$.
   USING $P(X \geq 1) = 1 - (1-p)^n$

2. $n > \dfrac{\log_e 0.05}{\log_e 0.9} \approx 28.43$. (The inequality flips because $\log_e 0.9 < 0$.)
   USING $a^n < b \Rightarrow n > \dfrac{\log_e b}{\log_e a}$ $(0 < a < 1)$

✓ $n = 29$. Always round *up* for "smallest $n$". Check: $1 - 0.9^{28} \approx 0.948$ (too small), $1 - 0.9^{29} \approx 0.953$ ✓.

### Worked example 3 TRICKY An integer from a rounded probability — 2 marks

$X \sim \text{Bi}(n, 0.1)$. For which values of $n$ is $P(X \geq 1) = 0.8$, correct to one decimal place?

1. $P(X \geq 1) = 1 - 0.9^n$. CAS: `seq(1-0.9^n,n,13,19)` gives 0.7458, 0.7712, 0.7941, 0.8147, 0.8332, 0.8499, 0.8649.
   USING rounded value $0.8 \iff [0.75, 0.85)$

✓ $n = 14, 15, 16, 17, 18$. $n = 18$ counts: 0.8499 rounds to 0.8. Give every value unless the question asks for the smallest or largest.

**Worked example 4** **Find n and p from the mean and variance** 2 marks

$X \sim \text{Bi}(n, p)$ with $E(X) = 6$ and $\text{Var}(X) = 4.2$. Find $n$ and $p$.

1. Divide: $\dfrac{np(1-p)}{np} = \dfrac{4.2}{6} \Rightarrow 1 - p = 0.7$.
   USING $E(X) = np$ $\text{Var}(X) = np(1-p)$

✓ $p = 0.3$, $n = \dfrac{6}{0.3} = 20$.
USING $n = \dfrac{E(X)}{p}$

**Finding p from a probability**

Write the probability out with $p$ and solve. E.g. $X \sim \text{Bi}(10, p)$ with $P(X = 0) = 0.2$: $(1-p)^{10} = 0.2$, so $p = 1 - 0.2^{0.1} \approx 0.1487$. CAS: `solve((1-p)^10=0.2,p)|0<p<1`. If CAS gives more than one value, keep those in $(0, 1)$.

**Turning the words into an inequality for X** TRICKY

**At least** $k$ is $X \geq k$. **More than** $k$ is $X \geq k + 1$. **At most** $k$ and **no more than** $k$ are $X \leq k$. **Fewer than** $k$ is $X \leq k - 1$. **Between** $a$ **and** $b$ **inclusive** is $a \leq X \leq b$. Because $X$ is a whole number, $P(X < 4) = P(X \leq 3)$: get the whole-number version *before* you type anything.

Then use the complement when it is shorter: "at least one" is $1 - P(X = 0)$, and "at least 3 out of 10" is $1 - P(X \leq 2)$. In CAS, `binomCdf(n,p,a,b)` is $P(a \leq X \leq b)$ and `binomCdf(n,p,b)` is $P(X \leq b)$.

**Trick alert: binomial traps**

- **"At least 1" =** $1 - P(X = 0)$. Never add up 1, 2, 3, ... by hand.
- **Discrete inequalities:** $P(X < 4) = P(X \leq 3)$, and $P(X > 4) = 1 - P(X \leq 4)$. Off-by-one errors are the most common binomial mistake.
- **Without replacement from a small population isn't binomial** (the probability changes each trial), and neither are independent trials with different probabilities: multiply along each arrangement instead (see Probability → p. 87).
- **Conditional binomial:** the numerator is the intersection. $P(X = 2 \mid X \geq 1) = \dfrac{P(X = 2)}{P(X \geq 1)}$ because $X = 2$ is inside $X \geq 1$.

## Summary

- Fixed $n$, two outcomes, constant $p$, independent: $\text{Bi}(n, p)$.
- $E = np$, $\text{Var} = np(1-p)$.
- "At least one": $1 - (1-p)^n$. For the smallest $n$, use logs and round up.
- Check every inequality sign: $<$ vs $\leq$ matters for discrete variables.

## Hard exam questions

**2023 Exam 2 · Section A · Question 8** © VCAA — 49% correct

**The question (not reprinted in this copy):** Probability of at least one green in 8 draws with replacement
Full question: 2023 Exam 2, Section A, Question 8 (page 4 of the paper).

**Answer: C** · A 12% · B 14% · C 49% · D 16% · E 9%

$1 - \Pr(\text{no green}) = 1 - \left(\frac{m}{n+m}\right)^8$.

DATA ANALYSIS, PROBABILITY & STATISTICS

# Continuous random variables (pdfs)

## Probability density functions

A continuous random variable takes any value in an interval. Its **probability density function** $f$ satisfies:

**Conditions for a pdf**

$$f(x) \geq 0 \text{ for all } x \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(a < X < b) = \int_a^b f(x)\,dx$$

$P(X = a) = 0$ for any single value, so $<$ and $\leq$ give the same probability. (For discrete variables they don't!)

**Summary statistics**

$$E(X) = \int x f(x)\,dx \qquad \mathrm{Var}(X) = \int x^2 f(x)\,dx - \mu^2 \qquad \int_{-\infty}^{m} f(x)\,dx = \frac{1}{2}\ (\text{median } m)$$

The $p$th percentile $q$ satisfies $\int_{-\infty}^{q} f(x)\,dx = \frac{p}{100}$. $E(aX+b) = aE(X)+b$ and $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$, exactly as for discrete variables. If no rounding is asked for, give exact values: for $f(x) = \frac{x^2}{9}$ on $[0,3]$, $\mathrm{Var}(X) = \frac{27}{80}$ and $\mathrm{sd}(X) = \frac{3\sqrt{15}}{20}$, not 0.581.

**The questions about a pdf are different integrals** TRICKY

**Probability** is an area: $\int_a^b f(x)\,dx$. **Mean** multiplies by $x$ first: $E(X) = \int x f(x)\,dx$. **Variance** needs $E(X^2) = \int x^2 f(x)\,dx$ (multiply by $x^2$, not square the whole product) and then subtract $\mu^2$. **Median** $m$ solves $\int_{-\infty}^{m} f(x)\,dx = \frac{1}{2}$: the $m$ that splits the *area* in half, which is not halfway along the domain and not where $f = \frac{1}{2}$.

For a piecewise pdf, add the area piece by piece. If the first piece has area less than $\frac{1}{2}$, the median is in the next piece: start that integral at the join and set the running total equal to $\frac{1}{2}$.

**Worked example 1** **Everything about f(x) = kx² on [0, 3]** 6 marks

$f(x) = kx^2$ for $0 \le x \le 3$, and 0 elsewhere. Find (a) $k$, (b) $P(X < 1)$, (c) $E(X)$, (d) the median, (e) $\mathrm{Var}(X)$.

1. (a) $\displaystyle\int_0^3 kx^2\,dx = 9k = 1 \Rightarrow k = \frac{1}{9}$.

   USING $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$

2. (b) $\displaystyle\int_0^1 \frac{x^2}{9}dx = \frac{1}{27}$.

   USING $P(a < X < b) = \displaystyle\int_a^b f(x)\,dx$

3. (c) $E(X) = \displaystyle\int_0^3 \frac{x^3}{9}dx = \frac{81}{36} = \frac{9}{4}$.

   USING $E(X) = \displaystyle\int x f(x)\,dx$

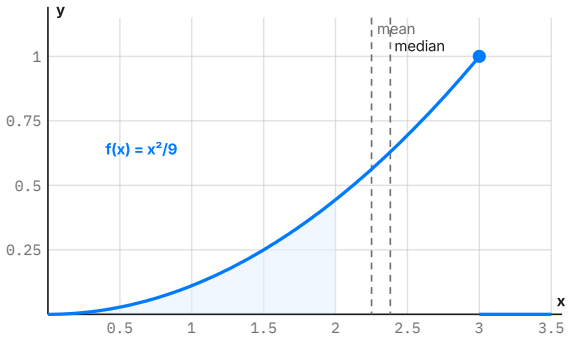
4. (d) $\displaystyle\int_0^m \frac{x^2}{9}dx = \frac{m^3}{27} = \frac{1}{2} \Rightarrow m = \frac{3}{\sqrt[3]{2}} \approx 2.381$.

   USING $\displaystyle\int_{-\infty}^{m} f(x)\,dx = \frac{1}{2}$

5. (e) $E(X^2) = \displaystyle\int_0^3 \frac{x^4}{9}dx = \frac{243}{45} = 5.4$; $\mathrm{Var} = 5.4 - \left(\frac{9}{4}\right)^2 = 0.3375$.

   USING $E(X^2) = \displaystyle\int x^2 f(x)\,dx$ $\quad \mathrm{Var}(X) = E(X^2) - [E(X)]^2$

✓ $k = \frac{1}{9}$; $P(X < 1) = \frac{1}{27}$; $E(X) = 2.25$; **median** $\approx 2.381$; $\mathrm{Var} = \frac{27}{80} = 0.3375$. The median is bigger than the mean here because the distribution is skewed left (bunched at high values).




$f(x) = \frac{x^2}{9}$ on $[0, 3]$. The shaded area is $P(X < 2) = \frac{8}{27}$ (the worked example finds $P(X < 1) = \frac{1}{27}$); the dashed lines mark the mean (2.25) and the median (≈ 2.38).

Probability

**Worked example 2** **A piecewise (triangular) pdf** 4 marks

$f(x) = \begin{cases} x, & 0 \le x \le 1 \\ 2 - x, & 1 < x \le 2 \\ 0, & \text{elsewhere} \end{cases}$. Find $P(X < 0.5)$, $E(X)$ and $P(X > 1.5 \mid X > 1)$.

1 $P(X < 0.5) = \int_0^{0.5} x\,dx = 0.125.$

USING $P(a < X < b) = \int_a^b f(x)\,dx$

2 The graph is symmetric about $x = 1$, so $E(X) = 1$ (and the median is 1).

USING $E(X) =$ line of symmetry

3 $P(X > 1.5) = \int_{1.5}^{2} (2 - x)\,dx = 0.125$ and $P(X > 1) = 0.5.$

USING $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

✓ **0.125; 1;** $\dfrac{0.125}{0.5} = 0.25$**.** For a piecewise pdf, integrate each piece over its own interval.

**Worked example 3** **Exponential-type pdf (CAS)** 3 marks

The time $T$ (minutes) between customer arrivals has pdf $f(t) = 0.5e^{-0.5t}$, $t \ge 0$. Find $P(T > 2)$ and the median time.

1 $P(T > 2) = \int_2^{\infty} 0.5e^{-0.5t}dt = \left[-e^{-0.5t}\right]_2^{\infty} = e^{-1}.$

USING $P(T > a) = \int_a^{\infty} f(t)\,dt$ $\quad e^{-kt} \to 0$ as $t \to \infty$

2 Median: $\int_0^m 0.5e^{-0.5t}dt = 1 - e^{-0.5m} = \dfrac{1}{2} \Rightarrow m = 2\log_e 2.$

USING $\int_0^m f(t)\,dt = \dfrac{1}{2}$

✓ $P(T > 2) = e^{-1} \approx 0.368$**; median** $= 2\log_e 2 \approx 1.386$ **min** (less than the mean of 2 min: the long right tail pulls the mean up).

**Worked example 4** TRICKY **Median of a piecewise pdf, and the mode** 3 marks

$X$ has pdf $f(x) = \dfrac{2x}{3}$ for $0 \le x \le 1$, $f(x) = \dfrac{3 - x}{3}$ for $1 < x \le 3$, and 0 elsewhere. Find the median and the mode.

1 $\int_0^1 \dfrac{2x}{3}\,dx = \dfrac{1}{3} < \dfrac{1}{2}$, so the median is in the second piece.

USING area of the first piece

2 $\dfrac{1}{3} + \int_1^m \dfrac{3 - x}{3}\,dx = \dfrac{1}{2} \Rightarrow 3m - \dfrac{m^2}{2} - \dfrac{5}{2} = \dfrac{1}{2} \Rightarrow m^2 - 6m + 6 = 0 \Rightarrow m = 3 \pm \sqrt{3}.$

USING $\dfrac{1}{3} + \int_1^m f(x)\,dx = \dfrac{1}{2}$

✓ **Median** $3 - \sqrt{3} \approx 1.27$ (the other root is outside $[1, 3]$); **mode 1**, where $f$ peaks at $\dfrac{2}{3}$.

**Transforming a pdf**

If $f$ is a pdf, then $g(x) = k\,f\left(\frac{x}{n}\right)$ (with $n > 0$) is a pdf only if $kn = 1$, because the dilation by $n$ from the $y$-axis multiplies the area by $n$. Then $g$ is the pdf of $nX$: its mean is $n$ times the mean of $f$, and its standard deviation is $n$ times the standard deviation of $f$.

**Trick alert: pdf traps**

- $f(x)$ **isn't a probability.** A pdf value can be bigger than 1; only areas are probabilities.
- **The median is where the area is** $\frac{1}{2}$, not where $f$ is $\frac{1}{2}$, and not halfway along the domain.
- **For a pdf, "find** $k$**" needs both conditions:** total area 1 *and* $f \geq 0$ (sometimes this rules out a solution).
- **Conditional:** $P(X > a \mid X > b) = \dfrac{P(X > a)}{P(X > b)}$ only when $a \geq b$ (so the first event is inside the second).

## Summary

- pdf: $f \geq 0$, total area 1; probabilities are areas.
- Mean $\int xf$, variance $\int x^2 f - \mu^2$, median: area to the left $= \frac{1}{2}$.
- Piecewise: integrate piece by piece. Conditional: divide areas.

**Also examined in:** 2023 Exam 2 B4 j (transforming a pdf) **(question p. 102; answer p. 103)**

### Hard exam questions

**2024 Exam 2 · Section A · Question 14** © VCAA — 53% correct

**The question (not reprinted in this copy):** $\Pr(X < 0.5)$ for a piecewise-linear pdf with unknown k
Full question: 2024 Exam 2, Section A, Question 14 (page 7 of the paper).

**Answer: B** · A 9% · B 53% · C 30% · D 7%

The total area $4k - 1 = 1$ gives $k = \frac{1}{2}$. Then $\Pr(X < 0.5) = 1 - \frac{1}{16} = \frac{15}{16}$.

**2024 Exam 2 · Section B · Question 4** © VCAA — state average 48% of marks

**The question (not reprinted in this copy):** Luggage masses (a cubic pdf), a discrete count of heavy bags per traveller, sample proportions and confidence intervals
Full question: 2024 Exam 2, Section B, Question 4 (page 21 of the paper).

**a.** Integral for Pr(heavy), heavy meaning > 23 kg (1 mark · state average 0.8 / 1, 80%)

- $\int_{23}^{30} \frac{1}{67500} x^2(30 - x)\,dx$

**Notes:** $\Pr(X > 23) = \int_{23}^{30} f(x)\,dx \approx 0.234$

**b.i.** Mean of X (1 mark · state average 0.9 / 1, 90%)

- 18 kg

**b.ii.** Standard deviation of X (2 marks · state average 1.5 / 2, 75%)

- Uses $E(X^2) - [E(X)]^2$ (var = 36)
- $\text{sd} = 6$ kg

**b.iii.** Pr(heavy | more than the mean) (3 d.p.) (2 marks · state average 1.2 / 2, 60%)

- $\dfrac{\Pr(X > 23)}{\Pr(X > 18)}$
- $\approx 0.446$

**c.i.** Show Pr(W = 2) = 0.027 (1 mark · state average 0.4 / 1, 40%)

- $0.5 \times 0.234^2 \approx 0.027$

**c.ii.** Complete the distribution of W (3 d.p.) (2 marks · state average 0.3 / 2, 15%)

- Pr(W = 0) = 0.1 + 0.4(0.766) + 0.5(0.766)² ≈ 0.700
- Pr(W = 1) = 0.4(0.234) + 0.5 × 2(0.234)(0.766) ≈ 0.273

**Notes:** Condition on the number of bags checked.

**d.i.** $\Pr(\hat{P} > 0.2)$ for n = 35 (3 d.p.) (2 marks · state average 0.9 / 2, 45%)

- Equivalent to $\Pr(X \geq 8)$, $X \sim \text{Bi}(35, 0.234)$
- $\approx 0.595$

**Notes:** $\hat{P} > 0.2 \Leftrightarrow X > 7$, i.e. $X \geq 8$: ≈ 0.595.

**d.ii.** Pr($\hat{P}$ within one sd of its mean), exact binomial (3 d.p.) (2 marks · state average 0.5 / 2, 25%)

- $\text{sd}(\hat{P}) = \sqrt{\dfrac{0.234 \times 0.766}{35}}$ gives X between 5.69 and 10.69, so X = 6, ..., 10
- $\approx 0.684$

**e.i.** Approximate 90% CI from 10 heavy out of 50 (3 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $(0.107,\ 0.293)$

**Notes:** $0.2 \pm 1.645\sqrt{\dfrac{0.2 \times 0.8}{50}}$

**e.ii.** Min and max number of heavy bags in a second sample of 50 giving a wider CI (1 mark · state average 0.1 / 1, 10%)

- 11 and 39

**Notes:** A wider CI needs a larger $\hat{p}(1 - \hat{p})$, so $\hat{p}$ closer to 0.5 than 0.2 is: 11 to 39. Only 10% got this.

## 2023 Exam 2 · Section A · Question 10 © VCAA

60% correct

**The question (not reprinted in this copy):** k with Pr(X < k) = 0.35 for a piecewise-linear pdf
Full question: 2023 Exam 2, Section A, Question 10 (page 5 of the paper).

**Answer: B** · A 8% · B 60% · C 14% · D 13% · E 5%

$\dfrac{(k-1)^2}{40} = 0.35 \Rightarrow k = 1 + \sqrt{14} \approx 4.7$, which lies in the first piece.

DATA ANALYSIS, PROBABILITY & STATISTICS

# The normal distribution

## The normal curve

$X \sim \mathrm{N}(\mu, \sigma^2)$ has a symmetric bell-shaped pdf centred at $\mu$ (mean = median = mode), with spread set by the standard deviation $\sigma$. Note that the second parameter is the **variance**: $\mathrm{N}(170, 64)$ means $\sigma = 8$.

**Standardising**

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1)$$

A $z$-score counts how many standard deviations a value is from the mean.

| Interval | $\mu \pm \sigma$ | $\mu \pm 2\sigma$ | $\mu \pm 3\sigma$ |
|---|---|---|---|
| Approx. probability | 68% | 95% | 99.7% |

## How $\mu$ and $\sigma$ change the graph

Changing $\mu$ translates the curve left or right without changing its shape. A larger $\sigma$ gives a lower, wider curve and a smaller $\sigma$ a taller, narrower one: the total area is always 1. The peak height is $\dfrac{1}{\sigma\sqrt{2\pi}}$, and the points of inflection are at $x = \mu \pm \sigma$. The standard normal $Z \sim \mathrm{N}(0, 1)$ has $\mu = 0$ and $\sigma = 1$.

**Worked example 1** **Probability and percentile** 3 marks

Heights of adult men are $X \sim \mathrm{N}(170, 8^2)$ cm. Find (a) $P(X > 180)$, (b) the height exceeded by only 10% of men.

1. (a) $z = \dfrac{180 - 170}{8} = 1.25$; $P(Z > 1.25) \approx 0.1056$.
   USING $Z = \dfrac{X - \mu}{\sigma}$
2. (b) Need $P(X > h) = 0.1$, i.e. $P(X < h) = 0.9$. Inverse normal: $z = 1.2816$, so $h = 170 + 1.2816 \times 8$.
   USING $P(X > h) = 0.1 \iff P(X < h) = 0.9 \quad x = \mu + z\sigma$

✓ **(a) 0.1056, (b) $h \approx 180.25$ cm.** CAS inverse normal uses the area to the *left* by default: convert "top 10%" into 0.9.

**Worked example 2** **Using the 68–95–99.7 rule (tech-free)** 2 marks

$X \sim \mathrm{N}(50, 16)$. Without technology, estimate $P(42 < X < 54)$.

1. $\sigma = 4$. $42 = \mu - 2\sigma$, $54 = \mu + \sigma$. By symmetry, $P(42 < X < 50) \approx \dfrac{95\%}{2} = 47.5\%$ and $P(50 < X < 54) \approx \dfrac{68\%}{2} = 34\%$.
   USING 68%, 95%, 99.7% within $1, 2, 3\,\sigma$ $\sigma = \sqrt{\mathrm{Var}(X)}$

✓ $\approx 0.815$ (81.5%). Draw the bell and split at the mean.

Probability

**Worked example 3** **Finding σ** 2 marks

The mass of a packet is normally distributed with mean 50 g, and 10% of packets weigh less than 40 g. Find the standard deviation.

1. $P(X < 40) = 0.1 \Rightarrow \dfrac{40 - 50}{\sigma} = -1.2816.$

   USING $Z = \dfrac{X - \mu}{\sigma}$ $P(Z < -1.2816) = 0.1$

✓ $\sigma = \dfrac{10}{1.2816} \approx 7.80$ **g.**

**Unknown μ and σ — standardise, then use the sign to check** TRICKY

Each given probability becomes one equation. For $P(X < a) = p$: find $z = \text{invNorm}(p)$, then $\dfrac{a - \mu}{\sigma} = z$. Two conditions give two equations that are linear in $\mu$ and $\sigma$: $a - \mu = z_1\sigma$ and $b - \mu = z_2\sigma$. Subtract to get $\sigma = \dfrac{b - a}{z_2 - z_1}$, then back-substitute for $\mu$.

Sign check before you trust the algebra: a probability **below 0.5 means** $z < 0$, so $a$ must be *less than* $\mu$; above 0.5 means $z > 0$ and $a > \mu$. If your $\mu$ lands on the wrong side of $a$, a sign slipped. And $\sigma$ must come out positive.

**Worked example 4** TRICKY **Finding μ and σ together (hard)** 4 marks

For a normal variable, $P(X < 30) = 0.2$ and $P(X > 60) = 0.1$. Find $\mu$ and $\sigma$.

1. $\dfrac{30 - \mu}{\sigma} = z_{0.2} = -0.8416$ and $\dfrac{60 - \mu}{\sigma} = z_{0.9} = 1.2816.$

   USING $Z = \dfrac{X - \mu}{\sigma}$

2. So $30 = \mu - 0.8416\sigma$ and $60 = \mu + 1.2816\sigma$. Subtract: $30 = 2.1232\sigma$.

   USING $x = \mu + z\sigma$ subtract to eliminate $\mu$

✓ $\sigma \approx 14.13$, $\mu \approx 30 + 0.8416 \times 14.13 \approx 41.89$.

**Worked example 5** TRICKY **Conditional normal, then binomial (Exam 2 chain)** 4 marks

Using $X \sim \text{N}(170, 8^2)$: (a) find $P(X > 180 \mid X > 170)$; (b) ten men are chosen at random. Find the probability that at least two are taller than 180 cm.

1. (a) $\dfrac{P(X > 180)}{P(X > 170)} = \dfrac{0.1056}{0.5} \approx 0.2113.$

   USING $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

2. (b) $Y \sim \text{Bi}(10, 0.1056)$: $P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1)$.

   USING $P(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}$ $P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1)$

✓ **(a) 0.2113; (b)** $\approx 0.2858$. Use the unrounded probability from the normal part in the binomial part.

**Trick alert: normal traps**

- $\mathrm{N}(\mu, \sigma^2)$: **the second number is the variance.** $\mathrm{N}(10, 9)$ has $\sigma = 3$.
- **Inverse normal = area to the left.** "Top 5%" means use 0.95.
- **Symmetry shortcuts:** $P(Z < -a) = P(Z > a)$, and $P(X > \mu) = 0.5$.
- **The 68–95–99.7 rule is approximate**, so use it only when told to (or tech-free).

## Summary

- $Z = \dfrac{X - \mu}{\sigma}$; remember $\mathrm{N}(\mu, \sigma^2)$ uses the variance.
- 68–95–99.7% for quick, tech-free estimates.
- Unknown $\mu$ or $\sigma$: turn each probability into a $z$-equation; two unknowns need two equations.
- Conditional: divide probabilities. Then feed an unrounded $p$ into a binomial if needed.

### Hard exam questions

**2024 Exam 2 · Section A · Question 19** © VCAA — 57% correct

**The question (not reprinted in this copy):** $\Pr(X < 12)$ given $\Pr(X < 10) = \Pr(X > 18) = 0.2$
Full question: 2024 Exam 2, Section A, Question 19 (page 10 of the paper).

**Answer: C** · A 10% · B 19% · C 57% · D 13%

By symmetry μ = 14, and $\sigma = \dfrac{4}{0.8416} = 4.75$. Then $\Pr(X < 12) \approx 0.337$.

**2023 Exam 2 · Section B · Question 4** © VCAA — state average 49% of marks

**The question (not reprinted in this copy):** Tennis-ball diameters (normal), fitting through a container, grading, a confidence level, and serving-speed pdfs
Full question: 2023 Exam 2, Section B, Question 4 (page 18 of the paper).

**a.** Pr(D > 6.8) (4 d.p.) (1 mark · state average 0.8 / 1, 80%)

- $\approx 0.1587$

**b.** Diameter exceeded by only 10% of balls (2 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $\approx 6.83$ cm

**Notes:** 90th percentile ≈ 6.83 cm

**c.** Pr(ball fits: D < 6.95) (4 d.p.) (1 mark · state average 0.8 / 1, 80%)

- $\approx 0.9938$

**d.** Pr(at least 3 of 4 fit) (4 d.p.) (2 marks · state average 1.3 / 2, 65%)

- Binomial with n = 4, p = Pr(D < 6.95), unrounded (≈ 0.9938)
- $\approx 0.9998$

**e.** Pr(grade A | fits) (4 d.p.) (2 marks · state average 1.2 / 2, 60%)

- $\dfrac{\Pr(6.54 < D < 6.86)}{\Pr(D < 6.95)}$
- $\approx 0.8960$

**f.** σ so that more than 99% are grade A (2 d.p.) (2 marks · state average 0.7 / 2, 35%)

- $\Pr\left(-\frac{0.16}{\sigma} < Z < \frac{0.16}{\sigma}\right) > 0.99 \Rightarrow \frac{0.16}{\sigma} > 2.5758$
- $\sigma < 0.0621$, so $\sigma \approx 0.06$ cm (2 d.p.)

**Notes:** Symmetric interval, so 0.5% in each tail.

**g.** Confidence level of the interval (0.7382, 0.9493) with n = 32 (2 marks · state average 0.5 / 2, 25%)

- $\hat{p} = 0.84375$, margin 0.10555, so $z \approx 1.645$
- 90%

**Notes:** 90%. Many found z correctly but then said 95%.

**h.** Pr(grade A serving speed > 50) (4 d.p.) (1 mark · state average 0.6 / 1, 60%)

- $\approx 0.1345$

**Notes:** $\int_{50}^{3\pi^2+30} f(v)\,dv \approx 0.1345$

**i.** Exact mean serving speed of grade A balls (1 mark · state average 0.6 / 1, 60%)

- $3\pi^2 + 12$

**j.** a and b in $g(w) = af\left(\frac{w}{b}\right)$ if the mean is $2\pi^2 + 8$ (2 marks · state average 0.2 / 2, 10%)

- Area stays 1, so $ab = 1$; the mean scales by b: $b(3\pi^2 + 12) = 2\pi^2 + 8$
- $b = \frac{2}{3}$, $a = \frac{3}{2}$

## 2025 Exam 2 · Section A · Question 12 © VCAA

58% correct

**The question (not reprinted in this copy):** Finding μ and σ from two normal probabilities
Full question: 2025 Exam 2, Section A, Question 12 (page 7 of the paper).

**Answer: D** · A 10% · B 16% · C 15% · D 58%

$\Pr(X > 200) = 0.325 \Rightarrow z = 0.454$, and $\Pr(X < 180) = 0.086 \Rightarrow z = -1.366$. Solve simultaneously: $\mu \approx 195$, $\sigma \approx 11$.

DATA ANALYSIS, PROBABILITY & STATISTICS

# Sampling & sample proportions

## Populations, samples and bias

A **parameter** describes a population (e.g. the true proportion $p$ of voters who support a policy). A **statistic** is calculated from a sample (e.g. the sample proportion $\hat{p}$). Different samples give different $\hat{p}$ values, so $\hat{p}$ is a random variable.

- **Simple random sample:** every group of $n$ members is equally likely to be chosen.
- **Bias** is a systematic tendency to over- or under-estimate. Examples: surveying only people at a gym about exercise habits (selection bias); voluntary online polls (self-selection bias); leading questions.
- A bigger sample reduces random variability, but it *doesn't* fix bias.

## The distribution of $\hat{p}$

**Sample proportion**

$$\hat{P} = \frac{X}{n},\ X \sim \text{Bi}(n, p) \qquad E(\hat{P}) = p \qquad \text{sd}(\hat{P}) = \sqrt{\frac{p(1-p)}{n}}$$

For large $n$ (a common guide is $np \geq 10$ and $n(1-p) \geq 10$), $\hat{P}$ is approximately $\text{N}\left(p, \frac{p(1-p)}{n}\right)$.

**Key idea**

Quadrupling the sample size **halves** the standard deviation of $\hat{P}$ (because of the $\sqrt{n}$). Diminishing returns: going from $n = 100$ to $n = 400$ helps as much as going from 400 to 1600.

## Simulating sample proportions

A simulation draws many random samples of size $n$ from a population with proportion $p$ and records $\hat{p}$ for each. A dot plot or histogram of the results shows the distribution of $\hat{P}$: centred near $p$, with spread close to $\sqrt{\frac{p(1-p)}{n}}$, and more bell-shaped as $n$ grows. A bigger $n$ shrinks the spread; with $p$ near 0 or 1, small samples give a skewed distribution.

Simulated confidence intervals show the other key idea: every sample gives a different interval, and about 95% of 95% intervals contain the true $p$. On the TI-Nspire CX II CAS, `randBin(100,0.3,50)/100` gives 50 simulated sample proportions for $n = 100$ and $p = 0.3$.

**Reading a simulation**

From a dot plot of simulated $\hat{p}$ values: **estimate a probability** by counting, e.g. if 7 of 50 simulated values are at least 0.4, $P(\hat{P} \geq 0.4) \approx \frac{7}{50} = 0.14$; **estimate** $p$ by the centre of the plot; **compare sample sizes** by spread (quadrupling $n$ halves the standard deviation). A simulation only estimates: the exact value comes from the binomial distribution of $X = n\hat{P}$.

**Worked example 1** **Mean and sd of p̂** 2 marks

30% of a town's households own a dog. Random samples of 100 households are taken. Find the mean and standard deviation of $\hat{P}$.

1. $E(\hat{P}) = p = 0.3$.
   USING $E(\hat{P}) = p$

✓ $\text{sd}(\hat{P}) = \sqrt{\dfrac{0.3 \times 0.7}{100}} \approx 0.0458$.
USING $\text{sd}(\hat{P}) = \sqrt{\dfrac{p(1-p)}{n}}$

**Worked example 2** TRICKY **Exact vs approximate (a favourite trap)** 4 marks

With $p = 0.3$ and $n = 100$, find $P(\hat{P} > 0.35)$ (a) exactly, using the binomial distribution, and (b) using the normal approximation.

1. (a) $\hat{P} > 0.35 \iff X > 35 \iff X \geq 36$. With $X \sim \text{Bi}(100, 0.3)$: $P(X \geq 36) \approx 0.1161$.
   USING $\hat{P} = \dfrac{X}{n}$, $X \sim \text{Bi}(n, p)$
2. (b) $\hat{P} \approx \text{N}(0.3, 0.0458^2)$: $P(\hat{P} > 0.35) = P(Z > 1.091) \approx 0.1376$.
   USING $\text{sd}(\hat{P}) = \sqrt{\dfrac{p(1-p)}{n}}$ $Z = \dfrac{\hat{p} - p}{\text{sd}(\hat{P})}$

✓ **(a) 0.1161 (exact); (b) 0.1376 (approximate).** Read which one the question asks for. For the exact version, the strict "$>$" matters: $X \geq 36$, not $X \geq 35$.

**Worked example 3** **Exact distribution of p̂ for a small sample (tech-free)** 3 marks

A fair coin is tossed 4 times and $\hat{P}$ is the proportion of heads. Find $P(\hat{P} \geq 0.75)$.

1. $\hat{P} \geq 0.75 \iff X \geq 3$ with $X \sim \text{Bi}(4, \frac{1}{2})$.
   USING $\hat{P} = \dfrac{X}{n}$, $X \sim \text{Bi}(n, p)$
2. $P(X = 3) + P(X = 4) = \dfrac{4}{16} + \dfrac{1}{16}$.
   USING $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$

✓ $\dfrac{5}{16}$. Small samples: convert to a count and use the binomial.

**Trick alert: sampling traps**

- $p$ **vs** $\hat{p}$: the formula for $\text{sd}(\hat{P})$ uses the population $p$ when it's known.
- **Convert proportions to counts carefully:** $\hat{P} \leq 0.2$ with $n = 50$ is $X \leq 10$.
- **Bigger samples don't remove bias.** A huge biased sample is still biased.
- **sd of** $\hat{P}$ **depends on** $\sqrt{n}$, not $n$.

## Summary

- $E(\hat{P}) = p$, $\mathrm{sd}(\hat{P}) = \sqrt{\dfrac{p(1-p)}{n}}$.
- Large $n$: approximately normal. Exact: convert to counts and use $\mathrm{Bi}(n, p)$.
- Random sampling controls bias; sample size controls variability.

> **Also examined in:** 2024 Exam 2 B4 d **(question p. 98; answer p. 99)**

DATA ANALYSIS, PROBABILITY & STATISTICS

# Confidence intervals for proportions

## The interval

**Approximate confidence interval for $p$**

$$\left(\hat{p} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}},\ \hat{p} + z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$$

$z = 1.96$ for 95% ($\approx 2$ tech-free), $z \approx 2.576$ for 99%, $z \approx 1.645$ for 90%. The **margin of error** is $M = z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, half the width.

**What "95% confident" means**

If we took many samples and built a 95% interval from each, about 95% of those intervals would contain the true $p$. The true $p$ is fixed: any single interval either contains it or doesn't.

**Worked example 1** Building a 95% interval — 3 marks

In a random sample of 200 students, 60 walk to school. Find an approximate 95% confidence interval for the proportion of all students who walk, to 4 decimal places.

1. $\hat{p} = \frac{60}{200} = 0.3$. Standard error $= \sqrt{\frac{0.3 \times 0.7}{200}} \approx 0.03240$.
   USING $\hat{p} = \frac{x}{n}$ $\quad \text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
2. $M = 1.96 \times 0.03240 \approx 0.0635$.
   USING $M = z \times \text{SE}$ $\quad z = 1.96$ (95%)

✓ $(0.2365,\ 0.3635)$.
USING $(\hat{p} - M, \hat{p} + M)$

**Worked example 2** A 99% interval is wider — 2 marks

In a sample of 400 voters, 248 support a proposal. Find an approximate 99% confidence interval for $p$.

1. $\hat{p} = 0.62$; SE $= \sqrt{\frac{0.62 \times 0.38}{400}} \approx 0.02427$; $M = 2.5758 \times 0.02427 \approx 0.0625$.
   USING $\text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\quad M = z \times \text{SE}$ $\quad z = 2.5758$ (99%)

✓ $(0.5575,\ 0.6825)$. The 95% interval is $(0.5724, 0.6676)$. More confidence needs a wider net.
USING $(\hat{p} - M, \hat{p} + M)$

**Reading an interval (a, b) backwards** — TRICKY

An approximate interval $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$ is symmetric, so from $(a, b)$ you can read three things: the **sample proportion** $\hat{p} = \frac{a+b}{2}$, the **margin of error** $M = \frac{b-a}{2}$, and then either the **confidence level** (find $z = M \div$ standard error, then the central area between $\pm z$) or the **sample size** (rearrange $M = z\sqrt{\hat{p}(1-\hat{p})/n}$ for $n$). The number of successes $n\hat{p}$ should be a whole number: use that to check $n$.

Probability

**Worked example 3** TRICKY **Working backwards from an interval (hard)** — 3 marks

An approximate 95% confidence interval for $p$ is $(0.52, 0.64)$. Find $\hat{p}$, the margin of error and the sample size.

1. $\hat{p}$ is the midpoint: $\dfrac{0.52+0.64}{2} = 0.58$. $M$ is half the width: $0.06$.
   USING $\hat{p} = \text{midpoint}$ $M = \dfrac{1}{2}(\text{width})$
2. $0.06 = 1.96\sqrt{\dfrac{0.58 \times 0.42}{n}} \Rightarrow n = \dfrac{1.96^2 \times 0.2436}{0.06^2} \approx 259.9$.
   USING $M = z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$

✓ $\hat{p} = 0.58$, $M = 0.06$, $n = 260$.

**Worked example 4** **Finding the confidence level** — 2 marks

A sample of 200 gives $\hat{p} = 0.3$ and the interval $(0.2467,\ 0.3533)$. Find the level of confidence.

1. $M = 0.0533$; SE $= \sqrt{\dfrac{0.3 \times 0.7}{200}} \approx 0.03240$; $z = \dfrac{0.0533}{0.03240} \approx 1.645$.
   USING $M = \dfrac{1}{2}(\text{width})$, $z = \dfrac{M}{\text{SE}}$

✓ $P(-1.645 < Z < 1.645) \approx 0.90$: **a 90% confidence interval**. On CAS: `normCdf(-1.645,1.645)`.

**Worked example 5** **Choosing a sample size** — 2 marks

A pollster wants a 95% margin of error of at most 3 percentage points and has no idea of $p$. How many people should be surveyed?

1. Use $\hat{p} = 0.5$ (which gives the largest possible margin): $1.96\sqrt{\dfrac{0.25}{n}} \leq 0.03 \Rightarrow n \geq \dfrac{1.96^2 \times 0.25}{0.03^2} \approx 1067.1$.
   USING $M = z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} = 0.5$ (most conservative)

✓ $n = 1068$ (round up). This is why national polls often survey about 1000 people.
USING $n$ rounds up

## Using an interval to judge a claim

If a claimed value of $p$ lies **inside** the interval, the data are consistent with the claim. If it's **outside**, the claim is doubtful (at that confidence level). An interval never *proves* the claim.

**Worked example 6** **How many intervals contain p?** — 2 marks

Twenty independent samples each give an approximate 95% confidence interval for $p$. Find the probability that all 20 contain $p$, and that at least 18 do.

1. $P(Y = 20) = 0.95^{20}$; $P(Y \geq 18)$ = `binomCdf(20,0.95,18,20)`.
   USING $Y \sim \text{Bi}(20, 0.95)$

✓ $0.95^{20} \approx 0.3585$; $P(Y \geq 18) \approx 0.9245$. On average 19 of the 20 contain $p$.

**Trick alert: confidence interval traps**

- **Don't say "there's a 95% probability that $p$ is in (0.24, 0.36)".** VCAA wants the "95% of such intervals" interpretation.
- **Higher confidence → wider interval; larger $n$ → narrower.** To halve the width, quadruple $n$.
- **Round sample sizes up when designing for a margin of at most $M$**, even from 1067.1. When recovering $n$ from a given (rounded) interval, round to the nearest whole number instead.
- **The interval is about the population**, not about future samples.

## Summary

- $\hat{p} \pm z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$ with $z = 1.96$ (95%).
- Midpoint = $\hat{p}$; half-width = margin of error.
- Sample size: rearrange for $n$, use $\hat{p} = 0.5$ if unknown, round up.
- Interpret as "95% of such intervals contain $p$".

**Also examined in:** 2024 Exam 2 B4 e **(question p. 98; answer p. 99)**; 2023 Exam 2 B4 g **(question p. 102; answer p. 103)**

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